Maritime Trade

Maritime Trade For Dummies

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What Is Maritime Trade?

Maritime trade refers to the movement of goods and commodities by sea using commercial ships. It encompasses not just the physical transport but also the entire ecosystem of loading, unloading, logistics, port operations, and related services like bunkering (fueling ships). It is so fundamental that it **accounts for over 80% of global trade by volume**.

Why it matters:

- A container of electronics from Shanghai to Rotterdam costs a fraction of air freight and moves in a predictable 28–35 days.
- A single large bulk carrier can move 400,000 tons of iron ore, lowering the per-unit cost to near zero.
- Without it, modern consumer economies and industrial supply chains wouldn't exist.

It's crucial to understand that maritime trade splits into two very different business models, **Liner Shipping** & **Tramp / Bulk Shipping**:

Feature	**Liner Shipping (Container)**	**Tramp / Bulk Shipping**
What it moves	Finished/Semi-finished goods in containers	Raw commodities (oil, ore, LNG, Coal, Grain)
How it operates	Fixed routes, fixed schedules (like a bus line). Publicly advertised.	No fixed schedule; goes where cargo is (like a taxi).
Contract	A Booking Note for slot space. Rate per container.	A Charter Party (a lease of the entire vessel or a cargo hold). Rate is paid as a lump sum (lump sum freight) or per ton.
Key Terms	FCL/LCL, demurrage & detention, FAK (Freight All Kinds).	Laytime (time for loading), demurrage (penalty if charterer delays ship), notice of readiness, Laycan, Despatch.
Pricing model	Complex "FAK" rates + a web of surcharges (BAF, CAF, PSS, etc.).	Driven purely by supply/demand for ships and commodities; extremely volatile. Baltic Dry Index (BDI) is the benchmark.

Bulk/Tramp Shipping can itself be split further by the type of its cargo:

- **Dry Bulk Trade**: The movement of unpackaged raw materials in large volumes. This includes major commodities like iron ore (for steel), coal (for energy), grain (for food), and cement. Ships used here are Bulk Carriers.

- **Liquid Bulk Trade**: The movement of liquid commodities. This is dominated by Crude Oil and refined petroleum products, but also includes Liquefied Natural Gas (LNG) and Chemicals. Ships used here are Tankers (VLCCs, ULCCs, etc.).
- **Breakbulk** and Specialized Trade: This involves cargo that cannot fit in a standard container or bulk hold. It includes heavy machinery, project cargo (like wind turbine blades), timber, steel pipes, and vehicles. Ships used here include Ro-Ro (Roll-on/Roll-off for cars), Heavy-Lift vessels, and traditional general cargo ships.

Shipping, usually bulk, also differs by the Commercial Charter Arrangement (how the ship is hired):

- **Voyage Charter**: The shipowner provides the vessel for a specific single trip (e.g., from Port A to Port B) for an agreed price. The shipowner pays for the voyage costs (fuel, canal fees, crew).
- **Time Charter**: The charterer hires the vessel for a specific period of time (weeks, months, or years). The charterer controls where the ship goes and what cargo it carries, but the shipowner retains operational control and provides the crew.
- **Bareboat Charter (Demise Charter)**: The charterer takes full possession and control of the vessel for a long period. The charterer provides the crew, pays for all operating costs, and essentially acts as the shipowner for the duration of the contract, returning the empty vessel at the end.

Furthermore, Shipping also differs by the route of a ship. In simple terms, distance and the regional waters the ship traverses:

- **Short-Sea Shipping (Coastal/Regional)**: This refers to the movement of cargo over relatively short distances, usually along a single continent's coastline or between neighboring countries (e.g., shipping goods from the UK to mainland Europe, or along the US East Coast). It often acts as a "feeder" service, collecting smaller cargo loads and funneling them to major hub ports.
- **Deep-Sea Shipping (Intercontinental/Ocean-Going)**: This involves crossing vast oceans to connect continents (e.g., Asia to Europe, North America to Asia, or the Middle East to the Americas). This is the main artery of global trade, involving massive vessels that sail for weeks at a time.
- **Cabotage**: A specific legal/geographic type of trade that refers to the transport of goods or passengers between two ports within the same country. Many countries heavily restrict cabotage to domestically flagged, owned, and crewed ships (e.g., the Jones Act in the United States).

In practice, these types overlap. For example, a massive container ship carrying electronics from Shanghai to Rotterdam is engaging in Liner Trade, carrying Containerized Cargo, via a Deep-Sea route. Conversely, a tanker carrying crude oil from the Middle East to India is engaging in Tramp Trade, carrying Liquid Bulk, on a Voyage Charter basis.

Who Is Involved?

Core Commercial Entities

Entity	Role	Seen in
Shipper / Cargo Owner / Exporter	Owns the goods and initiates the shipment. Could be a manufacturer, commodity trader, or government.	Both
Consignee / Importer / Receiver	The party entitled to take delivery of the cargo at destination.	Both
Carrier / Shipping Line / Vessel Operator	Owns or operates the ships and provides the sea transport. Sells slots (container) or leases the whole vessel (bulk).	Both (but role differs)
Freight Forwarder	The “architect” for small/medium shippers. Books space with carriers, consolidates cargo, handles documents, and coordinates the entire multimodal journey. Does not own ships.	Primarily Container (Liner)
NVOCC (Non-Vessel Operating Common Carrier)	Buys bulk container slots from carriers at wholesale and resells them to shippers. Issues its own Bill of Lading. Acts as a carrier to the shipper, and a shipper to the actual line.	Container only
Charterer	The party who hires an entire ship from a shipowner, usually under a Charter Party. A commodity trader, industrial company, or even another shipping line.	Primarily Bulk (Tramp)
Shipowner / Disponent Owner	The company that owns the vessel, or the one with the right to commercially operate it (the "disponent owner" in a chain of charters). Provides the ship and crew.	Bulk / Tramp (Some container chartering also uses brokers, but not for cargo bookings.)
Customs Broker	Licensed professional who files customs declarations, calculates duties, and ensures compliance with import/export regulations for the cargo.	Both
Shipping Agent (Owner’s/Charterer’s Agent)	The local representative of the vessel in a port. Handles port call formalities, orders pilots and tugs, arranges crew changes, provisions, and formally tenders the Notice of Readiness (NOR) in bulk.	Both (critical in bulk for NOR & SOF)

Port & Terminal Entities

Entity	Role	Seen in
Port Authority	The governing body that manages the port's infrastructure, waterways, and land. Leases terminals to operators and ensures safety/security. Usually public or semi-public.	Both
Terminal Operator	The company that runs a specific cargo-handling facility. Invests in cranes, conveyor belts, pipelines, and IT systems. Loads/discharges the ship and manages the yard. (E.g., DP World, APM Terminals, dedicated grain elevators.)	Both (but the equipment and pace differ dramatically)
Stevedores	Labour companies that physically load and discharge cargo. In modern ports, this function is often integrated into the terminal operator.	Both
Harbor Pilot	A local navigation expert who boards the ship and guides it safely through the port's channels to the berth. The captain retains command but follows the pilot's advice.	Both
Tugboat	Provides the muscle to push/pull large vessels onto the dock.	Both
Independent Marine Surveyor	Performs draft surveys to measure exact cargo weight (bulk), inspects cargo condition, hatch covers, and tanks. Their reports are the basis for freight payment and quality disputes.	Primarily Bulk (Tramp) (Container cargo is generally not weighed by draft survey; VGM is certified separately.)
Port Health / Phytosanitary / Agricultural Inspectors	Inspect bulk grain for pests, check crew health, and ensure compliance with food safety/agricultural regulations.	Primarily Bulk (especially grain, foodstuffs)

Maritime Operations at Sea

Entity	Role	Seen in
Master (Captain)	In charge of the vessel, crew, and safe navigation. In tramp, has crucial commercial responsibilities: tendering NOR, signing Bills of Lading, logging delays, and issuing Letters of Protest. In container shipping, the call is more automated and planned by others.	Both (greater commercial role in Bulk)
Chief Engineer & Crew	Operate and maintain the ship's machinery.	Both
P & I Club Correspondent	Local representative of the shipowner's liability insurer (Protection & Indemnity Club). Stands by to investigate accidents, pollution, cargo damage, or crew injury claims.	Both

Financial & Legal Entities

Entity	Role	Seen in
Marine Cargo Insurer	Covers the goods against physical loss or damage during the voyage. A policy is almost universally required.	Both
Hull & Machinery (H&M) Underwriter	Insures the ship itself against physical damage.	Both
P&I Club (Protection & Indemnity)	A mutual insurance association covering shipowners' third-party liabilities: cargo loss, collision, pollution, injury, wreck removal.	Both
Trade Finances Bank	Issues Letters of Credit (L/C), which guarantee payment to the exporter upon presentation of compliant documents (including the Bill of Lading). Bridges the trust gap between buyer and seller.	Both
Average Adjuster	A specialist who calculates the financial apportionment when a "general average" is declared (e.g., jettisoning cargo to save the ship, all parties proportionally share the loss).	Both (though more frequent in tramp)
Law Firm / Arbitrator	Maritime law is a specialist field. Disputes under Charter Parties or Bills of Lading are overwhelmingly settled by arbitration (LMAA in London, SMA in New York, etc.), not courts.	Both

Regulatory, Technical & Oversight Bodies

Entity	Role	Seen in
Flag State	The country where the vessel is registered. It has regulatory control and must enforce international conventions (SOLAS, MARPOL) on its ships. Many vessels fly “flags of convenience” (Panama, Liberia).	Both
Classification Society	An independent body that certifies that a ship is designed, built, and maintained to technical and safety standards. Issues a “class certificate” essential for insurance and port entry. (E.g., Lloyd’s Register, DNV, ABS.)	Both
Port State Control (PSC)	The inspection regime a coastal nation uses to examine foreign-flagged ships in its ports, detaining those that are substandard (e.g., Paris MoU, Tokyo MoU).	Both
IMO (International Maritime Organization)	The UN agency that sets global rules for safety, security, and environmental performance (SOLAS, MARPOL, ISPS Code).	Both
National Customs Authority	The government body that controls the flow of goods across borders, collects duties/taxes, and enforces trade rules.	Both

Real-World Scenarios

Scenario 1: The VLCC Crude Oil Run (Middle East to China)

- **Type**: Tramp Trade / Liquid Bulk / Voyage Charter
- **The Ship**: A Very Large Crude Carrier (VLCC) – 330 meters long, carrying 2 million barrels of oil.
- **The Cast**:
    - Saudi Aramco (Shipper/Charterer)
    - Greek Shipowner (owns the VLCC)
    - London-based Ship Broker (negotiated the single voyage)
    - P&I Club (highly vigilant due to pollution risk)
    - Offshore Marine Pilot
    - SPM (Single Point Mooring) terminal crew.
- **How it unfolds**:
    - **The broker** secures a **Voyage Charter** between **Aramco** and the **Greek owner**. The **charterer** pays a fixed price per ton of oil.
    - The ship doesn't go to a regular dock. It anchors offshore at Ras Tanura, Saudi Arabia. A harbor pilot doesn't board; instead, a specialized mooring master attaches the ship's bow to an underwater pipeline buoy (the **SPM**). Stevedores don't exist here; giant submersible pumps on the ship do the loading.
    - The ship sails through the Strait of Hormuz—a naval and geopolitical hot spot. The **Ship Manager** onshore tracks its route constantly.
    - Upon arrival in Ningbo, China, the local **Ship Agent** arranges for tugs. A **Port State Control inspector** boards to check the double-hull integrity. The **Customs Broker** clears the cargo based on the **Bill of Lading**, while the **Trade Finance Bank** releases the massive payment (often hundreds of millions of dollars) against the presented documents. The **P&I Club** remains intimately involved because a spill here would trigger their multi-billion-dollar liability coverage.

Scenario 2: FCL via Freight Forwarder

- **Type**: Liner Trade / Container (Full Container Load – FCL)
- **The Ship**: A 14,000 TEU Post-Panamax container vessel, ONE Stork, 366 meters long. She operates on a fixed weekly string between Yantian and Los Angeles.
- **The Cast & Their Roles:**
    - **Shipper (Exporter):** The electronics factory in Shenzhen. They pack the goods into a 40ft container and provide the commercial invoice and packing list.
    - **Consignee (Importer)**: The US retailer in Chicago, responsible for the import side.

  - **Freight Forwarder**: The master planner. The US buyer contracts them to handle everything from factory to warehouse. The forwarder books space with a carrier, arranges the truck in Shenzhen, files customs documents, and even coordinates the final rail move from Los Angeles to Chicago.
  - **Carrier (Shipping Line):** Ocean Network Express (ONE). The forwarder books a slot on ONE's Transpacific service, receiving a Booking Confirmation. ONE issues the Master Bill of Lading.
  - **Terminal Operator (Yantian International Container Terminals)**: Receives the full container, stacks it, and, following a stowage plan, loads it onto the ONE vessel using massive gantry cranes.
  - **Port Authority**: Shenzhen Port Authority, managing the waterfront infrastructure and ensuring the terminal operator follows safety/security rules.
  - **Harbor Pilot & Tugboat Operator:** A local pilot boards the container ship outside Yantian, guiding it to the berth. Two tugs assist.
  - **Customs Broker (Export side)**: A Shenzhen-based broker files the Chinese export declaration using the factory's documents and gets the customs release.
  - **Customs Broker (Import side):** A US broker files the Importer Security Filing (ISF) before sailing and clears the goods through US customs upon arrival in LA, paying duty.
  - **Shipping Agent (ONE's agent in LA)**: Oversees the vessel's port call, arranges the pilot for inbound, handles the crew, and ensures the manifest is transmitted to US Customs.
  - **Terminal Operator (Fenix Marine Services, LA)**: Discharges the container. The consignee's trucker picks it up after the import broker has electronically released it and the terminal receives the PIN.
  - **Rail Carrier (BNSF)**: The forwarder arranges a cross-country rail move from the LA intermodal yard to a Chicago railhead.
  - **Marine Insurer**: The US buyer's cargo policy covers the goods from factory to warehouse under an "all risks" policy.
- **How it unfolds**:
  - The US retailer, buying on FOB Shenzhen terms, instructs its contracted freight forwarder to handle everything.
  - The forwarder books a 40ft slot with ONE, receiving a Booking Confirmation. A trucker collects the empty container from the carrier's depot, and the electronics factory loads the smart hubs into it.
  - The export customs broker files the Chinese export declaration; once cleared, the full container is gated into Yantian terminal, weighed (VGM), and stacked in the yard.
  - ONE Stork arrives on schedule. The harbor pilot boards outside the breakwater, and tugs nudge the massive ship to the berth.
  - The terminal's automated gantry cranes execute the stowage plan, slotting the container into a stack destined for LA.
  - The shipping agent manages crew needs and transmits the electronic manifest to US Customs.

    - During the Pacific crossing, the forwarder's import broker files the Importer Security Filing (ISF).
    - Upon arrival in Los Angeles, the ship is piloted in again.
    - Fenix Marine Services terminal discharges the box.
    - The import broker completes customs clearance, duties are paid, and the terminal releases the container.
    - The forwarder's arranged BNSF rail move carries it to a Chicago rail yard, and a local trucker delivers it to the retailer's warehouse.
    - The marine insurer covers the cargo for the full multimodal journey under a single door-to-door policy.

Scenario 3: The Grain – Voyage Charter with Laytime Drama

- **Type**: Tramp Trade / Dry Bulk / Voyage Charter
- **The Ship**: A 58,000 DWT Supramax bulk carrier, Sea Harvest, 190 meters long, with four 30-tonne deck cranes and five large open holds. No fixed route; she ballasts empty across the globe to pick up cargo.
- **The Cast & Their Roles:**
    - **Charterer**: Cargill. Needs to fulfill the CIF sale and pays for the ocean freight. They are hiring the vessel.
    - **Shipbroker** (**Cargill's broker)**: In London, this broker sends out a "cargo order" to the market, gets offers from owners, and negotiates the fixture.
    - **Shipowner**: A Greek family-run company, Alpha Shipping, owns the Supramax bulk carrier "MV Sea Harvest." Their broker works the other side of the deal.
    - **Shipbroker** (**Owner's broker**): Negotiates on Alpha Shipping's behalf. They agree to a North American Grain Charter Party 1973 (NORGRAIN) with riders.
    - **Master (Captain of Sea Harvest)**: The pivotal commercial figure. When the ship arrives at Paranaguá pilot station, he hands the Notice of Readiness (NOR) to the agent. He logs every rain shower, every stoppage, and signs the Statement of Facts (SOF) at the end of the load.
    - **Shipping Agent (Owner's agent in Paranaguá)**: Appointed by Alpha Shipping. Tenders the NOR to the grain terminal on behalf of the Master. Compiles the detailed SOF from the ship's logs and terminal reports.
    - **Terminal Operator (Paranaguá Export Corridor)**: Operates the conveyor belts and ship-loader that pour soybeans into Sea Harvest's holds. Their hourly load rate is the basis for the SOF.
    - **Independent Marine Surveyor**: Boards the vessel on arrival for a draft survey before loading (reading the ship's draught marks and measuring ballast water). After loading, performs the final draft survey to determine the exact weight loaded: 60,447 MT. This figure becomes the freight bill.
    - **Port Authority (APPA)**: Manages the port, collects port dues from the shipowner (per the voyage charter), and maintains the navigable channel.

    - **P&I Club Correspondent (in Paranaguá)**: Standing by. When a stevedore's bulldozer accidentally dents the hatch coaming, the Master issues a Letter of Protest, and the correspondent surveys the damage for a potential claim.
    - **Consignee (Chinese feed mill)**: The end buyer. Presents the original Bill of Lading to the shipping agent in Qingdao to obtain a Delivery Order to get their soybeans.
    - **Laytime Clerk (for both Cargill and Alpha Shipping)**: Back offices take the signed SOF and calculate laytime used vs. allowed. A rain delay of 4h 35min is deducted. Cargill's demurrage bill comes to $3,200. They cross-check and settle.
- **How it unfolds**:
    - Weeks before loading, Cargill's shipbroker negotiates a voyage charter fixture with the owner's broker, using the NORGRAIN form. The main terms: 60,000 MT soybeans (±10% MOLOO), $45/MT freight, laytime 10,000 MT per day SHEX, WIBON, demurrage $15,000/day. The owner then orders Sea Harvest to ballast from Singapore to Paranaguá.
    - When the ship arrives at the pilot station, the Master immediately hands the Notice of Readiness (NOR) to the owner's agent, who tenders it to the terminal.
    - The berth is occupied, but because the C/P says "WIBON," the laytime clock starts ticking at 13:00 (since NOR was accepted before noon). Sea Harvest waits at anchor, burning Cargill's allowed time.
    - Once berthed, an independent marine surveyor boards and carries out the initial draft survey, measuring the ship's empty weight.
    - The terminal operator's conveyor belts pour soybeans into the holds.
    - The Master logs every event—rain stoppages, crane delays—which the agent compiles into the Statement of Facts (SOF) .
    - After loading, the final draft survey calculates exactly 60,447 MT. Freight is paid by Cargill to the owner based on this figure.
    - The Master signs the Bills of Lading, which Cargill's bank exchanges in a Letter of Credit transaction.
    - During loading, a stevedore's machine dents a hatch coaming. The Master issues a Letter of Protest, and the local P&I Club correspondent surveys the damage for a future claim.
    - At the destination of Qingdao, the Chinese feed mill presents an original Bill of Lading to the agent to obtain a Delivery Order.
    - After discharge, Cargill's and Alpha Shipping's laytime clerks take the SOF and calculate that 4 hours 35 minutes of rain time is deducted.
    - Cargill exceeded the allowed laytime by just over 5 hours, owing $3,200 in demurrage. The two sides reconcile and settle the statement.

Scenario 4: The Tanker – A VLCC Crude Oil Voyage

- **Type**: Tramp Trade / Liquid Bulk / Voyage Charter

- **The Ship**: A Very Large Crude Carrier (VLCC), Energy Glory, 330 meters long, with a cargo capacity of 2 million barrels in its central and wing tanks. Double-hulled, flying the Malaysian flag.
- **The Cast & Their Roles:**
    - **Charterer**: Shell International Trading and Shipping Company (STASCO). They control the cargo and hire the tanker.
    - **Shipbroker (Tanker broker)**: Specialist in dirty tankers. Negotiates the fixture on an amended BPVOY6 form. The deal is “WS (Worldscale) 45 flat, Ras Tanura to Rotterdam, one safe berth.”
    - **Shipowner**: A Malaysian tanker company owns “Energy Glory,” providing the crew and the vessel. They pay for fuel, port charges, and the Suez Canal transit (this is a voyage charter).
    - **Master**: Tenders NOR via the agent at Ras Tanura. Later, in Rotterdam, he issues a Letter of Protest if the shore-side pumping pressure is lower than agreed, delaying the discharge.
    - **Shipping Agent (at Ras Tanura)**: Coordinates with Saudi Aramco’s terminal, arranges the pilot, and formally tenders NOR.
    - **Terminal Operator (Ras Tanura Terminal)**: Pumps crude via flexible hoses into the VLCC’s massive tanks. An ullage report is jointly taken by the Chief Officer and the terminal operator to measure the empty space in each tank.
    - **Independent Marine Surveyor**: At both load and discharge, a surveyor measures cargo temperature, takes ullages, and calculates the gross and net standard volumes using the density. This determines the precise Bill of Lading quantity and the outturn at the refinery.
    - **Customs Broker (Dutch side)**: Files import declarations, arranging for the crude to be cleared into the bonded refinery circuit.
    - **P&I Club Correspondent (Rotterdam)**: Present if any oil spill is reported during the disconnection of hoses.
    - **Port Authority (Rotterdam)**: Assigns the berth, coordinates Vessel Traffic Services, and enforces port regulations.
    - **Receivers (Refinery)**: The ultimate consignee, the refinery that processes the crude. They sample the oil to ensure it meets specification.
- **How it unfolds**:
    - Shell’s chartering desk uses its tanker broker to fix Energy Glory on an amended BPVOY6 voyage charter: a single voyage from Ras Tanura to Rotterdam, freight expressed as WS (Worldscale) 45 flat. The shipowner pays for bunkers, port charges, and the Suez Canal transit.
    - At Ras Tanura, the ship does not go alongside a quay; it moors at an offshore SPM (Single Point Mooring) .
    - A mooring master boards and attaches the VLCC’s bow to the buoy. The Master tenders NOR, and laytime starts.
    - The terminal begins pumping Basrah Light crude through flexible hoses.

    - The Chief Officer and an independent surveyor jointly take ullage readings and measure cargo temperature. Using the oil's density, they compute the precise net barrel volume, which is recorded on the Bill of Lading.
    - This Bill of Lading is then used in Shell's trade finance process—a bank may release payment upon presentation of documents.
    - During the long voyage via the Suez Canal and around to Rotterdam, the shipowner's technical department monitors fuel consumption.
    - On arrival at the Rotterdam refinery berth, the local shipping agent arranges the pilot and tugs.
    - A Port State Control officer may inspect the ship's double-hull integrity and MARPOL compliance.
    - The P&I correspondent stands by as a precaution; any spill during hose disconnection triggers the P&I Club's multi-billion-dollar pollution cover.
    - The independent surveyor again measures the ullage on arrival.
    - The receiver's customs broker clears the crude into the bonded refinery circuit. The outturn shows a 0.2% shortage, which falls within the 0.5% customary tolerance, so no claim is filed—but the independent survey report is the absolute reference point.
    - The laytime clerks then calculate demurrage or despatch based on the pumping log SOF.

Scenario 5: The Wind Turbine – Project & Heavy-Lift Cargo

- **Type**: Project Cargo / Heavy-Lift / Part Charter
- **The Ship**: A 12,500 DWT heavy-lift vessel, MV Vera, 138 meters long, equipped with two 800-tonne cranes that can work in tandem for lifts up to 1,600 tonnes. She has a flat, open deck and unboxed holds, sailing from Aarhus to Chile.
- **The Cast & Their Roles:**
    - **Shipper/Manufacturer**: Vestas Wind Systems, Aarhus. They need to deliver the components to the Chilean project site, but their true expertise is manufacturing, not logistics.
    - **Freight Forwarder (Project Cargo Specialist)**: A niche forwarder like DHL Industrial Projects or deugro. They engineer the entire multimodal move: road transport from factory to quay, heavy-lift ship chartering, stowage planning, and sea-fastening design.
    - **Shipowner / Heavy-Lift Operator**: A company like SAL Heavy Lift, operating the MV "Vera," which has two 800-tonne cranes that can work in tandem. The forwarder will book "part charter" space on Vera, which is also carrying other heavy cargo.
    - **Marine Warranty Surveyor (MWS)**: A specialized third-party surveyor (often from a classification society or a firm like LOC Group) who reviews and approves the lifting plan, the stowage arrangement, and the sea-fastening (lashing and

welding to the deck) to ensure the cargo won't shift in a storm. Insurance is conditional on their approval.
    - **Terminal Operator (Port of Aarhus):** A dedicated heavy-lift berth with high load-bearing quay strength. The terminal coordinates the precise positioning of the vessel's cranes and the shore-side mobile cranes.
    - **Stevedores (Heavy-lift riggers)**: A highly skilled gang who attach the lifting slings and guide the massive nacelles onto the vessel's deck.
    - **Port Captain & Pilot**: A senior pilot coordinates the delicate maneuver. The harbor basin must be closed to other traffic during the lift.
    - **Marine Insurer**: Lloyd's syndicate provides a tailored "all risks" policy covering the cargo from the factory floor to the final site, including the extreme risk of a lift failure.
    - **Consignee (Chilean wind farm company)**: The developer. They have no capability to handle this. They rely entirely on the specialist forwarder's door-to-door contract. The Bill of Lading is a non-negotiable Sea Waybill for speed.
- **How it unfolds**:
    - The project freight forwarder is contracted by Vestas to deliver four 100-tonne nacelles and 50-meter blades to a remote Chilean wind farm.
    - They (Freight forwarder) engineer the entire door-to-port move and then book "part charter" space on MV Vera, which is also carrying other heavy cargo. A heavy-lift shipbroker may assist in the chartering.
    - Before any metal is lifted, the Marine Warranty Surveyor (MWS) reviews and approves the lift plan, the stowage arrangement, and the sea-fastening design. The marine insurer will not bind the cargo cover without an MWS Certificate of Approval.
    - At the Port of Aarhus heavy-lift terminal, the operation begins only within an approved weather window.
    - The ship's massive cranes swing into action, operated by the vessel's crew and assisted by specialist heavy-lift riggers on the quay. A spreader beam is attached to the first nacelle, and the lift is smooth and slow.
    - The MWS observes every step. The nacelle is placed onto a purpose-built steel cradle on deck. The riggers then weld and chain-lash it down.
    - The MWS inspects and signs the "Certificate of Approval for Sea-Fastening" before Vera can sail.
    - During the ocean passage, the Master constantly monitors lashing tensions.
    - On arrival at the Chilean port of Mejillones, the local shipping agent coordinates the berth.
    - The reverse operation begins: the MWS is present again, the cranes offload the nacelles onto specialized multi-axle trailers provided by the forwarder.
    - The Chilean developer receives the cargo under a non-negotiable Sea Waybill for speed.
    - The forwarder then arranges the inland heavy-haul transport to the wind farm site.

    - The entire project is financed by a letter of credit, with the forwarder's forwarding certificate and the sea waybill as key documents.

Scenario 6: The Car Carrier – Pure Car and Truck Carrier (PCTC)

- **Type**: Liner Trade / Roll-on/Roll-off (Ro-Ro) – Pure Car and Truck Carrier
- **The Ship**: A 200-meter long PCTC, Neptune Leader, with 12 decks, stern and side ramps, and a capacity of 7,000 CEU (Car Equivalent Units). She operates a fixed liner schedule from Nagoya to Zeebrugge.
- **The Cast & Their Roles:**
    - **Shipper**: Toyota Motor Corporation, via its in-house logistics division.
    - **Carrier / Vessel Operator**: "K" Line / Kawasaki Kisen Kaisha, operating a dedicated PCTC, "Neptune Leader," a floating 12-deck parking garage with specialized ramps.
    - **Shipping Agent ("K" Line agent in Nagoya)**: Coordinates the berth, sends loading lists to the terminal, and manages the crew. No complex laytime calculation here; the vessel is on a tight liner schedule.
    - **Terminal Operator (Nagoya Car Terminal)**: A dedicated Ro-Ro terminal. Their team of stevedores/drivers are the key: each one is a professional who drives the brand-new SUVs from the massive parking lot, up the internal ramps, and into a millimeter-precise position on the ship's decks. They secure each vehicle with lashings.
    - **Marine Surveyor (Vehicle Condition)**: Walks the terminal pre-shipment yard with a tablet, inspecting each car for pre-existing damage. Their report becomes the baseline for any transport damage claims. At Zeebrugge, another surveyor repeats the process on discharge, comparing against the "damage matrix."
    - **Customs Broker (Export/Import)**: A Japanese broker handles the export declaration and manages the "temporary import" customs regime for the ship's spare parts. In Belgium, a broker clears the vehicles under the EU's customs declaration.
    - **Consignee**: Toyota Motor Europe, who will then distribute the vehicles to national sales companies.
    - **Document**: A Sea Waybill, not a negotiable Bill of Lading. The cargo goes directly to the known consignee. No ownership transaction depends on it.
- **How it unfolds**:
    - Toyota's logistics division books space directly with "K" Line on the liner service. At the Nagoya Car Terminal, 1,500 brand-new SUVs are parked in a vast holding lot.
    - A team of independent surveyors walks the lot with tablets, photographing each car and recording any pre-existing damage; this creates the baseline "damage matrix."
    - As Neptune Leader calls on schedule, the shipping agent manages the port formalities.

- The terminal's professional stevedore drivers then take over. They drive each SUV up the ship's stern ramp and through the internal spiral ramps to their assigned decks. The cars are parked with only centimeters to spare and immediately lashed to the deck.
- The Chief Officer and the surveyor spot a 1cm scratch on a driver's door; it is photographed, logged, and signed by both parties, preventing a future cargo claim.
- No complex laytime calculation governs this; the ship operates to a tight liner schedule.
- The master issues a single Sea Waybill (not a negotiable B/L) for the entire consignment, listing Toyota Motor Europe as the consignee.
- The Japanese customs broker handles the export declaration electronically.
- Upon arrival in Zeebrugge, the ship is guided into the dedicated Ro-Ro terminal. Belgian customs brokers clear the vehicles.
- The same type of independent surveyor now inspects each car as it is driven off against the pre-shipment damage matrix. Any fresh damage is flagged as transport damage, and the marine insurer is notified.
- The vehicles are driven to a post-shipment holding area, from where Toyota Motor Europe distributes them to national sales companies across the EU.
- The entire process is swift, transparent, and built around the Ro-Ro principle: drive on, sail, drive off.

Entity Type	**Scenario 2: Container**	**Scenario 3: Grain Bulk**	**Scenario 4: Crude Tanker**	**Scenario 5: Wind Turbine**	**Scenario 6: Car Carrier**
Shipper/Charterer	Factory, buyer	Cargill	Shell	Vestas	Toyota
Freight Forwarder	✅	❌	❌	✅ (specialist)	❌ (in-house)
Carrier/Owner	ONE (line)	Alpha Shipping	Malaysian Tanker Co.	SAL Heavy Lift	"K" Line
Shipbroker	❌	✅	✅	Possibly	❌
Shipping Agent	✅	✅	✅	✅	✅
Terminal Operator	Yantian, Fenix	Paranaguá Corridor	Ras Tanura	Aarhus heavy-lift	Nagoya Ro-Ro
Harbor Pilot/Tugs	✅	✅	✅	✅	✅
Independent Marine Surveyor	❌ (VGM by shipper)	✅ (draft survey)	✅ (ullage)	✅ (Marine Warranty)	✅ (car condition)

Entity Type	**Scenario 2: Container**	**Scenario 3: Grain Bulk**	**Scenario 4: Crude Tanker**	**Scenario 5: Wind Turbine**	**Scenario 6: Car Carrier**
Customs Broker	✅ (both sides)	❌ (assumed)	✅ (import)	✅	✅
Laytime/Demurrage	(Detention/Demurrage on boxes)	**Central**	**Central**	Minor	Not present (liner)
Key Document	B/L	Charter Party, NOR, SOF	Charter Party, Ullage Report	MWS approval, Sea Waybill	Sea Waybill, Damage Matrix

Time Is Money—But Not in the Same Way

In all shipping, cargo uses expensive assets (a ship, a terminal, a container). When you hold that asset beyond the allowed time, you pay. But the logic, the names, and the documents are completely different depending on whether you're in bulk/tramp or container shipping.

Concept	**Bulk/Tramp Shipping**	**Container (Liner) Shipping**
What is the scarce asset?	The ship itself, and its time.	The container (equipment) and the terminal yard space.
Time allowance	**Laytime** – contractually agreed hours/days for loading and discharging.	**Free Time** – calendar days from container discharge (import) or empty pickup (export) until the box must be returned or the full container collected.
Penalty for exceeding time	**Demurrage** – paid by the charterer to the shipowner.	**Port Demurrage** (full container sits too long at terminal) and Detention (empty container not returned on time to the carrier's depot).
Reward for finishing early	**Despatch** – paid by the shipowner to the charterer (if agreed).	No despatch. You can sometimes buy extra free time in advance, but you're not rewarded for being early.
Governing document	**Charter Party** (negotiated bespoke contract).	**Carrier's tariff / service contract** (standard terms, non-negotiable for most).

Bulk / Tramp Shipping – Laytime, Demurrage & Despatch

This is the 150-year-old commercial framework where the ship itself is leased. The clock doesn't tick on a container; it ticks on the entire vessel.

Laytime: The Foundation

Laytime is the period agreed in the charter party during **which the charterer is allowed to load and discharge the cargo without paying additional freight**. It's essentially free time for the ship.

- **How it's expressed**: Usually as a daily rate, e.g., "10,000 metric tons per day SHEX". If the cargo is 60,000 MT, the allowed laytime = 60,000 / 10,000 = 6 days. It can also be a fixed number of "running hours" or "weather working days."
- **When it starts**: Laytime commences when a valid Notice of Readiness (NOR) has been tendered and accepted, according to the C/P. Usually, "if NOR tendered before noon, laytime starts at 13:00; if after noon, at 08:00 next working day."

- **When it ends**: Upon completion of loading (or discharge), when cargo is secured and hatches closed (or hoses disconnected). The exact moment is recorded in the Statement of Facts (SOF).

The charter party is riddled with clauses that modify how laytime is counted:

- **SHEX (Sundays and Holidays Excepted)**: Sundays and holidays are not counted, even if work takes place.
- **SHINC (Sundays and Holidays Included)**: Every day counts.
- **WIBON (Whether In Berth Or Not)**: NOR can be tendered even if the berth is occupied; laytime starts running while the ship waits at anchorage.
- **ATUTC (All Time Used To Count)**: All time actually used during excepted periods counts as laytime.
- **Weather Working Day (WWD)**: Time lost due to rain, storms, etc., is not counted.

All these exceptions are meticulously recorded in the Statement of Facts (SOF) by the master and port agent.

Demurrage (Bulk)

Demurrage is liquidated damages paid by the charterer to the shipowner for exceeding the allowed laytime. It's a contractual penalty that accrues for every day (or pro-rata fraction) the ship is delayed beyond the laytime.

- **Rate**: Agreed in the Charter Party (e.g., $15,000 per day or pro rata).
- **How it's triggered**: The moment total used laytime exceeds total allowed laytime, demurrage begins. There is usually no grace period; it's automatic.
- **Nature**: In English law, demurrage is payable "once on demurrage, always on demurrage." Even if a holiday or rain day occurs during the demurrage period, it counts because the demurrage clock is continuous and no exceptions apply unless explicitly stated.
- **Who pays?** The charterer. **The charterer may in turn try to pass this cost to the terminal, seller, or buyer depending on the sales contract, but the shipowner's claim is against the charterer**.
- **Documentation**:
  - **Charter Party** (the laytime and demurrage clauses).
  - **Notice of Readiness (NOR)**: the start trigger.
  - **Statement of Facts (SOF)**: the minute-by-minute log.
  - **Time Sheet / Laytime Statement**: a working document that translates the SOF into counted hours, based on C/P clauses. Both owner and charterer produce their own.
  - **Demurrage Invoice**: issued by the owner to the charterer, usually within a specific time bar (often 90 days after completion of discharge).

Scenario

A Supramax loads 60,000 MT soybeans. Allowed laytime 6 days SHEX. The SOF shows NOR tendered at 08:20, accepted immediately. The berth is occupied, but WIBON clause applies, so laytime starts at 13:00. Rain interrupts for 3 hours on day 3. After calculating all exceptions, the used laytime is 7 days 2 hours 15 minutes. That means demurrage accrues for 1 day 2 hours 15 minutes. At $12,000/day, the charterer owes $13,125.

Despatch

A **bonus paid by the shipowner to the charterer** for completing loading and/or discharge in less than the allowed laytime. It's the mirror image of demurrage and is meant to incentivize fast turnarounds.

- **Not automatic**: Despatch must be explicitly agreed in the charter party. Common phrase: "Despatch half demurrage" (the rate is half the demurrage rate) or "despatch same as demurrage".
- **Calculation**: Despatch is only earned for all time saved, not just "working time saved." So if the C/P says "despatch on all time saved," any time the ship is released early—including weekends—counts toward the bonus. If it says "despatch on working time saved," only the laytime working hours are counted.
- **Who pays/receives**: The shipowner pays the charterer. The charterer can keep it as pure profit or share it with the terminal.
- **Documentation**: Same documents as demurrage, but the resulting laytime statement will show a negative difference (saved time). The owner issues a Despatch Credit Note or deduction from freight.

Scenario

The same Supramax finishes loading in 4 days 10 hours. Allowed laytime was 6 days SHEX. Time saved = 1 day 14 hours. With despatch at half demurrage ($6,000/day), the charterer receives $9,750 from the owner.

Variants & Special Cases in Bulk

- **Tanker voyages**: The concept is identical, but laytime is often expressed in "**running hours**" and the triggering document is a Notice of Readiness followed by a **pumping log** (for discharge). Demurrage in tanker markets can be astronomically high—$50,000–$100,000 per day for a VLCC.
- **Time Charters**: **No laytime or demurrage** (as we defined them). **The charterer pays hire per day for the ship continuously**. If the ship is delayed, the charterer simply keeps paying hire. However, a time charter may include an "off-hire" clause if the ship breaks down. So the risk of delay is totally on the charterer.

- **Deadfreight**: Not demurrage, but related. If the charterer fails to supply the full contracted cargo quantity, the owner can claim deadfreight for the lost freight on the empty space.

Container / Liner Shipping – Detention & Demurrage (D&D)

In container shipping, you're not leasing the ship; you're using the carrier's equipment (the container) and occupying scarce terminal space. So the **penalties are about equipment usage and yard congestion**.

Port Demurrage & Detention

This is where everyone gets confused. In container shipping, "demurrage" and "detention" are two separate charges that apply to different phases:

- **Port Demurrage (Import)**: Charged by the terminal operator (or carrier on their behalf) when a full import container sits in the terminal beyond the allowed free time. It's a storage charge for clogging up the terminal yard.
- **Detention (Import/Export)**: Charged by the carrier when the shipper or consignee holds onto the empty container (import) or the loaded container before sailing (export) beyond the allowed free time. It penalizes you for keeping the carrier's equipment out of circulation.

There's a simple memory aid:

> **Demurrage = Full box, inside the port, sitting too long.**
> **Detention = Empty box (or full export), outside the port, in your possession too long.**

But note: in practice, many carriers and terminals bundle them together or use the terms loosely. For our purposes, we'll stick to the standard distinction.

Free Time: The Allowance

Each carrier publishes a Detention & Demurrage Tariff (or includes it in the service contract). It specifies:

- **Free Time for Import**: The number of calendar days from the container's discharge (or day of availability) until the full container must be picked up and the empty container must be returned. Typical: 5–7 days combined. Sometimes split into "terminal free time" (demurrage-free) and "equipment free time" (detention-free).
- **Free Time for Export**: The number of calendar days from the empty container pickup until the loaded container must be gated into the terminal. Often shorter—maybe 5 days.

Free time is usually calculated in calendar days, not working days. Weekends and holidays always count. This is harsher than bulk laytime.

How Charges Accrue

Charges are often structured in tiers that escalate the longer you hold:

- Days 1-4: Free
- Days 5-7: $50/day
- Days 8-10: $100/day
- Days 11+: $200/day

The carrier's system generates an invoice based on the events in its container tracking:

- **Discharge** date & time from the terminal.
- **Gate-out full** date & time (when the trucker picks up the import).
- **Gate-in empty** date & time (when the trucker returns the empty).
- **Gate-out empty** date & time (for export).
- **Gate-in full** date & time (when loaded export is delivered to terminal).

If the terminal or carrier's data is wrong (e.g., a "gate-out full" event is missing), you can get charged incorrectly. This is a fertile ground for disputes and a key part of what your app can audit.

Scenario

1. **Vessel discharges container** at terminal. The clock starts: that's the "availability date" (usually midnight of that day). The carrier sends an "Arrival Notice" to the consignee.
2. **Consignee clears customs** and arranges trucking.
3. **Trucker picks up the full container** before free time expires (say, within 5 days). If they pick it up on day 6, **port demurrage** begins from day 6 until the day of pickup.
4. **Consignee unloads the cargo** at their warehouse.
5. **Trucker returns the empty container** to the carrier's designated depot. The "free time" for detention is usually separate or combined, but if the combined total exceeds the free allowance, detention charges apply for each day the empty is held beyond the limit.
6. **Carrier issues D&D invoice**, often weeks later, aggregating all charges. The consignee has a short window to dispute (sometimes only 30 days).

Documentation

- **Booking Confirmation / Service Contract** – contains the D&D tariff.
- **Arrival Notice** – states free time expiry.
- **Equipment Interchange Receipt (EIR)** – paper or electronic record for each gate event (pickup, return). This is the equivalent of the SOF in the container world.
- **D&D Invoice** – the carrier's bill.
- **Dispute letter** – if you contest the charges, you need evidence (EIRs, terminal gate logs).

No Despatch, But Some Mitigation

There is no despatch in container shipping. You cannot earn a bonus for returning early. However, you can:

- **Pre-purchase extended free time** (e.g., 14 days instead of 7) at a discount, often at the time of booking.
- **Use Merchant Haulage** (consignee arranges own truck) rather than carrier's haulage to have control over timing.
- **In some contracts**, large BCOs negotiate D&D credits or waived charges as part of the annual tender.

Bulk Demurrage / Despatch Vs. Container Demurrage & Detention

Feature	Bulk Demurrage / Despatch	Container Demurrage & Detention
What's the asset?	The ship's time.	The container (equipment) and terminal yard space.
Who charges whom?	The shipowner charges the charterer.	The carrier charges merchants (shipper/consignee); terminals may charge carrier or merchant.
Allowance term	Laytime (in hours/days, with exceptions).	Free time (calendar days, no exceptions).
Exceptions	SHEX, rain, WIBON, etc. – heavily negotiated.	None. Weekends and holidays always count.
Penalty term	Demurrage (liquidated damages).	Demurrage (full box in port) / Detention (empty box outside).
Reward	Despatch (if agreed) – bonus for early completion.	None.
Trigger point	Notice of Readiness (NOR) accepted.	Container discharge / empty pickup date.
Clock stops	Completion of cargo ops (recorded in SOF).	Gate-in/gate-out events (recorded in EIR/terminal system).
Governing contract	Charter Party (bespoke).	Carrier's tariff / service contract (standard, but large shippers can negotiate).
Key document	SOF, NOR, Time Sheet.	EIR, D&D invoice, Arrival Notice.
Calculation	Manual, clause-heavy, prone to dispute.	Automated by carrier but error-prone (missing events, wrong free time).
Entities involved	Charterer, owner, broker, agent, master, terminal, surveyor, laytime clerks.	Consignee/shipper, carrier, terminal, trucker, depot, accounts payable.
Value per event	High ($5k–$500k+ per voyage).	Low per container ($10–$200/day), but massive aggregate (thousands of containers).

Real-World Scenarios

Scenario 1: Container Demurrage - Port Congestion in Los Angeles

- **Port**: Los Angeles, California (one of the busiest ports in the US)
- **Ship**: The "MSC Vanessa," a 14,000 TEU container vessel
- **Trade**: Container Liner Trade (Deep Sea)
- **Cast**:
  - MSC (Shipping Line): Carrier that owns and operates the vessel
  - Maersk (Shipping Line): MSC's partner in a vessel-sharing agreement; sells space on the vessel
  - ABC Electronics (BCO/Importer): A large US electronics retailer importing 50 containers of smartphones from China
  - Global Forwarding Inc. (Freight Forwarder): Hired by ABC Electronics to manage the shipment
  - Fenix Marine Services (Terminal Operator): Operates the terminal at the Port of LA where MSC Vanessa berths
  - Pacific Customs Brokers (Customs House Broker): Handles customs clearance for ABC Electronics
  - Port of Los Angeles (Port Authority): Manages the harbor, pilots, and overall operations
  - MSC Local Agent (Ship Agent): Represents MSC at the Port of LA
- **Documentation**:
  - Bill of Lading (B/L): MSC issues a Bill of Lading to ABC Electronics via Global Forwarding Inc. It shows the container numbers, describes the cargo, and includes a free time clause
  - Arrival Notice: MSC sends an arrival notice to ABC Electronics, stating the vessel's ETA and the demurrage free time period
  - Container Interchange Receipt (CIR): Issued when the container is discharged from the vessel and placed in the terminal; records the condition and start of the free time
  - Demurrage Invoice: MSC issues an invoice to ABC Electronics (or its forwarder) for the excess days
  - Proof of Delivery (POD) / Gate-Out Receipt: Issued when the container finally leaves the terminal; records the date and time of pickup
  - Dispute Letter: A formal letter from ABC Electronics disputing the demurrage charges, citing force majeure or port congestion
- **How it unfolds**:

Date	Event	Action/Document	Notes
Jan 1	MSC Vanessa arrives at Port of LA	Vessel is anchored outside the port; issues Notice of Readiness (NOR)	Port congestion: no available berth
Jan 5	Vessel berths at Fenix Marine Services terminal	Containers begin discharging; CIR issued for ABC's containers	5-day delay at anchorage
Jan 6	ABC's containers are discharged	Free time starts; MSC sends Arrival Notice	ABC has **5 free days** (Jan 6-10)
Jan 10	Free time expires	ABC has not arranged trucking or customs clearance	Demurrage begins accruing
Jan 15	ABC finally picks up the containers	Gate-Out Receipt issued; demurrage ends	Delay due to customs inspection and trucker shortage
Jan 20	MSC issues Demurrage Invoice	Invoice: 5 excess days × $200/day × 50 containers = **$50,000**	ABC receives the bill

- **The Dispute**
  ABC Electronics disputes the demurrage invoice. Their grounds:
    - **Port Congestion**: The vessel itself was delayed 5 days at anchorage. ABC argues this was force majeure beyond their control.
    - **Trucker Shortage**: They provide evidence that their booked trucking company was unable to pick up containers due to an industry-wide shortage.
    - **Customs Delay**: Pacific Customs Brokers documents show the cargo was selected for an intensive customs inspection that delayed clearance.
- **The outcome**:
    - After negotiation and presenting their evidence, MSC reduces the bill by 50% to $25,000.
    - ABC Electronics pays the reduced amount but improves their processes: they now pre-book trucking and submit customs documentation earlier.
    - Global Forwarding Inc. implements an internal alert system to monitor free times and prevent future issues.

- **Lessons**:
    - **Proactive Alerts**: The system could have alerted ABC Electronics on Jan 8 (2 days before free time expired) that their containers were at risk.
    - **Automated Evidence Gathering**: It would have collected the customs inspection notice and trucker communication automatically, making the dispute easier.
    - **Dispute Template**: Generated a professionally formatted dispute letter using the gathered evidence.

Scenario 2: Container Detention - Trucker Delay in Chicago

- **Port:** Chicago, Illinois (Inland intermodal hub)
- **Ship:** A rail-borne shipment from the Port of Newark to Chicago
- **Trade:** Container Liner Trade (Intermodal)
- **Cast:**
    - CMA CGM (Shipping Line)**:** Carrier that issued the Bill of Lading and provides the container
    - Midwest Retail Co. (BCO/Importer): A large retail chain importing apparel from Vietnam
    - Quick Haul Logistics (Trucking Company): Hired by Midwest Retail to move containers from the rail ramp to their warehouse
    - CSX Transportation (Rail Operator): Moves the containers from Newark to Chicago
    - Union Pacific (Rail Operator): Operates the intermodal ramp in Chicago where the container arrives
- **Documentation**:
    - Bill of Lading: CMA CGM's B/L includes detention terms and free time
    - Intermodal Rail Receipt: Issued when the container is transferred from ship to rail; starts the clock for detention
    - Interchange Receipt (Gate-Out): Issued when the container leaves the rail ramp; starts the clock for detention
    - Interchange Receipt (Gate-In): Issued when the empty container is returned to the carrier's depot; ends the clock
    - Detention Invoice: CMA CGM invoices Midwest Retail for exceeding the detention free time
    - Trucker Logs: Quick Haul Logistics provides logs showing when they arrived and departed
- **How it unfolds**:

Date	Event	Action/Document	Notes
Mar 1	Container arrives at Union Pacific intermodal ramp in Chicago	Intermodal Rail Receipt issued; free time starts	Free time for demurrage is separate

Date	Event	Action/Document	Notes
Mar 5	Midwest Retail arranges pickup with Quick Haul Logistics	Trucker picks up the full container; Gate-Out Receipt issued	**Detention free time starts** (5 days)
Mar 6-9	Midwest Retail unloads the container at their warehouse	Container is being unpacked	They face an unexpected inventory backlog
Mar 10	Detention free time expires	Container is still full at the warehouse	Detention begins accruing
Mar 12	Trucker returns the empty container to CMA CGM's depot	Gate-In Receipt issued	CMA CGM issues detention invoice
Mar 15	Midwest Retail receives Detention Invoice	Invoice: 2 excess days × $150/day × 1 container = **$300**	Relatively small amount, but multiplied across thousands of containers

- **The Dispute**
  Midwest Retail disputes the detention invoice:
    - **Trucker Delay:** Quick Haul Logistics experienced a mechanical breakdown on Mar 8, delaying the pickup by 1 day. They provide service records.
    - **Warehouse Congestion:** Midwest Retail had a temporary backlog that delayed unloading. They claim this was due to unexpected inventory arrivals.
- **The outcome**:
    - CMA CGM agrees to waive 1 day of detention due to the trucker's mechanical failure, but upholds the remaining 1 day charge.
    - Midwest Retail pays $150.
    - They implement a new policy: inventory managers must notify logistics teams 48 hours before container arrival to ensure warehouse capacity is available.
    - They also add a clause in their trucking contracts that penalizes carriers for delays that cause detention costs.
- **Lessons**:
    - **Warehouse Integration:** The system would integrate with Midwest Retail's warehouse management system to predict when unloading capacity will be available.
    - **Trucker Performance** Tracking: It would track trucker pickup times and automatically flag when trucker delays are likely to cause detention.

    ○ **Automated Claims:** Generate a claim against the trucking company for the cost of detention caused by their breakdown.

Scenario 3: Bulk Voyage Charter - Demurrage and Despatch in South America

- **Port:** Santos, Brazil (Exporting soybeans)
- **Destination:** Ningbo, China (Importing soybeans)
- **Ship:** The "Star Harvest," a 80,000 DWT Panamax bulk carrier
- **Trade:** Bulk Voyage Charter (Tramp Trade)
- **Cast**:
    - Pacific Grains Co. (Charterer): A large commodity trader that chartered the vessel to move soybeans
    - Greek Shipowners Ltd. (Shipowner): Owns the Star Harvest; operates the vessel
    - Noble Chartering (Ship Broker): The broker who negotiated the voyage charter
    - Santos Terminal SA (Terminal Operator): Operates the grain terminal in Santos
    - SGS Brazil (Cargo Surveyor): Inspects and certifies the cargo quantity and quality
    - Wilson Sons (Ship Agent): Represents the vessel in Santos and prepares the Statement of Facts
    - P&I Club (Insurance): Provides liability coverage for the shipowner
    - COFCO (Cargo Receiver): The Chinese buyer of the soybeans
- **Charter Party Terms**:

Term	Clause
Laytime	**5 days** for loading at Santos
Demurrage Rate	**$15,000 per day** (pro rata for part of a day)
Despatch Rate	**50% of demurrage rate** = $7,500 per day
Laytime Exceptions	"Weather working days" only; weekends and holidays excluded if not worked; strikes excluded

- **Documentation**:
    - **Charterparty (Gencon Form)**: The contract between Greek Shipowners Ltd. and Pacific Grains Co. Contains laytime, demurrage, despatch terms
    - Notice of Readiness (NOR): Issued by the Master upon arrival at Santos; triggers the start of laytime
    - Statement of Facts (SOF): Prepared by Wilson Sons; records all port activities, including start/end of loading, delays, weather

    - Cargo Manifest / Mate's Receipt: Documents the quantity loaded; certified by SGS
    - Bill of Lading: Issued by the Master to Pacific Grains; includes "freight payable as per charterparty"
    - Laytime Calculation: Prepared by the shipowner's chartering department; calculates demurrage or despatch
    - Demurrage Invoice or Despatch Credit Note: issued by the relevant party to settle the final amount

Document	Purpose
Charterparty (Gencon Form)	The contract between Greek Shipowners Ltd. and Pacific Grains Co. Contains laytime, demurrage, despatch terms
Notice of Readiness (NOR)	Issued by the Master upon arrival at Santos; triggers the start of laytime
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Laytime Calculation	Prepared by the shipowner's chartering department; calculates demurrage or despatch
Demurrage Invoice or Despatch Credit Note	Issued by the relevant party to settle the final amount

- **How it unfolds:**
  Loading at Santos, Brazil:

Date	Event	Action/Document	Notes
Jun 1 08:00	Star Harvest arrives at Santos	Vessel anchors; Master issues NOR	Vessel is ready to load
Jun 1 10:00	NOR accepted by Pacific Grains	**Laytime starts**	

Date	Event	Action/Document	Notes
Jun 1-3	Weather delays	Rain stops loading twice, totaling 12 hours	These are "non-weather working days" and are excluded
Jun 3-5	Loading continues	Loading rate: ~15,000 tons per day	
Jun 6 10:00	Loading completes	Total loading time: **5 days 2 hours** (120 hours, plus 12 hours weather exclusion)	

- **Laytime Calculation:**

Item	Calculation
Laytime allowed	5 days (120 hours)
Actual time used	5 days, 2 hours (122 hours)
Less weather exclusion	12 hours
Net time used	**110 hours** (4 days 14 hours)
Time saved	120 - 110 = **10 hours**

- **Result:** The loading was completed **10 hours early**. Pacific Grains is entitled to **despatch**.
- **Despatch Calculation:**
    - Despatch rate = $7,500 / 24 hours = $312.50 per hour
    - 10 hours × $312.50 = **$3,125**
    - Greek Shipowners Ltd. issues a Credit Note to Pacific Grains for **$3,125**
- **The Dispute:**
  Pacific Grains disputes the despatch calculation:
    - **NOR Timing:** They argue that the NOR was tendered at 08:00 but laytime should not start until 10:00, which is already accounted for.
    - **Weather Exclusion:** They argue that an additional 4 hours of rain should be excluded because the terminal declared the port "inoperative" even though the actual rain was light.
- **The outcome**:

    - Greek Shipowners Ltd. reviews the SGS weather logs and the terminal's official declarations.
    - They agree to exclude an additional 4 hours, resulting in 14 hours of time saved.
    - The despatch payment increases to **$4,375** ($312.50 × 14).
    - Pacific Grains receives the higher despatch.
- **Lessons**:
    - **Laytime Calculation Automation:** Automatically calculates laytime used vs. allowed, including weather exclusions.
    - **NOR Tracking:** Monitors NOR tender and acceptance times to ensure accurate laytime start.
    - **Exception Management:** Flags potential weather exceptions and prompts the user to gather evidence.
    - **Despatch Maximization:** Alerts the charterer when despatch is likely and suggests actions to maximize it.

Scenario 4: A Complex Dispute - NVOCC vs. Shipping Line vs. Cargo Owner

Ocean Logistics Inc. (an NVOCC) booked 20 containers of furniture from China to Germany for Euro Imports GmbH. They issued a House Bill of Lading to Euro Imports and secured a Master Bill of Lading from Maersk.

- **Port:** Rotterdam, Netherlands (One of Europe's busiest ports)
- **Ship:** The "Emma Maersk" (Maersk container vessel)
- **Trade:** Container Liner Trade (Deep Sea)
- **Cast**:

Entity	Role
Maersk (Shipping Line)	Carrier operating the vessel and issuing the Master Bill of Lading
Ocean Logistics Inc. (NVOCC)	Issued a House Bill of Lading to the cargo owner; bought space from Maersk
Euro Imports GmbH (BCO/Importer)	The German importer; contracted with Ocean Logistics Inc.
Rotterdam World Gateway (Terminal Operator)	The terminal where Emma Maersk berths
Dutch Customs (Government Agency)	Performed a random X-ray inspection that delayed cargo
Maersk Local Agent (Ship Agent)	Represented Maersk at Rotterdam

- **Documentation:**

Document	Issued By	Purpose
Master Bill of Lading	Maersk	Contract between Maersk and Ocean Logistics; includes demurrage terms
House Bill of Lading	Ocean Logistics	Contract between Ocean Logistics and Euro Imports; has its own demurrage terms
Arrival Notice	Maersk	Sent to Ocean Logistics; includes free time details
Demurrage Invoice	Maersk	Sent to Ocean Logistics for 10 excess days
Demurrage Invoice	Ocean Logistics	Sent to Euro Imports for 10 excess days + administrative fees
Customs Inspection Notice	Dutch Customs	Shows the date and duration of the inspection

- **How it unfolds:**

Date	Event	Action/Document	Notes
Oct 1	Emma Maersk arrives; containers discharged	Free time starts; Arrival Notice sent to Ocean Logistics	Free time: **5 days**
Oct 5	Free time expires	Ocean Logistics has not arranged pickup	Demurrage begins accruing
Oct 2-6	Dutch Customs performs random X-ray inspection	Containers are held in the customs area; inspection completes Oct 6	Ocean Logistics was not aware of the inspection
Oct 10	Euro Imports picks up containers	Gate-Out Receipt issued	5 excess days
Oct 15	Maersk issues Demurrage Invoice	Invoice to Ocean Logistics: 5 days × \$250 × 20 = **\$25,000**	

Date	Event	Action/Document	Notes
Oct 20	Ocean Logistics issues Demurrage Invoice	Invoice to Euro Imports: 5 days × $300 × 20 = **$30,000** + admin fees = **$33,000**	Ocean Logistics includes a markup

- **The Dispute - Multiple Layers:**
  - Dispute 1: Ocean Logistics vs. Maersk
    Ocean Logistics' Argument:
    - The delay was caused by Dutch Customs, a government authority beyond their control.
    - They were not notified by Maersk that the containers were selected for inspection.
    - They request a full waiver of demurrage.

    Maersk's Response:
    - The inspection is between the customs authority and the cargo owner; it's not the carrier's responsibility.
    - The Bill of Lading clearly states that demurrage accrues regardless of the cause unless it's a carrier-specific issue.
    - They offer a **50% reduction** to $12,500 as a goodwill gesture, but no full waiver.

    Outcome:
    - Ocean Logistics accepts the 50% reduction and pays $12,500.
    - They learn to proactively monitor customs status for their shipments.

  - Dispute 2: Euro Imports vs. Ocean Logistics
    Euro Imports' Argument:
    - The demurrage was caused by a customs inspection, which was not their fault.
    - Ocean Logistics charged a higher rate ($300 vs. Maersk's $250) and added administrative fees.
    - They request a 50% reduction to match Maersk's settlement.

    Ocean Logistics' Response:
    - The $300 rate is clearly stated in the House Bill of Lading.
    - The administrative fees cover their time and effort in managing the dispute.
    - They offer to reduce the invoice to **$20,000** as a compromise (matching Maersk's settlement + small admin fee).

    Outcome:

- Euro Imports pays $20,000.
- They are unhappy with Ocean Logistics and may consider using a different forwarder in the future.

- **Final Result:**

Entity	Paid	Received	Net Position
Maersk	-	$12,500	+$12,500
Ocean Logistics	$12,500 (to Maersk)	$20,000 (from Euro)	+$7,500 (covers admin and dispute costs, leaves a small profit)
Euro Imports	$20,000 (to Ocean)	-	-$20,000 (significant unexpected cost)

- **Lessons**:
  - **Visibility Across the Chain:** Ocean Logistics could have used the tool to track customs inspection status, alerting them earlier.
  - **Audit Trail:** The system would have documented all communications, making disputes more transparent.
  - **Contract Comparison:** It would compare the Master and House B/L terms, flagging discrepancies (like the $50/day difference) for Ocean Logistics to address proactively.
  - **Dispute Automation:** Generate formal dispute letters for both parties with all relevant evidence attached.
  - **Cost Recovery:** Help Ocean Logistics create a claim against Euro Imports' shipment insurance if applicable.

Summary Table of Scenarios

Scenario	Type	Key Charge	Outcome	Key Documentation
1. LA Port Congestion	Container Demurrage	$50,000 (reduced to $25,000)	Dispute partially successful	NOR, CIR, Demurrage Invoice, Dispute Letter
2. Chicago Trucker Delay	Container Detention	$300 (reduced to $150)	Partial waiver	Intermodal Rail Receipt, Gate-In/Out

Scenario	Type	Key Charge	Outcome	Key Documentation
				Receipts, Trucker Logs
3. Brazil Soybeans	Bulk Voyage Charter	**Despatch of $3,125** (increased to $4,375)	Despatch received; dispute resolved	Charterparty, NOR, SOF, Laytime Calculation
4. Rotterdam Dispute	Container Demurrage (NVOCC chain)	$25,000 (reduced to $12,500); $33,000 (reduced to $20,000)	Multiple negotiations; all parties compromised	Master B/L, House B/L, Customs Inspection Notice, Invoices

Demurrage Defender

Target Customer Groups

The application will cater two groups of customers:

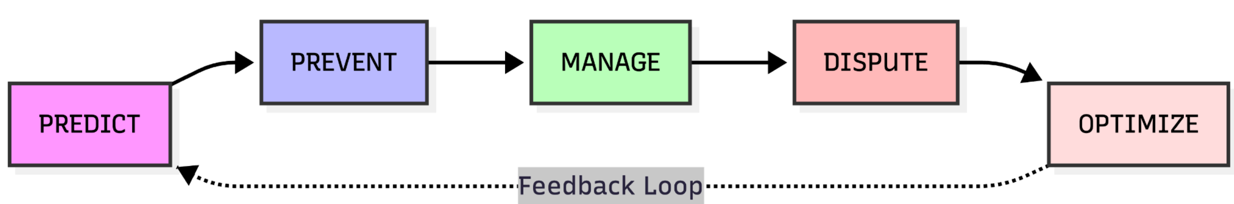
- **Bulk Charterers** - Commodity trading houses (Cargill, Bunge, Vitol, Trafigura, Glencore), oil majors (Shell, BP), mining companies (Rio Tinto, Vale), large grain exporters/importers, steel mills, and power utilities that buy spot coal or LNG.
- **Large BCOs/NVOCCs** - Global retailers (Walmart, IKEA, Target), electronics manufacturers, large auto parts importers, and the top 20 global freight forwarders/NVOCCs (Kuehne+Nagel, DSV, DHL Global Forwarding)

While the groups are distinct, every customer will get access to the same system, same set of features. This keeps the application more focused and development streamlined.

- **Unified pain, different anatomy**: Both segments bleed money through time-based penalties. One from ships, one from boxes. The emotional journey is identical—frustration, manual work, missed deadlines, invoice shock. A single platform that understands both worlds is a moat: no competitor does this well today.
- **Cross-pollination potential**: Many large commodity traders also move containerized goods (e.g., grains in containers, metals, packaged commodities) and already manage both types. Giving them one login for all demurrage/detention is a real convenience play.
- **Data synergy**: A port congestion indicator is valuable to both. A terminal that delays bulk vessels also delays container gate operations. Your benchmarking data gets richer when you see both sides.
- **Brand power**: "Demurrage Defender" instantly means "I protect you from time penalties"—whether that penalty is called demurrage, detention, or even despatch (by recovering it). It's a single promise.

Core Value Loop

Our application will be built around the following Loop:



Each phase in this loop answers a specific question:

- **PREDICT** - What will happen, and how much will it cost me?
- **PREVENT** - How do I stop the bad thing from happening in real time?
- **MANAGE** - What's happening now, and how do I keep control?
- **DISPUTE** - How do I challenge what I'm being charged?
- **OPTIMIZE** - How do I make smarter decisions next time?

From the POV of the customer groups these would look like:

Phase	Bulk Charterer Lens
PREDICT	Before fixing, simulate laytime risk at a port based on historical AIS and terminal data. Forecast demurrage cost for the upcoming voyage.
PREVENT	Real-time alert when vessel arrives at pilot station, track NOR and laytime clock. If waiting at anchorage due to congestion, flag force majeure or WIBON arguments.
MANAGE	Automated laytime calculation from uploaded SOF and digital C/P clauses. Live dashboard of all vessels, demurrage accruals, and remaining allowed time.
DISPUTE	Generate a fully reasoned, audit-ready laytime statement as a counter to the owner's claim. One-click generation of despatch invoice.
OPTIMIZE	Post-voyage analytics: port/terminal performance, owner demurrage claim accuracy, despatch recovery rate. Feed into next fixture negotiation.

PREDICT

Capability Categories:

- **Port & Terminal Risk Scoring**
  - **Laytime**: Probability of congestion, average anchorage waiting time, berth turnaround efficiency for the specific cargo type.
  - **Free Time**: Historical D&D accrual rates at a terminal, typical gate delays, depot availability.

- **Voyage / Shipment Simulation**

- **Laytime**: Given a proposed C/P clause set, cargo parcel, and port pair, simulate probable laytime used, demurrage cost, and despatch likelihood based on historical patterns and current conditions.
- **Free Time**: Predict the risk of exceeding free time for a given port/consignee combination based on past shipment data, allowing pre-booking "extended free time" purchases at optimal cost.

- **Counterparty / Carrier Behavior Forecast**
  - **Laytime**: How aggressively does this specific shipowner claim demurrage? What's their dispute resolution record?
  - **Free Time**: Which container carriers have the highest D&D invoice error rate or most frequent unwarranted charges?

- **Market Condition Alerts**
  - **Laytime**: Rising demurrage rates in a region, port strikes, upcoming weather windows that will slow loading/discharge.
  - **Free Time**: Carrier announcements of reduced free time, terminal tariff changes, depot gate hour reductions during holidays.

PREVENT

- **Real-Time Event Detection & Alerting**
  - **Laytime**: AIS-based detection of vessel arrival at pilot station → NOR auto-tender logged. Anchorage departure → berth proximity alert. Inactivity at berth beyond expected time → flag for immediate action.
  - **Free Time**: Container discharge detected, free-time clock starts. 72h/48h/24h warnings before expiry. Overstay warning if container not gated out.

- **Operational Recommendations**
  - **Laytime**: "Charterer, your vessel has been waiting 2 days. Based on your C/P's WIBON clause, laytime is already counting. Consider declaring force majeure due to terminal congestion under Clause X." Or, "You can still save laytime if you order night loading; current rate allows completion within allowed time if you start before 22:00."
  - **Free Time**: "Return to Depot B (open until 20:00) instead of Depot A (closes 17:00) to avoid $150 detention." Or, "Schedule dual transaction pickup/drop-off to combine moves and save one day."

- **Collaborative Workflow Triggers**
    - Automatically notify the port agent to chase berthing or the trucking company to expedite pickup based on impending deadlines.

MANAGE

- **Automated Time Calculation Engine**
    - **Laytime**: Ingest SOF/NOR, interpret C/P clauses, compute allowed vs. used laytime in real time, display running demurrage/despatch balance.
    - **Free Time**: Aggregate container milestones from carrier/terminal, compute per-diem charges according to tariff tiers, show total accrued D&D cost per container/shipment.

- **Unified Operational Dashboard**
    - A single-pane-of-glass showing all live vessel calls and their demurrage status, plus all active import/export containers and their free-time health. Filterable by port, trader, cargo, carrier.

- **Document & Audit Trail Management**
    - Central repository for all SOFs, NORs, C/Ps, EIRs, gate tickets, Arrival Notices. Full versioning and timestamps for every manual correction, ensuring audit readiness for arbitration.

- **What-If Scenario Testing (Live)**
    - "If the rain stops in 2 hours and loading resumes at 600tph, will we finish within laytime or go on demurrage?" The system instantly recalculates based on remaining cargo and current weather forecast.

DISPUTE

- **Automated Claim / Counter-Claim Generation**
    - **Laytime**: One-click generation of a fully reasoned laytime statement (Time Sheet) ready to send to the owner, with all clause interpretations cited. Automatic generation of a despatch invoice when time is saved.
    - **Free Time**: Automated reconciliation of carrier D&D invoice against actual gate events and applicable tariff. Flag every discrepancy (wrong date, wrong tier, duplicate charge) and generate a dispute letter with attached evidence.

- **Dispute Management Workflow**
    - Track the lifecycle of a dispute: opened, evidence submitted, carrier response received, negotiated settlement, final resolution. Assign tasks to team members.

- **Legal & Clause Knowledge Base (Chatbot Integration)**
    - Natural language query: "Does the 'always accessible' clause in our C/P mean the owner pays for the tug delay on arrival?" The chatbot provides a legal summary and suggested dispute language based on precedent and the specific clause.

- **Settlement Analytics**
    - Track win rates, average recovery per dispute, and carrier/owner dispute behavior. This data feeds back into PREDICT and OPTIMIZE.

OPTIMIZE

- **Post-Event Performance Analytics**
    - **Laytime**: Actual laytime vs. allowed per port/cargo/terminal, demurrage paid vs. despatch earned, owner claim accuracy, demurrage recovery rate.
    - **Free Time**: D&D spend by carrier, port, trade lane, and responsible party; free-time utilization percentage; detention avoidance rate.

- **Contract & Procurement Intelligence**
    - **Laytime**: Based on actual data, recommend optimal laytime rates, demurrage/despatch splits, and exception clauses for the next fixture. Identify which C/P clauses are most often misinterpreted and cost you money.
    - **Free Time**: Recommend negotiated free-time allowances per carrier and port for the next service contract tender. Identify which carriers consistently overbill so you can either demand better terms or switch.

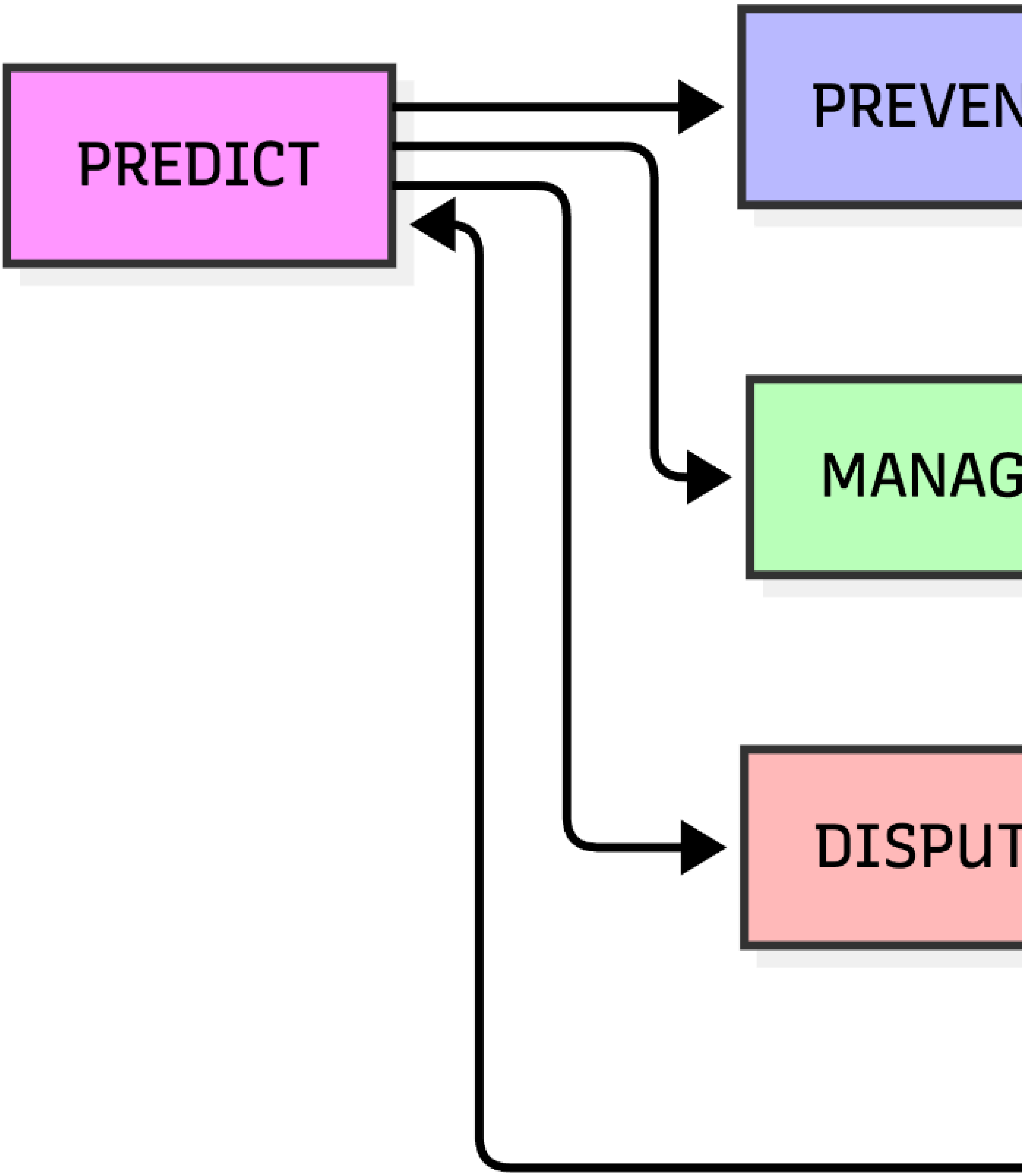
- **Network & Routing Optimization**
    - **Laytime**: "For your Brazilian soybeans to China, 65% of your vessels at Paranaguá incur demurrage. Consider a COA with a terminal at Santos where your data shows only 20% demurrage probability, despite a slightly longer inland haul."
    - **Free Time**: "Your Chicago imports via LA/LB have a 22% detention rate. Routing via Savannah and intermodal rail yields only 8% detention, saving an estimated $340,000 annually."

- **Benchmarking (Internal & External)**

- Compare your demurrage/despatch performance against your own historical baseline, and (anonymized) against industry peers if data sharing is enabled. This turns the platform into a continuous improvement engine.

Dataflow Loop: The Self-Improving System

The more a company uses the application, the smarter and more personalized its predictions and recommendations become.


PREDICT
PREVEN
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DISPUT

Model Retraining

Every completed voyage or container delivery updates the machine learning models that predict laytime/free-time risk. A voyage that went on demurrage because of unexpected port congestion becomes a data point that sharpens the Port Risk Score for that terminal. The system learns that "rain in July in Paranaguá during a Supramax soybean load typically adds 1.2 days" and adjusts its simulation accordingly.

Dispute Outcome Learning

When a carrier's D&D invoice is successfully disputed because of a systematic "gate-out" timestamp error, that carrier's behavior profile is updated. Future predictions for that carrier automatically flag higher likelihood of incorrect charges, and the dispute engine can pre-emptively apply the same challenge logic.

Contract Effectiveness Analysis

The OPTIMIZE phase reveals that Charter Party clause "Rider Clause 12 – Shifting Time" is consistently interpreted against the charterer by owners, costing an average $4,500 per occurrence. This feeds into PREDICT's contract simulation, which now warns users before fixing: "This clause has a high dispute rate. Consider renegotiating to BIMCO standard wording."

Operational Pattern Mining

If OPTIMIZE reveals that containers routed through a specific depot after 4 PM always incur detention because of next-day gate-in, the system bakes that rule into PREVENT alerts.

AI / Co-pilot / Chatbot

Phase	Defender AI Capabilities
PREDICT	"What is the demurrage risk for a Capesize loading iron ore at congestion, giving a quantified risk score with explanation. "Si AI runs the simulation and presents probabilistic demurrage co
PREVENT	"Why am I getting a demurrage alert?" → AI explains the spec AI suggests operational actions (order night shift, invoke force

Phase	Defender AI Capabilities
MANAGE	“Explain how the laytime calculation reached 6 days 3 hours.” counting rule. “What if loading finishes 4 hours earlier?” → AI r
DISPUTE	“Draft a demurrage counter-claim email to the owner for $23,4 amounts. “Is the owner’s argument that rain during meal break *“Happy Day”*) and provides a reasoned opinion with citations.
OPTIMIZE	“Which three ports cause the most demurrage for our grain sh “Propose a reworded demurrage clause for our next COA to re

Data Sources

The AI’s value is directly proportional to the breadth and quality of its accessible data:

- **Company-Specific Data (Private, per customer)**: All historical SOFs, NORs, Charter Parties, time sheets, dispute correspondence, carrier D&D invoices, internal SOP documents, and operational notes. This is the AI’s institutional memory.
- **Platform-Generated Data (Cross-customer, anonymized)**: Aggregated port performance metrics, carrier/owner behavior scores, market demurrage indices. (Only if data-sharing consent is given.)
- **Industry Knowledge Base (Curated, static and updated)**:
  - Full text of standard Charter Party forms (GENCON, NORGRAIN, BPVOY, SHELLVOY, etc.).
  - BIMCO clauses and explanatory notes.
  - Key maritime law cases and commentaries (e.g., Scrutton on Charterparties, Voyage Charters, LMAA arbitration rules).
  - INCOTERMS 2020 definitions.
  - Published carrier D&D tariff documents (periodically scraped).
- **External Real-Time Feeds**: AIS vessel data, weather forecasts, port authority announcements, carrier schedule changes.

Interaction Modes

- **Embedded Contextual Help**: A chat panel inside every screen. When viewing a time sheet, the user can ask, “Why was this rain period deducted?” and the AI answers with reference to the specific C/P clause and SOF line.
- **Proactive Alerts**: The AI pushes notifications not just of deadlines, but of anomalies: “Unusual pattern detected: The owner’s demurrage claim for M/V Sea Harvest includes 2

hours of alleged waiting time not recorded in the SOF. This has occurred on 3 of your last 5 voyages with this owner. Recommend automatic dispute."

- **Document Generation & Workflow Initiation**: Users can instruct the AI in natural language: "Issue a despatch invoice for the completed voyage M/V Ocean Glory," and the AI populates the template, attaches the final laytime statement, and sends it for approval.

Scope Boundaries (What the AI Does NOT Do)

- It does not provide unqualified legal advice; all legal interpretations are flagged as "guidance for review by qualified counsel."
- It does not automatically execute financial transactions or bind the company to settlements without human approval.
- It does not replace the need for a human to verify the accuracy of OCR-extracted SOF data before it enters the laytime calculation (though it will flag low-confidence extracts).

Modules & Features

PREDICT

Risk Explorer - Port, Terminal & Route Risk Assessment

User Stories

"As a charterer, I want to see the historical demurrage probability for a Supramax loading grain at Paranaguá in August so that I can decide whether to nominate that port or route via another terminal."

Features

1. Interactive map and searchable database showing demurrage/free-time risk scores for ports, terminals, and trade lanes.
2. Risk scores are calculated from aggregated historical AIS, SOF/D&D data (anonymized across tenants if permitted), weather patterns, and port authority announcements.
3. Drill-down to see root causes: average anchorage waiting time, terminal crane availability, labor strike history, seasonal congestion patterns.

Laytime Simulator” (Bulk) / “Free Time Forecaster” (Container) - Voyage / Shipment Simulation

User Stories

“As a charterer, I want to input my draft NORGRAIN C/P with 10,000 MT SHEX terms and compare it against an alternative with 12,000 MT SHINC, to see which one is projected to result in lower demurrage for my specific cargo.”

Container:
“As an importer, I want to input a planned shipment of 200 containers from Shanghai to Chicago via LA/LB and see the probability of detention charges exceeding $5,000, so I can pre-purchase extended free time or re-route via another gateway.”

Features

4. Upload or select a draft Charter Party (or carrier service contract) and cargo details, then run a probabilistic simulation that outputs expected laytime/free-time usage, demurrage/detention cost, and despatch likelihood.
5. Upload or select a draft Charter Party (or carrier service contract) and cargo details, then run a probabilistic simulation that outputs expected laytime/free-time usage, demurrage/detention cost, and despatch likelihood.
6. Side-by-side comparison of multiple scenarios (e.g., two different load ports, or two different laytime clauses).

Counterparty Intel - Counterparty & Carrier Behavior Analytics

User Stories

“As a demurrage analyst, I want to check if a specific shipowner we haven't worked with before has a reputation for inflated demurrage claims, so I can price that risk into the fixture.”

Features

7. A profile page for each shipowner/container carrier showing demurrage claim frequency, average overcharge percentage, dispute win rates, and payment promptness.
8. Data sourced from the company’s own historical interactions (and optionally, anonymized cross-industry benchmarks).

Market Pulse - Market Watch & Early Warning

User Stories

"As a logistics manager, I want to receive an alert if a typhoon is forecast to hit a port where I have cargo due to discharge next week, so I can plan for potential demurrage."

Features

9. Configurable alerts for rising port congestion (AIS-based anchorage counts), tariff changes, upcoming strikes, or weather events that could impact laytime/free-time.
10. A news feed aggregating shipping and commodity trade publications, filtered by the user's trade lanes.

PREVENT

Market Pulse - Real-Time Deadline Tracking & Alerting

User Stories

"As an operations user, I want to see all my import containers approaching their free-time expiry in the next 48 hours in one view, so I can prioritize pickups and avoid detention."

Features

11. A single dashboard of all active vessels (bulk) and containers (liner) with color-coded countdown timers showing time remaining in allowed laytime/free-time.
12. Tiered, escalating notifications (push, email, SMS) at configurable intervals: 72h, 48h, 24h, and upon expiry.
13. Integration with AIS for vessels and carrier/terminal APIs for container milestones to automatically start the clock.

"Defender AI Assistant" (Preventive mode) - Actionable Recommendation Engine

User Stories

"As a trader, when my vessel is at anchorage and demurrage is running, I want the system to tell me if I have a contractual right to stop the clock, and draft the communication to exercise it."

Features

14. When a risk is detected, the AI proactively suggests specific operational actions to avoid or reduce the penalty.
15. For bulk: “Vessel at anchorage for 36h. WIBON clause active. You can request the terminal to accept NOR earlier based on your C/P rider clause 8. Draft email generated.”
16. For container: “Return to Depot B (open until 20:00) instead of Depot A to save $150/day detention. Map and directions included.”

Smart Alerts & Automations - Automated Workflow Triggers

User Stories

“As an importer, I want the system to automatically email my trucking company when a container has less than 24 hours of free time remaining, including the exact container number and recommended return depot.”

Features

17. Rules engine that, when a deadline breach is imminent, can automatically notify a pre-defined list of contacts (trucker, port agent, depot).
18. Integration with common communication tools (email, Slack, Teams) to send alerts directly into operational workflows.

MANAGE

DocIn - Automated Document Ingestion & Digitization

User Stories

“As an analyst, I want to drag-and-drop a scanned SOF PDF into the system and have all event timestamps extracted and ready for calculation within seconds, with low-confidence values flagged for review.”

Features

19. Upload PDF, scanned, or email-based SOFs, NORs, Arrival Notices, EIRs, and carrier invoices. AI-powered OCR extracts all timestamps and structured data with confidence scores.
20. Manual override with full audit trail for any misread fields.

21. Bulk upload for container D&D invoices (PDF/Excel) with automatic line-item extraction.

“Laytime Engine” (Bulk) / “Free-Time Engine” (Container) - Laytime & Free-Time Calculation Engine

User Stories

“As an analyst, I want the system to calculate the exact demurrage amount for a completed voyage, showing me every minute that contributed, and let me adjust a disputed rain period to see how it changes the final figure.”

Features

22. Rules engine that interprets Charter Party clauses (SHEX, WIBON, ATUTC, rain exceptions, etc.) or container tariff free-time rules, and computes allowed time, used time, demurrage, and despatch.
23. Visual timeline view that overlays the SOF/gate events with calculated laytime/free-time, clearly showing when the clock is running, stopped, or on demurrage.
24. What-if slider: “What if rain stopped 1 hour earlier?”—instantly recalculates.

FleetView - Unified Operational Dashboard

User Stories

“As a head of operations, I want to open a single dashboard and immediately see the total financial exposure from demurrage and detention across all my ships and containers right now.”

Features

25. Real-time map showing all active vessel positions (AIS) and their current laytime status.
26. Table view of all active containers with free-time status, filterable by port, carrier, consignee.
27. At-a-glance KPI cards: “Estimated Demurrage Exposure Today,” “Containers At Risk,” “Despatch Earned YTD.”

DocVault - Centralized Document & Audit Trail

User Stories

"As a legal user, I need to retrieve the original SOF and the complete laytime calculation for a voyage that happened 18 months ago, with a clear record of who modified what and when, because we're going to arbitration."

Features

28. Every document (SOF, NOR, C/P, EIR, invoice) is automatically filed and linked to its corresponding voyage or container.
29. Complete, immutable audit log of every data extraction, manual correction, calculation, and user action.

DISPUTE

ClaimBuilder - Automated Claim & Counter-Claim Generation

User Stories

"As an analyst, after reviewing the AI's laytime calculation, I want to generate a final Laytime Statement with a single click that is ready to send to the shipowner as my formal counter-claim."

Features

30. One-click generation of a professionally formatted, fully reasoned Laytime Statement (Time Sheet) for bulk demurrage/despatch, with all clause references and SOF line citations.
31. One-click generation of a despatch invoice when laytime is saved.
32. Automated reconciliation of a carrier D&D invoice against actual gate events and applicable tariff. Outputs a "Dispute Package" listing every overcharge with evidence.

Dispute Tracker - Dispute Workflow & Case Management

User Stories

"As a finance user, I want to see all ongoing D&D disputes with Maersk, their total value, and the stage of each, so I can prioritize high-value cases and ensure we meet response deadlines."

Features

33. A case file for each dispute, tracking status (open, evidence submitted, in negotiation, resolved), assigned users, and all correspondence.
34. Automated reminders for response deadlines and escalation triggers.
35. Dashboard of all active disputes with financial value at stake.

"Defender AI Assistant" (Dispute mode) - AI-Powered Legal & Clause Reasoning

User Stories

"As a charterer, when an owner claims demurrage for a rainy Sunday under a SHEX term, I want the AI to draft a response explaining why the time is excluded, citing both the clause and relevant legal precedent."

Features

36. Natural language question answering on maritime law and charter party clauses. "Is the owner's claim for time lost during a crew change valid under GENCON 94?"
37. AI drafts dispute letters with specific legal arguments and attaches the evidence package.
38. Research mode that finds relevant LMAA arbitration awards or English court precedents that support the user's position.

OPTIMIZE

Insights Hub - Post-Event Performance Analytics

User Stories

"As a CFO, I want a quarterly report that shows our total demurrage and detention spend, broken down by root cause, and how that compares to the previous quarter and the same quarter last year."

Features

39. Dashboards and reports showing demurrage/despatch performance over time: by port, by vessel, by cargo type, by trader, by carrier.

40. Detention/D&D spend analysis: total costs, free-time utilization, detention rate by carrier and trade lane.
41. Trend identification: "Your demurrage costs at Port Hedland have increased 18% YoY due to growing anchorage times."

Contract Optimizer - Contract & Procurement Optimization

User Stories

"As a chartering manager, before my next annual COA negotiation, I want the system to show me which three C/P clauses consistently lead to the most disputes and demurrage costs, and propose revised wording."

Features

42. Based on historical performance data, the system recommends specific C/P clause amendments (e.g., "Increase laytime allowance from 10,000 MT to 12,000 MT; your data shows this would have saved $145,000 last year with minimal impact on fixture competitiveness").
43. For container contracts, it recommends negotiated free-time extensions, guaranteed depot return windows, or carrier-specific D&D caps.

Route Optimizer - Network & Routing Optimization

User Stories

"As a supply chain director, I want to know which import gateway has the lowest average detention cost for our electronics shipments, so I can make data-driven routing decisions for next year's carrier tender."

Features

44. Compares actual demurrage/despatch and D&D outcomes across different ports and routes, identifying lower-cost alternatives for future shipments.
45. Scenario modeling: "What would our total landed cost (including D&D) have been if we routed all Chicago-bound cargo via Savannah instead of LA/LB last year?"

Benchmark - Internal and External Benchmarking

User Stories

“As a head of operations, I want to see how our detention rate for import containers compares to the industry average for retailers of similar size, so I can identify if we have a process problem or if our carriers are charging us unfairly.”

Features

46. Compare the company’s demurrage/detention KPIs against its own historical baseline (internal benchmarking) and, with consent, against anonymized industry averages (external benchmarking).
47. Leaderboards and performance scores by team, region, or individual trader (internal) to drive accountability.

SRS

Software Requirements Specification for Demurrage Defender

Prepared by: Person
Date: Date

1. Introduction

The introduction of the Software Requirements Specification (SRS) provides an overview of the entire document, establishing its purpose, scope, and basic terminology.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the complete functional and non-functional requirements for Demurrage Defender, a unified, AI-augmented maritime cost-management platform. This document describes the system's behavior, interfaces, data, and constraints in sufficient detail to guide architecture, development, and quality assurance.

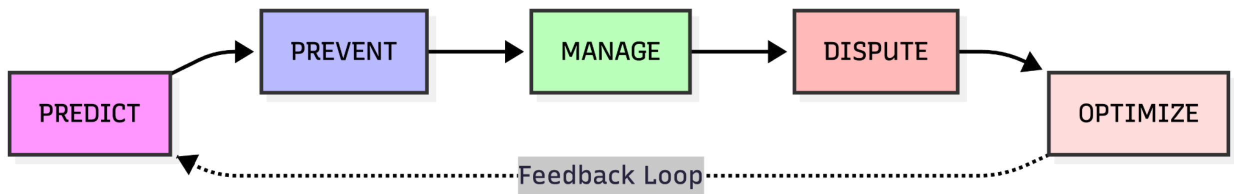
Demurrage Defender automates the prediction, prevention, management, dispute, and optimization of time-based penalties and rewards—specifically demurrage, detention, and despatch—for two primary user segments:

- Bulk Charterers (commodity traders, oil majors, miners, grain houses, steel mills, utilities) who manage vessel laytime under Charter Parties.
- Large Beneficial Cargo Owners (BCOs) and NVOCCs (global retailers, manufacturers, freight forwarders) who manage container free-time and associated detention & demurrage charges.

The system is sold as a single subscription per company, granting all users within that company unrestricted access to all platform capabilities, irrespective of their industry segment.

1.2 Scope

Demurrage Defender is a full-scale, cloud-based SaaS platform. The system comprises two interlocking operational modules and a pervasive AI assistant, all built around the Core Value Loop:



1.2.1 In-Scope

1.2.1.1 Module A - Laytime Defender (Bulk / Tramp Shipping)

	Core Value Loop Phase(s)
ess (NOR) and Statement of Facts (SOF) via OCR.	**Manage**
clauses (e.g., SHEX, WIBON, ATUTC) and computes allowed	**Manage**
etection and continuous laytime clock monitoring.	**Prevent, Manage**
s based on historical congestion, weather, and performance	**Predict**
h outcomes under user-defined Charter Party terms before	**Predict**
Sheets), demurrage counter-claims, and despatch invoices.	**Dispute**
and deadline alerts.	**Dispute**
t and claim letter drafting.	**Dispute**
spatch performance by port, vessel, cargo, and trader.	**Optimize**
ents based on historical outcomes.	**Optimize** (feeds into **Predict** via feedback loop)
e efficiency.	**Optimize**
y, dispute tendency).	**Predict, Dispute, Optimize**

	Core Value Loop Phase(s)
n laytime is at risk (e.g., “invoke force majeure clause”).	**Prevent**

1.2.1.2 Module B – Detention Defender (Container Shipping)

	Core Value Loop Phase(s)
scharge, availability) via API/EDI.	**Manage**
rier tariffs and service contract terms.	**Manage**
time free-time status and accrued D&D charges.	**Manage**
pproaches expiry.	**Prevent**
rnate depot with later hours, dual-transaction moves).	**Prevent**
g and optimal gateway selection.	**Predict**
s against actual gate events and tariff rules.	**Dispute**
or overcharges.	**Dispute**
	Dispute
oring.	**Optimize**
xtensions and carrier-specific D&D caps for tender.	**Optimize** (feeds into **Predict**)
e-time breach rates.	**Predict**

1.2.1.3 Module C – Defender AI (Co-pilot / Chatbot)

This module operates across all phases and data sources.

	Core Value Loop Phase(s)
arter Party clauses, carrier tariffs, and platform data.	**All**
ytime/free-time calculations.	**Manage, Dispute**
threaten demurrage/detention.	**Prevent**
dispute letters, and suggested C/P amendments.	**Dispute, Optimize**
ments.	**Manage, Predict**

	Core Value Loop Phase(s)
ecific documents and curated industry knowledge base	**All**

1.2.1.4 Underlying Platform Capabilities

These cross-cutting system features support the entire loop.

	Core Value Loop Phase(s)
osure across both bulk and container operations.	**Manage**
t trail for every voyage and container.	**Manage, Dispute**
/Dispute automatically retrain Predict models and refine	**Optimize → Predict**
rrier APIs, terminal/port authority systems, weather services,	**All**
s.	**All**

1.2.2 Out Of Scope

1.3 Definitions, acronyms, and abbreviations

Provide the definitions of all terms, acronyms, and abbreviations required to properly interpret the SRS.

1.4 References

Provide a complete list of all documents referenced elsewhere in the SRS, including title, date, and publishing organization.

1.5 Overview

This SRS is organized as follows:

- Section 1: **Introduction** – Provides purpose, scope, definitions, and this overview.
- Section 2: **Overall Description** – Describes product perspective, operating environment, user characteristics, design constraints, and assumptions.

- Section 3: **System Features and Functional Requirements –** The core of the document. For each phase of the Core Value Loop (Predict, Prevent, Manage, Dispute, Optimize) and for the Defender AI, detailed feature specifications are provided, including inputs, processing, outputs, and acceptance criteria. Both Laytime Defender (bulk) and Detention Defender (container) functionalities are covered within each phase where applicable.
- Section 4: **External Interface Requirements –** Specifies user interfaces, hardware interfaces, software APIs (AIS, carrier, terminal, weather, ERP), and communication protocols.
- Section 5: **Data Requirements –** Defines the logical data model, data retention policies, security, and tenant isolation.
- Section 6: **Non-Functional Requirements –** Addresses performance, scalability, availability, security, auditability, and usability.
- Section 7: **AI/ML Requirements –** Details the specific requirements for the OCR/NLP document ingestion, laytime rules engine, predictive models, and the Defender AI chatbot.
- Section 8: **Appendices –** Includes glossary, references to industry standards, and traceability to the Business Requirements Document.

2. Overall description

This section describes the general factors that affect the product and its requirements but does not state specific requirements.

2.1 Product perspective

Demurrage Defender is a standalone, cloud-native Software-as-a-Service (SaaS) platform that integrates into the existing technology ecosystems of large commodity trading houses, industrial charterers, beneficial cargo owners (BCOs), and non-vessel operating common carriers (NVOCCs). It is not a subsystem of a larger product but rather a central hub for maritime time-cost management, designed to interoperate with external systems through well-defined APIs.

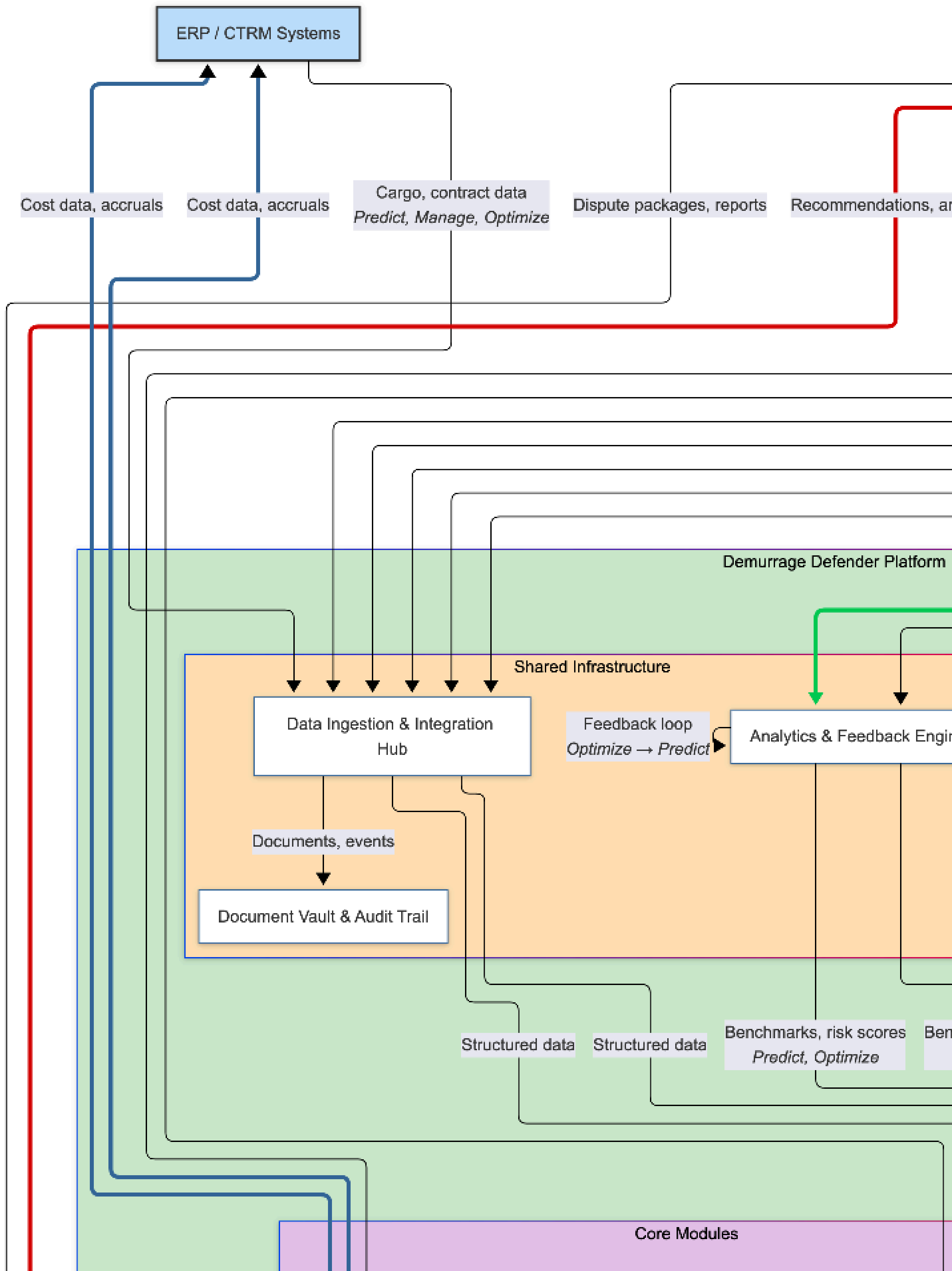
The platform sits at the intersection of three operational domains:

1. **Voyage Operations (Bulk)**: It ingests real-time vessel positions (AIS), port call documents (NOR, SOF), and charter party contracts to manage laytime, demurrage, and despatch.
2. **Container Logistics (Liner)**: It ingests container milestone events from ocean carriers and terminals, alongside tariff and service contract data, to manage free-time, detention, and port demurrage.
3. **Financial & Legal Workflow**: It produces auditable laytime statements, demurrage counter-claims, despatch invoices, and D&D dispute packages that feed into the company's accounts payable, treasury, and legal departments, and can integrate with ERP/CTRM systems.

The platform is decomposed into three interlocking modules, all accessible under a single company subscription:

- **Laytime Defender –** Bulk/Tramp shipping cost management.
- **Detention Defender –** Container shipping cost management.
- **Defender AI –** An AI co-pilot and chatbot that operates across both modules.

The following high-level context diagram illustrates the system's boundaries and external interfaces:


ERP / CTRM Systems
Cost data, accruals
Cost data, accruals
Cargo, contract data
Predict, Manage, Optimize
Dispute packages, reports
Recommendations, ar
Demurrage Defender Platform
Shared Infrastructure
Data Ingestion & Integration Hub
Feedback loop
Optimize → Predict
Analytics & Feedback Engi
Documents, events
Document Vault & Audit Trail
Structured data
Structured data
Benchmarks, risk scores
Predict, Optimize
Ben
Core Modules

2.2 Operating Environment

The platform operates as a multi-tenant, web-based SaaS application with the following characteristics:

- Client Environment: Users access the system via modern web browsers (Chrome, Firefox, Safari, Edge – latest two major versions) on desktop and tablet devices. A mobile-responsive design ensures core alerting and dashboard functionality is available on smartphones, though complex document review and calculation screens are optimized for larger screens.
- Server Environment: The application backend is hosted on a major cloud provider (e.g., AWS, Azure, GCP) using containerized microservices. The system supports horizontal scaling to handle high data ingestion volumes and concurrent user sessions.
- Real-Time Communication: The platform utilizes WebSocket connections for live clock updates, AIS map tracking, and instant alert delivery. Asynchronous task queues handle long-running operations such as OCR processing, model retraining, and bulk invoice audits.
- Data Storage:
    - Relational Database: Stores transactional data (voyages, container records, user management, dispute cases, audit logs).
    - Document Storage (Object Store): Holds uploaded SOFs, NORs, C/Ps, EIRs, invoices, and generated reports (PDF, Excel).
    - Time-Series Database: Optimized for high-frequency AIS tracking data and container milestone event streams.
    - Vector Database: Stores embeddings of Charter Party clauses, maritime case law, carrier tariff documents, and company-specific operational notes. Used by the Defender AI's Retrieval-Augmented Generation (RAG) engine to retrieve relevant context for answering user queries and drafting documents.
- AI/ML Infrastructure:
    - Model Training Environment: Dedicated GPU-enabled compute instances for periodic retraining of predictive models (port risk scoring, demurrage forecasting, counterparty behavior) and fine-tuning of document understanding models (OCR/NLP).
    - Inference Serving: Low-latency endpoints for real-time AI features: document extraction (OCR), laytime clause interpretation, chatbot responses, and

predictive risk scores. The Defender AI uses a combination of hosted large language models (LLMs) via secure APIs and proprietary fine-tuned models, with all data processed within the tenant's logical boundary.
    - Feedback Loop: The Analytics & Feedback Engine continuously collects actual outcomes (e.g., final demurrage paid, dispute resolution) and retrains models on a scheduled basis, improving prediction accuracy and AI recommendations over time.

2.3 User characteristics

The system is procured by a Company (the legal entity subscribing to the platform). There are no pre-defined, role-restricted user profiles within the application. Every user account associated with a subscribing company has unrestricted access to all features, data views, and actions across both Laytime Defender and Detention Defender modules.

The platform does not differentiate between user types (e.g., analyst, manager, finance) at the permission level. Instead, it provides a consistent, unified interface where the full breadth of functionality is available. The system does not enforce any role-based access control (RBAC) limitations on feature access; any user can perform any operation, including uploading documents, running calculations, issuing disputes, viewing financial dashboards, and configuring integrations.

The assumed user is a professional working in maritime operations, chartering, logistics, finance, or legal within the target customer organization. The system's usability will cater to this knowledgeable audience, assuming familiarity with industry terminology and workflows without requiring extensive training.

2.4 Constraints

The following constraints guide the system's architecture and development:

- **Auditability & Legal Admissibility**: Every data extraction, user modification, calculation, and document generation must be logged in an immutable, tamper-proof audit trail. The output laytime statements, dispute packages, and time sheets must be generated in a format suitable for presentation in LMAA, SMA, or other maritime arbitration proceedings.

- **Data Accuracy**: The AI-powered document ingestion (OCR) must achieve field-level extraction accuracy sufficient for financial and legal use, with clear confidence flags for manual review. The deterministic laytime/free-time rules engine must produce outputs that match a manual expert calculation under all documented Charter Party clauses and tariffs.
- **Multitenancy & Data Isolation**: Each subscribing company's data (documents, contracts, voyages, container records, financials) must be logically isolated to prevent cross-tenant access. Anonymized, aggregated benchmarking data is only derived from tenants that explicitly opt-in to data sharing.
- **Integration Breadth**: The system must be designed to integrate with a wide variety of external APIs and EDI feeds (multiple AIS providers, at least 10 major container carriers, terminal systems, weather services) without requiring core code changes. An integration framework with configurable adapters is required.
- **Regulatory Compliance**: The platform must comply with relevant data protection regulations (GDPR for European users, CCPA for California, as applicable) and provide mechanisms for data residency where required.
- **Language Support**: The initial release targets English-language documents and UI. The OCR engine must be capable of processing SOFs written in English (and, in later phases, commonly encountered languages such as Spanish, Portuguese, and French). The Defender AI shall respond in English.

2.5 Assumptions and dependencies

2.5.1 Assumptions

1. **Document Quality**: Users will upload Statements of Facts, Notice of Readiness forms, and Equipment Interchange Receipts that are legible, predominantly in English, and not excessively degraded. The OCR engine's accuracy is bounded by the quality of the source documents.
2. **Contractual Data Availability**: Charter Party clauses and container tariff rules can be digitized either through user input (clause tagging), selection from a standard library, or structured upload. The rules engine requires a complete and correct representation of the governing contractual terms to function accurately.

3. **AIS Coverage**: The chosen AIS data provider offers sufficient global coverage and update frequency (terrestrial and satellite) to reliably detect vessel arrivals at pilot stations and anchorage zones for all ports of interest.
4. **Carrier/Terminal Cooperation**: Container carriers and terminal operators will provide the necessary API or EDI access for milestone and tariff data, or users will provide such data via structured upload. The system does not rely on screen scraping.
5. **User Expertise**: Users possess sufficient domain knowledge to validate AI outputs, interpret contractual clauses, and make final commercial decisions. The system is a decision-support and automation tool, not a fully autonomous agent.

2.5.2 Dependencies

1. **External Service Availability**: The real-time alerting and tracking features are dependent on the uptime and latency of third-party services (AIS providers, carrier status APIs, weather services). The system will implement caching and graceful degradation where possible.
2. **Legal/Regulatory Stability**: The AI's legal knowledge base is dependent on the continued availability and accuracy of maritime law sources (BIMCO, case law databases). Updates to international conventions (e.g., IMO regulations) must be incorporated into the knowledge base as they occur.
3. **Data Partnerships**: For the external benchmarking feature, a sufficient number of customers must opt-in to anonymized data sharing. The feature's value scales with the size of the contributing pool.

3. System Features and Functional Requirements

3.1 Predict Phase

The Predict phase provides the intelligence layer before a fixture or shipment is committed. It uses historical operational data, external market feeds, and machine learning models to forecast demurrage, detention, and despatch outcomes, enabling data-driven decision-making on port, terminal, contract terms, and routing.

The below features collectively enable users to shift from reactive to proactive management of time-cost risks. The output of Predict flows directly into the Prevent phase (alerts triggering operational action) and the Manage phase (initial baseline for laytime/free-time clocks). The feedback loop ensures that actual outcomes from Manage and Dispute continuously improve Predict accuracy.

3.1.1 Port, Terminal & Route Risk Assessment

Risk Explorer

Feature ID	Feature Name	Description	Inputs
FR-PRED-01	Port Risk Dashboard	Interactive map and table displaying a quantified risk score for demurrage (bulk) or detention (container) for each port/terminal, filterable by vessel type, commodity, season, and cargo volume.	User-selected filters (port, vessel size/type, commodity, date range); historical AIS data (anchorage times, berth occupancy); historical SOF/D&D outcomes; weather patterns; port authority announcements.
FR-PRED-02	Terminal Drill-Down	When a port is selected, the system displays a ranked list of terminals at that port with individual risk scores and historical performance charts.	Same as FR-PRED-01 plus terminal-level data (berth specifications, crane availability, historical SOF events).
FR-PRED-03	Route Risk Assessment	Calculates risk for a specific trade lane (origin → destination) considering cumulative risks at load port, discharge port, and seasonal weather patterns along the route.	Origin port/terminal, destination port/terminal, vessel type, planned laycan window.

User Stories:

- "I want to see the demurrage risk for loading iron ore at Port Hedland in August, so I can decide whether to accept a CIF delivery there or push for a different discharge port."
- "I want to compare the detention risk at Los Angeles/Long Beach vs. Savannah for my electronics imports from Asia, so I can route shipments to the lower-risk gateway."

3.1.2 Voyage / Shipment Simulation

Laytime Simulator (Bulk) / Free Time Forecaster (Container)

This module enables users to create "what-if" scenarios using draft contracts and cargo details, generating probabilistic forecasts of demurrage, detention, and despatch outcomes.

Feature ID	Feature Name	Description	Inputs
FR-PRED-04	Laytime Simulator (Bulk)	Simulates a complete loading and discharge operation based on a draft Charter Party, cargo parcel, and port pair, forecasting laytime used, demurrage cost, and despatch likelihood.	Cargo quantity (MT) and tolerance; load port & discharge port (with berth); draft C/P clauses (laytime rate, SHEX/SHINC, WIBON, demurrage rate, despatch terms); planned laycan window.
FR-PRED-05	Scenario Comparison (Bulk)	Side-by-side comparison of up to three different simulation scenarios, highlighting the financial impact of different C/P terms or port choices.	Multiple saved scenarios from FR-PRED-04.
FR-PRED-06	Free Time Forecaster (Container)	Forecasts the probability and cost of detention and port demurrage for a planned container shipment, given port of discharge, consignee, carrier, and historical free-time performance.	Number of containers; carrier/service contract free-time rules; discharge port/terminal; consignee (or historical pickup/return patterns if new consignee, use terminal averages).
FR-PRED-07	Gateway/Route Comparison (Container)	Compares total landed D&D cost estimates for the same shipment routed through different import gateways.	Same as FR-PRED-06, but with multiple discharge port/terminal options.

User Stories:

- "Before fixing a vessel on a 10,000 MT SHEX NORGRAIN C/P, I want to run a simulation to see if I should instead offer 12,000 MT SHINC to reduce my demurrage exposure."

- "I want to input my planned shipment of 50 containers from Shanghai to Chicago and see whether routing via Savannah reduces detention costs compared to Los Angeles, given my consignee's warehouse patterns."

3.1.3 Counterparty & Carrier Behavior Analytics

Counterparty Intel

This module profiles shipowners (for bulk) and container carriers (for liner) based on historical interactions, helping users anticipate claim aggressiveness and invoice accuracy.

Feature ID	Feature Name	Description	Inputs
FR-PRED-08	Owner/Carrier Profile Page	A dedicated profile page for each shipowner or container carrier the company has interacted with, summarizing behavior metrics.	Company's own historical voyage and D&D data: demurrage claims received, dispute outcomes, invoice overcharge %, average time to settle despatch payment promptness.
FR-PRED-09	Counterparty Risk Alert	An alert integrated into the voyage simulation and fixture workflow that flags high-risk owners or carriers before a commitment is made.	Integration with FR-PRED-04/06; if selected owner/carrier has a negative behavior profile, the system flags it.

User Stories:

- "I want to check whether a shipowner we haven't worked with before has a history of inflated demurrage claims, based on our internal data, so I can price that risk into the fixture."
- "Before signing a new carrier service contract, I want to see which carriers have the highest D&D invoice error rates, so I can demand better audit rights or more favorable free-time terms."

3.1.4 Market Watch & Early Warning

Market Pulse

This module provides a continuous feed of external events and conditions that could impact laytime and free-time, enabling proactive planning.

Feature ID	Feature Name	Description	Inputs
FR-PRED-10	Congestion & Disruption Alerts	Configurable alerts for port congestion, strikes, weather events, and carrier free-time policy changes on user-selected trade lanes.	AIS data (anchorage counts), weather services (storm warnings), port authority announcements, carrier tariff updates, news feeds.
FR-PRED-11	Market Intelligence Dashboard	A curated news feed and trend chart showing macro indicators relevant to demurrage/detention (e.g., Baltic Dry Index, container spot rates, port throughput stats).	Third-party maritime data providers, news APIs, public port statistics.

User Stories:

- "I want to receive an alert if a typhoon is approaching a port where I have vessels due to load next week, so I can warn the terminal and estimate potential laytime overruns."
- "I want a monthly summary of how port congestion in my key trade lanes is trending, so I can adjust my laytime assumptions in upcoming fixtures."

3.2 Prevent Phase

The Prevent phase transforms raw operational data (AIS positions, container gate events, document uploads) into a live, intelligent watchtower. It automatically detects when a laytime or free-time clock is active or at risk, alerts responsible parties, and provides concrete, actionable recommendations to avoid penalties. The phase integrates tightly with the Manage phase (which provides the calculation backbone) and the Defender AI (which powers recommendations).

The Prevent phase converts predictive intelligence and live data into immediate, action-oriented interventions. Features FR-PREV-01 through FR-PREV-04 establish the real-time monitoring and alerting backbone. FR-PREV-05 through FR-PREV-07 layer on AI-powered recommendations that turn alerts into actionable strategies. FR-PREV-08 through FR-PREV-10 close the loop by automating communication and task management, ensuring that no deadline is missed due to manual oversight.

All Preventive actions generate data that flows into the Manage phase (for continuous clock tracking) and the Optimize phase (for measuring alert effectiveness and user response patterns).

3.2.1 Real-Time Deadline Tracking & Alerting

Live Clock

This module provides a single, consolidated view of all active time-sensitive commitments, with color-coded countdowns and tiered notifications.

Feature ID	Feature Name	Description	Inputs
FR-PREV-01	Unified Deadline Dashboard	A real-time dashboard showing all active vessels (bulk) and containers (liner) with their current time status: within allowed time (green), approaching expiry (amber, configurable threshold), on demurrage/detention (red).	For bulk: AIS vessel position, NOR status from Manage (FR-MAN-01), allowed laytime from Laytime Engine (FR-MAN-06). For container: container milestones from carrier/terminal APIs (FR-MAN-03), free-time expiry from Free-Time Engine (FR-MAN-07).
FR-PREV-02	Escalating Alert Rules	Configurable, tiered notifications that trigger when a vessel or container approaches or exceeds its allowed time.	User-defined thresholds (e.g., 72h, 48h, 24h before free-time expiry; immediately upon NOR tender for bulk; immediately upon demurrage commencement). Notification channels: in-app, email, SMS, Slack/Teams.
FR-PREV-03	Bulk Arrival Detection & NOR Alert	Automatically detects when a tracked vessel enters the pilot station or anchorage zone of its designated port and alerts the user to tender or expect NOR, triggering the laytime clock.	AIS position data; geofence definitions for anchorage and pilot station zones; voyage schedule from Manage.

Feature ID	Feature Name	Description	Inputs
FR-PREV-04	Container Free-Time Countdown Alert	Detects when a container's free time is approaching expiry and sends alerts at configurable intervals.	Container milestones (discharge date, gate-out full, gate-in empty) and free-time rules from Manage.

User Stories:

- "I want to see all my import containers that will breach free time in the next 48 hours in a single list, so I can prioritize trucking resources to avoid detention."
- "I want my phone to buzz the moment one of my vessels arrives at the Paranaguá pilot station, so I can immediately check if the NOR has been tendered and start tracking laytime."

3.2.2 Actionable Recommendation Engine

Defender AI Assistant (Preventive Mode)

When a risk is detected, the AI proactively suggests specific, contract- and context-aware operational actions to avoid or reduce the penalty.

Feature ID	Feature Name	Description	Inputs
FR-PREV-05	Bulk Demurrage Prevention Advisor	When a vessel is at anchorage, on berth with stoppages, or approaching laytime exhaustion, the AI analyzes the C/P and real-time situation to suggest mitigating actions.	Current SOF events, remaining laytime, C/P clauses (WIBON, force majeure, shifting, demurrage rate), real-time weather, port congestion status.
FR-PREV-06	Container Detention Prevention Advisor	When a container's free time is about to expire, the AI suggests the most cost-effective action based on	Container ID, current location, free-time expiry, carrier tariff (detention tiers, depot locations

Feature ID	Feature Name	Description	Inputs
		current depot rules, gate hours, and the carrier's tariff tiers.	and hours), trucker appointment slots, user's preferred depot list.
FR-PREV-07	What-If Intervention Simulator	User can interactively test potential operational decisions (e.g., "What if we start loading at 22:00 instead of 06:00?") to see the impact on demurrage/despatch in real time.	Current vessel/container state from Manage; user-specified changed event.

User Stories:

- "My vessel is on demurrage and it's raining at the port. I want the AI to tell me if I can claim the rain time as an exception under my C/P, and if so, draft the dispute note for the SOF."
- "A container's free time expires tomorrow at noon. I want the system to tell me which return depot is open latest today and how much detention I'll save by using it."

3.2.2 Automated Workflow Triggers

Smart Automations

This module automates routine communications and task creation, reducing manual effort in the critical hours before a deadline is breached.

Feature ID	Feature Name	Description	Inputs
FR-PREV-08	Automated External Notifications	Automatically sends pre-configured notifications to external parties (truckers, port agents, terminals, depots) when a time-critical event is detected.	User-defined rules: "If container free time expires in < 24h, email trucker@xyz.com with container number and recommended depot." Or "If vessel demurrage starts, message agent@shipowner.com to request SOF update."

Feature ID	Feature Name	Description	Inputs
FR-PREV-09	Task Creation & Assignment	When an alert requires human follow-up, the system automatically creates a task within the platform and assigns it to a designated user or group.	Same as FR-PREV-08, but output is an internal task.
FR-PREV-10	Integration with External Workflow Tools	Pushes alerts and tasks directly into the company's existing workflow systems (e.g., Jira, ServiceNow, Asana) via API, for teams not logged into Demurrage Defender.	User-provided webhook URL or API credentials for target system.

User Stories:

- "I want the system to automatically email my trucking company's dispatch desk when any of my import containers has less than 24 hours of free time remaining, including the exact container number and recommended return depot, so I don't have to manually call them."
- "When my Capesize vessel goes on demurrage, I want a task automatically created for my demurrage analyst with the current cost and a link to the live laytime clock, so they can immediately start the dispute process."

3.3 Manage Phase

The Manage phase transforms raw operational inputs—documents, AIS feeds, container milestones—into structured, real-time records of time used, time allowed, and money accruing. It provides the single source of truth for every active voyage and container, ensuring that users always know exactly where they stand against their contractual obligations and can drill down to the finest detail.

The Manage phase is the platform's operational core. It transforms raw data (documents, AIS, carrier feeds) into structured, continuously updated records. The Laytime Engine and Free-Time Engine are the algorithmic centerpieces, delivering authoritative, defensible calculations. The FleetView dashboard gives instant oversight, while the DocVault ensures that every supporting document and every user action is preserved in an immutable, arbitration-ready format. The outputs of Manage—time sheets, accrual records, audit trails—flow directly into the Dispute phase for claim generation, and into the Optimize phase for performance analysis.

3.3.1 Automated Document Ingestion & Digitization

DocIn

This module is the entry point for all operational documents and data feeds. It accepts multiple formats, uses AI-powered extraction to digitize unstructured content, and allows manual correction with full auditability.

Feature ID	Feature Name	Description	Inputs
FR-MAN-01	SOF/NOR Upload & OCR	Users can upload scanned or electronic copies of Statement of Facts (SOF) and Notice of Readiness (NOR) documents. The system extracts all timestamped events, vessel details, and port information using AI-powered OCR and NLP.	PDF, JPEG, PNG, or TIFF files; email-forwarded attachments; structured EDI/XML (for future agent integration).
FR-MAN-02	Bulk Email/Agent Data Ingestion	Automatically processes SOF and NOR data sent by email (PDF attachment or inline text) from agents, or via direct API from port agent systems.	Emails to a dedicated system address; agent API data feed (future).
FR-MAN-03	Container Milestone Ingestion	Connects to ocean carrier APIs and/or EDI feeds to automatically ingest container status events (gate-in, gate-out, discharge, load, empty return, availability).	Carrier API (Maersk, MSC, CMA CGM, etc.) or EDI 214/322 messages; manual CSV upload for non-integrated carriers.
FR-MAN-04	Manual Event Entry & Override	For any extracted or ingested event, a user can manually edit the timestamp, event type, or remarks.	User selects event, modifies field(s) enters a reason for change.

Feature ID	Feature Name	Description	Inputs
		All changes are logged with justification.	

User Stories:

- "I want to drag-and-drop a scanned, handwritten SOF PDF and have the system pull out every timestamp and event so I don't have to type it manually."
- "I want container milestones from Maersk and MSC to flow directly into the platform without any manual entry, so my free-time clocks are always accurate."

3.3.2 Laytime & Free-Time Calculation Engine

Laytime Engine (Bulk) / Free-Time Engine (Container)

The core calculation engines. They interpret contractual terms against operational events to produce a complete, defensible record of allowed time, used time, demurrage, and despatch.

Feature ID	Feature Name	Description	Inputs
FR-MAN-05	Charter Party Clause Library & Tagging	A searchable library of standard Charter Party forms (GENCON 94/2022, NORGRAIN, BPVOY, SHELLVOY, etc.) and common rider clauses. Users can upload their own C/P text and tag key clauses for the engine.	User uploads C/P PDF or text; selects standard form as base; manually highlights or tags clauses (laytime rate, SHEX/SHINC, WIBON/WIPON, demurrage rate, despatch terms, shifting, force majeure).
FR-MAN-06	Laytime Engine (Bulk)	Deterministic rules engine that computes allowed laytime, used laytime, and the resulting demurrage or despatch from the SOF events and tagged C/P clauses.	Structured SOF events (FR-MAN-01), tagged C/P clauses (FR-MAN-05), cargo quantity, any user-defined exceptions.

Feature ID	Feature Name	Description	Inputs
FR-MAN-07	Free-Time Engine (Container)	Tracks free-time usage for each container based on carrier tariff rules, computing detention and port demurrage accruals.	Container milestones (FR-MAN-03), carrier tariff / service contract free-time rules (days, per-diem tiers, calendar day counting, exceptions).
FR-MAN-08	Timeline Visualization & What-If Slider	Visual timeline that overlays SOF events (or container milestones) with the computed laytime/free-time clock, clearly showing running, stopped, and demurrage/detention periods. An interactive what-if slider allows users to drag event times and see instant recalculation.	Events and calculated periods from FR-MAN-06/07.

User Stories:

- "I want the system to calculate the exact demurrage for my voyage, showing me exactly when the clock was stopped for rain, and let me see how much I would save if that rain period was 30 minutes shorter."
- "I want to see a visual countdown for each of my import containers showing exactly how many free days remain and what the charge will be if I don't return it today."

3.3.3 Unified Operational Dashboard

FleetView

A single-screen overview of all active operations, giving instant visibility into financial exposure and operational status across both bulk and container.

Feature ID	Feature Name	Description	Inputs
FR-MAN-09	Bulk Voyage Dashboard	Map-centric view showing all active voyage charter vessels, their positions, laytime status, and demurrage/despatch exposure.	AIS positions, voyage data, laytime status from FR-MAN-06.
FR-MAN-10	Container Operations Dashboard	Table and summary view of all active import/export containers with their free-time status, days remaining, and accrued D&D charges.	Container milestone data and free-time status from FR-MAN-07.
FR-MAN-11	Combined Financial Exposure View	A roll-up dashboard that consolidates the total demurrage, detention, and despatch exposure across all active bulk voyages and containers, with trend indicators.	Financial data from FR-MAN-06, FR-MAN-07, and historical benchmarks from Optimize.

User Stories:

- "I want to open a single screen and immediately see that my total exposure to demurrage and detention right now is $142,000, and that it's 15% higher than this time last month."
- "I want to see all my vessels at anchor in the US Gulf on a map, and know which ones are on demurrage without clicking into each one."

3.3.4 Centralized Document & Audit Trail

DocVault

An immutable, searchable repository that links every document to its voyage or container and provides a complete, tamper-proof history of every action taken on the platform.

Feature ID	Feature Name	Description	Inputs
FR-MAN-12	Document Storage & Auto-Linking	Every uploaded document (SOF, NOR, C/P, EIR, Arrival Notice, carrier invoice) is securely stored and	Uploaded files (FR-MAN-01, etc.) with extracted or user-provided voyage/container references.

Feature ID	Feature Name	Description	Inputs
		automatically linked to the correct voyage or container record based on metadata extraction or manual association.	
FR-MAN-13	Immutable Audit Log	A chronological, non-editable log that records every system action and user modification: data extractions, field overrides, calculations, manual edits, user logins, and dispute submissions.	All system events generated by other modules.

User Stories:

- "I need to find the original signed SOF and the final laytime statement for a voyage from 18 months ago because the owner has filed for arbitration, and I need to show the arbitrator exactly what was calculated and when."
- "I want to know who changed the rain stoppage time from 14:00 to 14:15 and why, because it affected our demurrage bill by $3,000."

3.4 Dispute Phase

The Dispute phase transforms laytime statements, free-time calculations, and carrier D&D invoices into actionable financial outcomes. It automates the generation of time sheets and despatch invoices, audits incoming charges, tracks the full lifecycle of each dispute, and provides an AI-powered legal assistant for clause interpretation and correspondence drafting.

The Dispute phase closes the financial loop. ClaimBuilder automates the production of professional, evidence-backed documents. Dispute Tracker ensures that nothing falls through the cracks, with full visibility into the status and value of every claim. Defender AI acts as a junior maritime paralegal, available 24/7 to research clauses, draft correspondence, and build legal arguments. The outcomes of resolved disputes—settlement amounts, root causes, clause interpretations—flow into the Optimize phase, strengthening the feedback loop for future predictions and contract negotiations.

3.4.1 Automated Claim & Counter-Claim Generation

ClaimBuilder

This module turns verified calculations from the Manage phase into professional, audit-ready documents that can be sent directly to shipowners, carriers, or trading counterparties.

Feature ID	Feature Name	Description	Inputs
FR-DIS-01	Laytime Statement & Demurrage Counter-Claim (Bulk)	Generates a complete, fully reasoned Laytime Statement (Time Sheet) based on the system's final laytime calculation. This serves as the charterer's formal position on demurrage/despatch for a voyage.	Final laytime calculation from FR-MAN-06; SOF events; tagged C/P clauses; cargo quantity; demurrage/despatch rates.
FR-DIS-02	Despatch Invoice (Bulk)	When a voyage completes with time saved, the system automatically generates a despatch invoice to be presented to the shipowner.	Laytime calculation showing despatch earned; C/P despatch rate and terms; beneficiary details.
FR-DIS-03	D&D Invoice Auditor (Container)	Automatically reconciles a carrier-issued D&D invoice against the system's own free-time calculation and gate event records, flagging every discrepancy.	Carrier D&D invoice (PDF, Excel, or API data); container milestone data from FR-MAN-03; free-time calculation and tariff from FR-MAN-07.
FR-DIS-04	Dispute Package Generator (Container)	One-click generation of a comprehensive dispute package for carrier D&D overcharges, ready for	Discrepancy Report from FR-DIS-03 supporting EIRs, gate logs, and tariff excerpts.

Feature ID	Feature Name	Description	Inputs
		submission to the carrier's billing department.	

User Stories:

- "After reviewing the AI's laytime calculation, I want to generate the final Laytime Statement with a single click and send it to the owner as my formal demurrage counter-claim, knowing it includes all necessary clause citations."
- "I've received a $45,000 D&D invoice from the carrier. I want to upload it and instantly see that $12,000 of it is overcharged because they used the wrong gate-out dates on 15 containers."

3.4.2 Dispute Workflow & Case Management

Dispute Tracker

This module manages the entire lifecycle of a dispute—from initial identification to final resolution—providing visibility, accountability, and deadline management.

Feature ID	Feature Name	Description	Inputs
FR-DIS-05	Dispute Case Creation & Tracking	A user can create a formal dispute case from any identified overcharge or counter-claim, or the system can auto-create one upon invoice audit or despatch detection.	Discrepancy report or claim document; user-assigned owner and priority.
FR-DIS-06	Correspondence & Evidence Management	Within a dispute case, users can log all correspondence with the counterparty (email, letters) and link additional evidence documents.	User-uploaded emails/letters; generated dispute packages; any additional files.
FR-DIS-07	Deadline & Escalation Alerts	The system tracks key deadlines for dispute resolution (e.g., contractual time bars for claims, carrier response	User-defined deadlines per case type; default time-bar rules from common C/Ps and carrier tariffs.

Feature ID	Feature Name	Description	Inputs
		windows) and alerts users to avoid missing critical dates.	
FR-DIS-08	Dispute Dashboard & Settlement Tracking	A high-level dashboard showing all active disputes, their financial value at stake, aging, and resolution rates. Includes fields to record final settlement amounts and reasons.	All active and closed dispute cases.

User Stories:

- "I want to see all ongoing disputes with a specific shipowner, how much money is at stake, and when our next response is due, all in one place."
- "When a carrier sends an email rejecting our dispute, I want to forward it into the case file so that my entire team can see the latest response and we can plan our next move."

3.4.3 AI-Powered Legal & Clause Reasoning

Defender AI Assistant (Dispute Mode)

The AI assistant, already used in Prevent and Manage, takes on a specialized legal advisory role in the Dispute phase. It leverages the full knowledge base of maritime law, Charter Parties, and company history to assist users in building strong arguments.

Feature ID	Feature Name	Description	Inputs
FR-DIS-09	Clause Interpretation & Legal Q&A	Users can ask natural language questions about contractual clauses and their application to the dispute at hand.	User query (e.g., "Does the rain stoppage during meal hours count as laytime under our GENCON 94 SHEX clause?"); the specific C/P text; industry knowledge base (BIMCO commentaries, case law like *The "Happy Day"*).
FR-DIS-10	Dispute Letter Drafting	The AI drafts a formal dispute letter or email, incorporating the specific	Dispute case details (overcharge type, amounts, event

Feature ID	Feature Name	Description	Inputs
		facts of the case, the legal/contractual basis for the claim, and the evidence references.	discrepancies, relevant C/P or tariff clauses); user instructions (e.g., "focus on the incorrect gate-out date for containers MSKU123 and MSKU456").
FR-DIS-11	Research Memo Generation	For complex disputes, the AI can generate a short research memo summarizing the legal position and precedents that support or undermine the company's claim.	Same as FR-DIS-09, but with a broader scope and structured output.

User Stories:

- "The owner argues that rain during the night shift counts as laytime because the terminal was 'working' on other cargo. I want to ask the AI if that argument is valid under English law and get a draft response letter."
- "I have a complex dispute with multiple issues. I want the AI to produce a legal memo I can share with my in-house counsel to get their opinion faster, rather than explaining the whole context from scratch."

3.5 Optimize Phase

The Optimize phase unlocks the strategic value of the platform. It analyzes all historical operational and financial data to identify trends, root causes, and improvement opportunities across ports, contracts, carriers, and routes. The outputs are dashboards for performance review, recommendations for contract negotiation, and routing alternatives that reduce future time-cost exposure. All insights feed back into the Predict phase, retraining models and refining risk assessments.

The Optimize phase completes the cycle. Every resolved dispute, every completed voyage, and every returned container enriches the data lake. The Insights Hub provides retrospective clarity, the Contract Optimizer translates past pain into future protection, and the Route Optimizer steers cargo toward the paths of least financial resistance. All findings are systematically fed back into the Predict phase (via automated model retraining and risk score recalibration), making the entire platform smarter with every transaction.

3.5.1 Post-Event Performance Analytics

Insights Hub

This module provides a comprehensive analytics suite for reviewing past performance, understanding cost drivers, and tracking key metrics over time.

Feature ID	Feature Name	Description	Inputs
FR-OPT-01	Demurrage & Despatch Analytics (Bulk)	Dashboards and reports showing demurrage paid, despatch earned, and laytime efficiency across all voyages, sliceable by port, terminal, vessel, cargo type, trader, and time period.	Completed voyage data from Manage (FR-MAN-06): allowed laytime, used laytime, demurrage/despatch amounts, port, cargo, vessel, C/P terms, SOF events.
FR-OPT-02	Detention & Demurrage Analytics (Container)	Dashboards showing D&D spend by carrier, port, trade lane, consignee, and month. Highlights detention vs. port demurrage split, free-time utilization, and charge tier distribution.	Completed container shipment data from Manage (FR-MAN-07): D&D charges, free-time used vs. allowed, carrier, port, consignee, service contract.
FR-OPT-03	Root Cause Analysis & Trend Detection	The system automatically identifies the most significant causes of demurrage/detention over a selected period and surfaces emerging trends.	All data from FR-OPT-01/02; anomaly detection models.

User Stories:

- "I want a quarterly report that breaks down our $2.1M demurrage spend by root cause, so I can see that 45% of it was due to anchorage waiting times at three specific ports."
- "I want to see the trend of our detention costs per container over the last 12 months, and have the system alert me if there's a sudden increase that needs investigation."

3.5.2 Contract & Procurement Optimization

Contract Optimizer

This module translates historical performance data into concrete, data-backed recommendations for future Charter Party and service contract negotiations.

Feature ID	Feature Name	Description	Inputs
FR-OPT-04	C/P Clause Impact Analyzer (Bulk)	Analyzes which Charter Party clauses have historically led to the most demurrage, disputes, or despatch outcomes, and simulates the financial impact of alternative clause wordings.	Historical C/P clauses, linked laytime outcomes, dispute data, and demurrage/despatch amounts from FR-OPT-01.
FR-OPT-05	Service Contract D&D Optimizer (Container)	Evaluates current carrier service contract free-time terms against actual performance and recommends negotiated adjustments (e.g., extended free time, higher free-time tiers, specific depot guarantees).	Carrier service contract terms, actual D&D outcomes from FR-OPT-02, port/depot performance data.
FR-OPT-06	Counterparty Negotiation Brief	Generates a one-page summary for a specific shipowner or carrier, compiling their historical performance, dispute behavior, and financial impact on your company, to be used as leverage in negotiations.	Counterparty Intel data (FR-PRED-08), dispute history, demurrage/despatch outcomes.

User Stories:

- "Before my annual COA negotiation, I want to see which three C/P clauses led to the most disputes and demurrage costs last year, with a suggested alternative wording that BIMCO recommends."

- "I want to know which carriers overcharge us on D&D the most, so I can prioritize auditing their invoices and demand better free-time terms in our next tender."

3.5.3 Network & Routing Optimization

Route Optimizer

This module compares actual cost outcomes across different ports, routes, and gateways to identify lower-cost alternatives for future shipments.

Feature ID	Feature Name	Description	Inputs
FR-OPT-07	Port/Route Cost Comparison (Bulk)	Compares the total landed demurrage/despatch cost for the same commodity on different port pairs, combining historical data with current risk forecasts.	Historical voyage outcomes per port/route (FR-OPT-01); current port risk scores (FR-PRED-01).
FR-OPT-08	Container Gateway Optimization	Analyzes D&D and transit time outcomes for import containers through different gateways, recommending the most cost-effective routing for a given origin/destination pair.	Container shipment outcomes by gateway (FR-OPT-02), transit time data, inland cost models.

User Stories:

- "I want to compare the total demurrage cost of shipping iron ore through Port Hedland vs. Dampier over the last two years, to see if we should push our supplier to use a different load port."
- "For our LA/LB imports to Chicago, I want to see if routing through Savannah or Houston would reduce our overall detention costs enough to justify any increase in rail cost."

3.5.3 Internal & External Benchmarking

Benchmark

This module enables performance comparison—internally across teams, time periods, or business units, and externally against anonymized industry peers (with opt-in).

Feature ID	Feature Name	Description	Inputs
FR-OPT-09	Internal Benchmarking	Compares demurrage/detention KPIs across different traders, operational teams, regions, or business units within the same company.	Performance data from FR-OPT-01/02, tagged with organizational dimensions (trader, team, region).
FR-OPT-10	External Industry Benchmarking	Compares the company's anonymized performance against an aggregated industry benchmark, derived from other Demurrage Defender customers who have opted into data sharing.	Company's own data (anonymized and aggregated); pooled, anonymized data from consenting customers.

User Stories:

- "I want to see how our detention rate for electronics imports compares to the industry average, so I can tell if we have a systemic problem or if the whole market is seeing the same congestion issues."
- "I want to benchmark my individual traders against each other on demurrage cost per ton, so I can identify my top performers and share their best practices."

3.6 Feedback & Learning Loop

This module is the platform's continuous improvement engine. It operates in the background, automatically collecting outcomes from Manage, Dispute, and Optimize, then feeding them back into the Predict models and the Defender AI's knowledge base. This loop ensures that risk scores, simulations, and recommendations become more accurate and personalized over time.

Feature ID	Feature Name	Description	Inputs
FR-FL-01	Outcome Data Collection Pipeline	Automatically captures structured outcomes from completed voyages and container shipments, resolved disputes, and user feedback signals.	Completed laytime calculations (FR-MAN-06), final demurrage/despatch amounts, dispute resolution data (FR-DIS-08), container D&D outcomes (FR-MAN-07), user feedback on AI answers (thumbs up/down), manual overrides to predictions.
FR-FL-02	Predictive Model Retraining Scheduler	Orchestrates periodic retraining of all ML models used in the Predict phase (Port Risk, Laytime Simulator, Free-Time Forecaster, Counterparty Behavior).	Curated datasets from FR-FL-01; hyperparameter configurations; previous model performance metrics.
FR-FL-03	AI Knowledge Base Update	Incorporates resolved disputes, new case law, updated BIMCO clauses, and user corrections into the Defender AI's retrieval corpus and fine-tuning datasets.	New arbitration awards (user-uploaded or from subscribed legal databases), BIMCO clause updates, user-flagged incorrect AI answers, final dispute outcomes.
FR-FL-04	Feedback Signal Integration (In-Loop)	Enables users to provide direct feedback on predictions and recommendations, which is used to fine-tune future outputs.	Thumbs up/down on Predict risk scores, simulation results, and AI recommendations (FR-PRED, FR-PREV). User corrections to predicted demurrage (e.g., "actual was $12k, predicted was $8k").
FR-FL-05	Feedback Loop Monitoring Dashboard	A system health dashboard that visualizes the performance of the feedback loop itself—model accuracy trends, data recency, retraining	Logs from FR-FL-01 through FR-FL-04; model performance metrics.

Feature ID	Feature Name	Description	Inputs
		status, and knowledge base freshness.	
FR-FL-01	Outcome Data Collection Pipeline	Automatically captures structured outcomes from completed voyages and container shipments, resolved disputes, and user feedback signals.	Completed laytime calculations (FR-MAN-06), final demurrage/despatch amounts, dispute resolution data (FR-DIS-08), container D&D outcomes (FR-MAN-07), user feedback on AI answers (thumbs up/down), manual overrides to predictions.

3.7 Defender AI (Standalone Capabilities)

The Defender AI module is the persistent, intelligent assistant layer that spans the entire Demurrage Defender platform. It provides a natural language interface, retrieval-augmented generation (RAG) over proprietary and industry knowledge, and the ability to orchestrate tasks across multiple phases of the Core Value Loop from a single conversation. This section specifies the AI capabilities that are not confined to a single phase but serve as the unified "brain" of the application.

This module is the connective tissue of Demurrage Defender. It leverages a unified RAG infrastructure and cross-phase orchestration to provide a seamless, intelligent user experience. By embedding the AI deeply into every workflow—while maintaining full explainability and an unalterable audit trail—the platform ensures that every user, regardless of their role, can access the full power of the system through natural, conversational interaction.

3.5.1 Post-Event Performance Analytics

Insights Hub

This module provides a comprehensive analytics suite for reviewing past performance, understanding cost drivers, and tracking key metrics over time.

Feature ID	Feature Name	Description	Inputs
FR-AI-01	Unified Conversational Interface	A persistent chat panel accessible from any screen within the application. Users can ask questions, request actions (simulations, document drafting, data retrieval), and receive proactive notifications. The interface supports free-text input, clickable suggestion chips, and rich responses (tables, charts, links).	User text or voice input; current screen context (e.g., voyage ID, container number); user profile and preferences.
FR-AI-02	Knowledge Base & RAG Infrastructure	The AI retrieves relevant information from a curated knowledge base to ground its answers. This includes company-specific documents (Charter Parties, SOFs, carrier tariffs, internal SOPs) and industry sources (BIMCO clauses, case law summaries, INCOTERMS 2020, maritime regulations).	User query; embedded representations of knowledge base documents; metadata filters (e.g., restrict to a specific voyage or contract).
FR-AI-03	Context-Aware Assistance	The AI automatically inherits the user's current working context (active voyage, selected container, open dispute case) and tailors responses accordingly, reducing the need for repetitive detail.	Current screen context (e.g., Voyage V-2026-045, C/P = GENCON 94); user's recent actions.
FR-AI-04	Cross-Phase Task Orchestration	The AI can execute a sequence of actions spanning multiple Core Value Loop phases in response to a single user command.	Natural language command (e.g., "Compare the laytime risk for the upcoming Paranaguá voyage under SHEX vs. SHINC, and if SHINC is lower risk, generate a draft C/P clause for the broker.")

Feature ID	Feature Name	Description	Inputs
FR-AI-05	Explainability & Audit Logging	Every AI-generated answer, recommendation, or document draft is accompanied by an explanation of the reasoning process and links to source data. All AI interactions are logged immutably.	Internal chain-of-thought reasoning; source documents and data used; model version.
FR-AI-06	Personalization & Feedback Learning	The AI learns from user feedback and corrections to better align with the company's specific terminology, risk appetite, and operational preferences over time.	User feedback signals (thumbs up/down, corrections, "Regenerate requests), integrated via the Feedback Loop (FR-FL-04). Company-specific settings (e.g., preferred demurrage threshold, typical clause preferences).

User Stories:

- "While viewing a disputed demurrage invoice, I want to ask the AI, 'Why is this charge $12,000?' and get an explanation referencing the specific SOF events and C/P clauses that caused it."
- "I want to tell the AI, 'Run a laytime simulation for M/V Sea Harvest with the alternative clause my broker suggested, and if it saves more than $5,000, draft an email to the broker accepting the change.'"

4. External Interface Requirements

4.1 User Interfaces

The platform provides a web-based graphical user interface (GUI) accessible via modern browsers. The UI must support the following high-level characteristics:

Interface Element	Description
Web Application	Single-page application (SPA) built with responsive design principles, ac tablet. Core functionality available on mobile via adaptive layouts.

Interface Element	Description
Map View	Interactive map (e.g., OpenStreetMap, Mapbox) displaying vessel positi and geofences.
Dashboard & Reporting	Customizable widget-based dashboards; tabular data views with sortin
Document Viewer	Side-by-side view for original documents (SOF, NOR, EIR) and extracted
Chat Interface	Persistent chat panel for Defender AI, with conversation history and cli
Notification Center	In-app notification bell with real-time alerts, grouped by type (deadline
Accessibility	The UI should meet WCAG 2.1 Level AA standards.

4.2 Software Interfaces

The platform must integrate with a variety of external software systems and data providers. The following interfaces are required:

4.2.1 AIS Data Providers

Requirement ID	Description
SI-AIS-01	Vessel Position & Voyage Data
SI-AIS-02	Geofence Event Detection
SI-AIS-03	Supported Providers
SI-AIS-04	Data Format & Protocol
SI-AIS-05	Authentication

4.2.2 Ocean Carrier & Terminal Systems (Container)

Requirement ID	Description
SI-CC-01	Container Milestone Events
SI-CC-02	Supported Carriers
SI-CC-03	Integration Methods
SI-CC-04	Free-Time & Tariff Data
SI-CC-05	Authentication & Frequency

4.2.3 Weather Services

Requirement ID	Description
SI-WX-01	Current & Forecast Weather
SI-WX-02	Supported Providers
SI-WX-03	Data Usage
SI-WX-04	Protocol

4.2.4 Port & Terminal Systems

Requirement ID	Description
SI-PT-01	Berth Schedules & Gate Events
SI-PT-02	Integration
SI-PT-03	Scope

4.2.5 Enterprise Systems (ERP / CTRM)

Requirement ID	Description
SI-ERP-01	Cargo & Contract Data Import
SI-ERP-02	Financial Data Export
SI-ERP-03	Integration Protocol
SI-ERP-04	Authentication

4.2.6 Communication & Collaboration Tools

Requirement ID	Description
SI-COM-01	Email Notifications
SI-COM-02	Messaging Platforms
SI-COM-03	Content Format

4.2.7 Legal & Maritime Knowledge Bases

Requirement ID	Description
SI-LAW-01	BIMCO Clause Library
SI-LAW-02	Case Law & Arbitration Awards
SI-LAW-03	INCOTERMS & Regulations
SI-LAW-04	Data Format

4.3 Communication Interfaces

Requirement ID	Description
CI-01	Web Application Protocol
CI-02	Real-Time Updates
CI-03	Data Formats
CI-04	Asynchronous Processing
CI-05	API Versioning
CI-06	Rate Limiting & Throttling

5. Data Requirements

5.1 Logical Data Model

The following describes the core entities, their key attributes, and their relationships. This is a high-level conceptual model; physical implementation details are deferred to design documents.

5.1.1 Bulk / Tramp Shipping Domain

Entity	Description	Key
Company	The subscribing legal entity. All data is scoped to a single company.	cor
Vessel	A physical ship, identified by IMO number.	ves
Voyage	A single chartered voyage from load port to discharge port.	voy dis

Entity	Description	Key
Charter Party (C/P)	The contractual document for the voyage, stored as original PDF and structured clauses.	cp_
C/P Clause	A single tagged clause extracted from the C/P.	clau des
SOF Document	The uploaded Statement of Facts file.	sof
SOF Event	A single timestamped event extracted from the SOF.	eve sta is_r
NOR Document	Notice of Readiness document.	nor
Laytime Calculation	The computed result of allowed vs. used time.	calc des
Calculation Period	A segment of time within the laytime calculation (e.g., running, stopped for rain).	per der
Dispute Case (Bulk)	A formal dispute or despatch claim for a voyage.	disp sta
Counterparty	A shipowner or charterer.	cou

5.1.2 Container / Liner Domain

Entity	Description	Key
Container	A single intermodal container identified by its number.	con
Shipment	A logical grouping of containers under one Bill of Lading or booking.	ship ves
Container Milestone	A status event for a container.	mil eve
Carrier Tariff / Service Contract	The D&D free-time rules for a carrier, often specific to a service contract.	tari con

Entity	Description	Key
Free-Time Clock	The live or historical tracking of free-time usage for a container.	clo der
D&D Invoice	A carrier-issued invoice for D&D charges.	inv
D&D Invoice Line Item	A single charge on an invoice.	line clai
Dispute Case (Container)	A formal dispute of D&D charges.	disp fina

5.1.3 Cross-Cutting Entities

Entity	Description	Key
Port / Terminal	A reference entity for ports and terminals, used for risk scores and analytics.	por
Risk Score	A historical or predicted risk assessment for a port, terminal, route, or counterparty.	risk risk
Audit Log Entry	Immutable record of every action.	aud nev
AI Interaction Log	Record of user queries and AI responses.	inte mo
Feedback Signal	User-provided corrective feedback on AI outputs or predictions.	fee (thu
Knowledge Base Document	A chunk of text from the RAG corpus, stored as embedding.	doc em

5.2 Data Storage Strategy

The platform uses a polyglot persistence approach, matching storage technology to data access patterns:

	Storage Technology
a (Voyages, Containers, Users, Disputes, Tariffs,	Relational Database (e.g., PostgreSQL) with row-level security per comp
NORs, C/Ps, EIRs, Invoices, generated PDFs)	Object Storage (e.g., AWS S3, Azure Blob) with metadata indexing in the DB.
s (High-frequency, time-ordered)	Time-Series Database (e.g., TimescaleDB, InfluxDB).
nes (High-volume, append-only)	Time-Series Database or Event Store.
n Logs (Immutable, high-volume, append-only)	Separate log-optimized database or append-only ledger with cryptogra
s (Knowledge base chunks for RAG)	Vector Database (e.g., pgvector, Pinecone, Weaviate).
ack Loop Data (Aggregated metrics, model	Data Warehouse / Data Lake (e.g., Snowflake, BigQuery, or HDFS-basec

5.3 Data Security & Tenant Isolation

Requirement ID	Description
DS-01	All data at rest must be encrypted using AES-256 or equ
DS-02	Multi-tenant data isolation: Each company's data must b permissible.
DS-03	Anonymization for benchmarking: Data used for externa container numbers). Differential privacy techniques sho
DS-04	Access controls: Since all users within a company have fu single sign-on (SSO) for enterprise integration.
DS-05	API security: All external-facing APIs must use OAuth 2.0
DS-06	AI data privacy: The Defender AI's LLM inference must b guarantees zero data retention and is covered by a data

5.4 Data Retention & Archiving

Requirement ID	Description
DR-01	Operational data (voyage statutory requirements.
DR-02	Audit logs and AI interacti
DR-03	After the active retention
DR-04	Users can request perma
DR-05	A "legal hold" flag must b

5.5 Data Quality & Integrity

Requirement ID	Description
DQ-01	All extracted SOF/NOR timestamps must carry a confide
DQ-02	The laytime and free-time calculation engines must pro
DQ-03	Carrier invoice audit must reconcile down to the individu logs).
DQ-04	The system must enforce referential integrity in the data
DQ-05	Regular data integrity checks must be performed to ens

6. Non-Functional Requirements

6.1 Performance

Requirement ID	Description
NF-PERF-01	**UI Responsiveness:** All standard page loads and naviga
NF-PERF-02	**Dashboard Data Loading:** Analytical dashboards (Fleet actions must return results within 2 seconds.

Requirement ID	Description
NF-PERF-03	**Document Ingestion:** Automated OCR and event extrac
NF-PERF-04	**Laytime Calculation:** The deterministic laytime engine near-instantaneous (<1 second).
NF-PERF-05	**Predict Simulation:** A voyage laytime simulation of 1,00
NF-PERF-06	**Container D&D Audit:** Reconciliation of a 500-line carri
NF-PERF-07	**AI Response Time:** The Defender AI chat interface must
NF-PERF-08	**Real-Time Alerts:** AIS-based arrival detection must trigg event being ingested.

6.2 Scalability

Requirement ID	Description
NF-SCALE-01	**Horizontal Scaling:** All backend microservices (API, inge downtime.
NF-SCALE-02	**Tenant Volume:** The platform must handle a single larg defined in 6.1.
NF-SCALE-03	**Data Volume Growth:** The data architecture must supp partitioning strategies.
NF-SCALE-04	**Concurrent Users:** The platform must support at least !
NF-SCALE-05	**ML Model Training:** The model retraining pipeline must batch window.

6.3 Availability & Reliability

Requirement ID	Description
NF-AVAIL-01	**Overall Uptime:** The platform must achieve 99.9% upti
NF-AVAIL-02	**Planned Maintenance:** Scheduled maintenance windo

Requirement ID	Description
NF-AVAIL-03	**Disaster Recovery:** Recovery Time Objective (RTO) of 4
NF-AVAIL-04	**Graceful Degradation:** If an external data source (AIS, c staleness. Real-time alerting dependent on the unavaila
NF-AVAIL-05	**Fault Tolerance:** The platform must be deployed across
NF-AVAIL-06	**Backup & Restore:** Full daily backups of all databases a

6.4 Security

Requirement ID	Description
NF-SEC-01	**Authentication:** The platform must support multi-facto
NF-SEC-02	**Authorization:** Although all users within a company hav
NF-SEC-03	**Encryption:** All data in transit must be encrypted using
NF-SEC-04	**Vulnerability Management:** Regular penetration testin
NF-SEC-05	**Secrets Management:** API keys, database credentials, a
NF-SEC-06	**API Security:** All external APIs must enforce rate limiting
NF-SEC-07	**Logging & Monitoring:** All security-relevant events (faile
NF-SEC-08	**Data Privacy:** The platform must comply with GDPR and
NF-SEC-09	**AI Security:** Prompts sent to external LLM APIs must no available.

6.5 Auditability & Compliance

Requirement ID	Description
NF-AUD-01	**Immutable Audit Trail:** Every user action and system e timestamp (UTC), user ID, action type, affected entity, ol
NF-AUD-02	**Tamper Evidence:** The audit log must be protected aga

Requirement ID	Description
NF-AUD-03	**Retention:** Audit logs must be retained for a minimum
NF-AUD-04	**Compliance Standards:** The platform architecture and
NF-AUD-05	**Arbitration Readiness:** The system must be able to exp and correspondence, in a format acceptable for submis

6.6 Usability

Requirement ID	Description
NF-US-01	**Domain Language:** The UI must use standard maritime
NF-US-02	**Onboarding:** First-time users must be able to upload a
NF-US-03	**Error Handling:** All user-facing errors must be displayed
NF-US-04	**Accessibility:** The web application must meet WCAG 2.1 generation).
NF-US-05	**Mobile Responsiveness:** Core alerting, dashboard, and landscape tablet or desktop.

6.7 Maintainability & Extensibility

Requirement ID	Description
NF-MAINT-01	**Modular Architecture:** The platform must be compose Defender, Defender AI).
NF-MAINT-02	**Integration Adapter Framework:** A standardized adap logic.
NF-MAINT-03	**Clause & Tariff Configurability:** The laytime engine and and modified by users or administrators.
NF-MAINT-04	**API Versioning:** All external-facing APIs must be version

Requirement ID	Description
NF-MAINT-05	**Testability:** All calculation engines, model inference enc calculation, free-time engine, D&D audit).

6.8 Operational & Environment

Requirement ID	Description
NF-OPS-01	**Infrastructure as Code:** All cloud infrastructure must b
NF-OPS-02	**Monitoring & Alerting:** System health, service availabili receive alerts for any service degradation.
NF-OPS-03	**Continuous Deployment:** The platform must support z
NF-OPS-04	**Cost Observability:** Infrastructure costs must be tagged

7. AI/ML Requirements

7.1 General AI/ML Principles

Requirement ID	Description
AI-GEN-01	All ML models must be versioned, and their performance m
AI-GEN-02	AI outputs that affect financial calculations or legal argumen
AI-GEN-03	All AI/ML components must operate within the data isolatio
AI-GEN-04	Model inference latency must meet the performance requir
AI-GEN-05	The platform must support both cloud-hosted LLM APIs (wi

7.2 Document Understanding & Extraction (OCR/NLP)

Requirement ID	Description
AI-DOC-01	**OCR Engine:** Must accurately extract printed and legible h and French must be present.

Requirement ID	Description
AI-DOC-02	**Field Extraction Accuracy:** For clean, typed SOFs, field-l
AI-DOC-03	**Confidence Scoring:** Every extracted field must carry a co
AI-DOC-04	**Event Classification:** The NLP model must correctly class extensible.
AI-DOC-05	**Handling Variability:** The model must handle SOFs from d
AI-DOC-06	**Training Data:** The system must be initially trained on a cu over time.

7.3 Predictive Models (PREDICT Phase)

Requirement ID	Description
AI-PRED-01	**Port Risk Scoring Model:** A supervised regression/classif include historical AIS anchorage times, SOF-based laytime
AI-PRED-02	**Laytime Simulation Engine:** A discrete-event simulation r terms and cargo parcels. Must support 1,000 iterations in u
AI-PRED-03	**Free-Time Forecaster:** A classification/regression model t company's own container history.
AI-PRED-04	**Counterparty Behavior Model:** A classifier that predicts a
AI-PRED-05	**Model Accuracy Monitoring:** All predictive models must h classification, and calibration error for probability scores. Ar
AI-PRED-06	**Feature Freshness:** The retraining pipeline must track fea review.

7.4 Recommendation & Alerting Engine (PREVENT Phase)

Requirement ID	Description
AI-REC-01	**Contract-Aware Reasoning:** The recommendation engine actions that would violate the contract.
AI-REC-02	**Cost-Benefit Analysis:** Every operational recommendatio
AI-REC-03	**Contextual Prioritization:** When multiple risks are present

Requirement ID	Description
AI-REC-04	**Feedback Integration:** User acceptance or rejection of rec precision.

7.5 Defender AI Chatbot & RAG (Cross-Phase)

Requirement ID	Description
AI-BOT-01	**Large Language Model (LLM):** The system must integrate window of at least 32,000 tokens to accommodate lengthy
AI-BOT-02	**Retrieval-Augmented Generation (RAG):** All knowledge- summaries, INCOTERMS 2020, and IMO regulations. Chu
AI-BOT-03	**Citation Enforcement:** Every factual claim, legal interpreta found, the AI must respond with "I cannot find a source for
AI-BOT-04	**Multi-Turn Conversation:** The AI must maintain context a
AI-BOT-05	**Hallucination Prevention:** The system must employ retrie confidence is low, a disclaimer must be displayed.
AI-BOT-06	**Legal Disclaimer:** All AI outputs that could be construed a *with your legal counsel before taking action."* The disclaime
AI-BOT-07	**Drafting Tone & Style:** The AI must support configurable t
AI-BOT-08	**Guardrails:** The AI must refuse to answer queries unrelate security. A list of blocked topics must be maintained.

7.6 Model Lifecycle & Governance

Requirement ID	Description
AI-ML-01	**Model Registry:** All models must be registered with metad
AI-ML-02	**Human-in-the-Loop for Legal Updates:** Updates to the A and applicability.
AI-ML-03	**Bias Testing:** Predictive models must be tested for bias ac
AI-ML-04	**A/B Testing Framework:** For AI features (chatbot respons resolution).

Requirement ID	Description
AI-ML-05	**Incident Response:** Any instance of harmful or severely e version available within 15 minutes.

8. Supporting information

The supporting information makes the SRS easier to use.

8.1. Appendices

8.1.1 Glossary

A consolidated list of key terms used throughout this SRS. (See also Section 1.3 for the complete definitions and acronyms table.)

Term	Definition
AIS	Automatic Identification System – vessel tracking system.
BCO	Beneficial Cargo Owner – the entity that owns the goods.
BIMCO	Baltic and International Maritime Council – maritime stand
B/L	Bill of Lading – document of title.
C/P	Charter Party – vessel hire contract.
CTRM	Commodity Trading and Risk Management – enterprise so
D&D	Demurrage & Detention – combined container time penalt
Demurrage (Bulk)	Penalty for exceeding vessel laytime.
Demurrage (Container)	Penalty for full container sitting too long in terminal.
Despatch	Bonus for completing cargo operations faster than allowe
Detention	Penalty for holding container (empty or full export) beyon

Term	Definition
EDI	Electronic Data Interchange – standardised messaging (e.g
EIR	Equipment Interchange Receipt – container handover rec
Free Time	Allowed calendar days before container D&D charges star
GENCON	Standard uniform general charter party form.
Laycan	Laydays and Cancelling date – vessel arrival window.
Laytime	Allowed time for loading/discharging without penalty.
LLM	Large Language Model – AI foundation model.
LMAA	London Maritime Arbitrators Association.
MOLOO	More or Less in Owner's Option – cargo quantity tolerance
NLP	Natural Language Processing.
NOR	Notice of Readiness – triggers laytime commencement.
NORGRAIN	Standard grain voyage charter party form.
NVOCC	Non-Vessel Operating Common Carrier.
OCR	Optical Character Recognition.
P&I Club	Protection and Indemnity – shipowner liability insurance.
RAG	Retrieval-Augmented Generation – AI technique for groun
SHEX	Sundays and Holidays Excepted – laytime counting rule.
SHINC	Sundays and Holidays Included – laytime counting rule.
SMA	Society of Maritime Arbitrators (New York).
SOF	Statement of Facts – port call event log.
TEU	Twenty-foot Equivalent Unit – container measure.
VLCC	Very Large Crude Carrier.

Term	Definition
WIBON	Whether In Berth Or Not – clause allowing NOR even if no
WWD	Weather Working Day – laytime counts only if weather per

8.1.2 References

The following external documents and standards are referenced within this SRS.

Ref ID	Document / Standard
REF-01	Business Requirements Document – Demurrage Defender
REF-02	BIMCO Standard Charter Party Forms (GENCON, NORGRAIN, BPVOY, SHI
REF-03	INCOTERMS 2020
REF-04	IMO Conventions (SOLAS, MARPOL, ISPS Code)
REF-05	LMAA Arbitration Rules
REF-06	SMA Arbitration Rules
REF-07	GDPR (Regulation (EU) 2016/679)
REF-08	CCPA (California Consumer Privacy Act)
REF-09	WCAG 2.1 Level AA
REF-10	OWASP Top 10 Web Application Security Risks
REF-11	SOC 2 Type II Trust Services Criteria
REF-12	ISO 27001 Information Security Management

8.1.3 Requirements Traceability Matrix

This matrix maps each functional feature in the SRS to the Core Value Loop phase(s) and the corresponding user story or capability described in the BRD. Non-functional requirements are traced to the BRD's platform-wide objectives.

8.1.3.1 Predict Phase

SRS Feature ID	Feature Name	Loop Phase
FR-PRED-01	Port Risk Dashboard	Predict
FR-PRED-02	Terminal Drill-Down	Predict
FR-PRED-03	Route Risk Assessment	Predict
FR-PRED-04	Laytime Simulator (Bulk)	Predict
FR-PRED-05	Scenario Comparison (Bulk)	Predict
FR-PRED-06	Free Time Forecaster (Container)	Predict
FR-PRED-07	Gateway/Route Comparison (Container)	Predict
FR-PRED-08	Owner/Carrier Profile Page	Predict
FR-PRED-09	Counterparty Risk Alert	Predict
FR-PRED-10	Congestion & Disruption Alerts	Predict
FR-PRED-11	Market Intelligence Dashboard	Predict

8.1.3.2 Prevent Phase

SRS Feature ID	Feature Name	Loop Phase
FR-PREV-01	Unified Deadline Dashboard	Prevent
FR-PREV-02	Escalating Alert Rules	Prevent
FR-PREV-03	Bulk Arrival Detection & NOR Alert	Prevent
FR-PREV-04	Container Free-Time Countdown Alert	Prevent
FR-PREV-05	Bulk Demurrage Prevention Advisor	Prevent
FR-PREV-06	Container Detention Prevention Advisor	Prevent
FR-PREV-07	What-If Intervention Simulator	Prevent
FR-PREV-08	Automated External Notifications	Prevent

SRS Feature ID	Feature Name	Loop Phase
FR-PREV-09	Task Creation & Assignment	Prevent
FR-PREV-10	Integration with External Workflow Tools	Prevent

8.1.3.3 Manage Phase

SRS Feature ID	Feature Name	Loop Phase
FR-MAN-01	SOF/NOR Upload & OCR	Manage
FR-MAN-02	Bulk Email/Agent Data Ingestion	Manage
FR-MAN-03	Container Milestone Ingestion	Manage
FR-MAN-04	Manual Event Entry & Override	Manage
FR-MAN-05	Charter Party Clause Library & Tagging	Manage
FR-MAN-06	Laytime Engine (Bulk)	Manage
FR-MAN-07	Free-Time Engine (Container)	Manage
FR-MAN-08	Timeline Visualization & What-If Slider	Manage
FR-MAN-09	Bulk Voyage Dashboard	Manage
FR-MAN-10	Container Operations Dashboard	Manage
FR-MAN-11	Combined Financial Exposure View	Manage
FR-MAN-12	Document Storage & Auto-Linking	Manage
FR-MAN-13	Immutable Audit Log	Manage

8.1.3.4 Dispute Phase

SRS Feature ID	Feature Name	Loop Phase
FR-DIS-01	Laytime Statement & Demurrage Counter-Claim (Bulk)	Dispute
FR-DIS-02	Despatch Invoice (Bulk)	Dispute

SRS Feature ID	Feature Name	Loop Phase
FR-DIS-03	D&D Invoice Auditor (Container)	Dispute
FR-DIS-04	Dispute Package Generator (Container)	Dispute
FR-DIS-05	Dispute Case Creation & Tracking	Dispute
FR-DIS-06	Correspondence & Evidence Management	Dispute
FR-DIS-07	Deadline & Escalation Alerts	Dispute
FR-DIS-08	Dispute Dashboard & Settlement Tracking	Dispute
FR-DIS-09	Clause Interpretation & Legal Q&A	Dispute
FR-DIS-10	Dispute Letter Drafting	Dispute
FR-DIS-11	Research Memo Generation	Dispute

8.3.5 Optimize Phase

SRS Feature ID	Feature Name	Loop Phase
FR-OPT-01	Demurrage & Despatch Analytics (Bulk)	Optimize
FR-OPT-02	Detention & Demurrage Analytics (Container)	Optimize
FR-OPT-03	Root Cause Analysis & Trend Detection	Optimize
FR-OPT-04	C/P Clause Impact Analyzer (Bulk)	Optimize
FR-OPT-05	Service Contract D&D Optimizer (Container)	Optimize
FR-OPT-06	Counterparty Negotiation Brief	Optimize
FR-OPT-07	Port/Route Cost Comparison (Bulk)	Optimize
FR-OPT-08	Container Gateway Optimization	Optimize
FR-OPT-09	Internal Benchmarking	Optimize
FR-OPT-10	External Industry Benchmarking	Optimize

8.3.6 Feedback & Learning Loop

SRS Feature ID	Feature Name	Loop Phase
FR-FL-01	Outcome Data Collection Pipeline	Feedback Loop
FR-FL-02	Predictive Model Retraining Scheduler	Feedback Loop
FR-FL-03	AI Knowledge Base Update	Feedback Loop
FR-FL-04	Feedback Signal Integration (In-Loop)	Feedback Loop
FR-FL-05	Feedback Loop Monitoring Dashboard	Feedback Loop

8.3.7 Defender AI (Standalone)

SRS Feature ID	Feature Name	Loop Phase
FR-AI-01	Unified Conversational Interface	All
FR-AI-02	Knowledge Base & RAG Infrastructure	All
FR-AI-03	Context-Aware Assistance	All
FR-AI-04	Cross-Phase Task Orchestration	All
FR-AI-05	Explainability & Audit Logging	All
FR-AI-06	Personalization & Feedback Learning	All

Database

Database Design Document for Demurrage Defender

Prepared by: Person
Date: Date

1. Introduction

1.1 Purpose

The Database Design Document (DDD) defines the complete physical data model for the Demurrage Defender platform. It translates the logical data requirements from the Software Requirements Specification (SRS) into concrete, implementable database structures tailored to PostgreSQL with TypeORM in a NestJS multi-tenant environment.

This document serves as the authoritative reference for the backend development team. By reading it, a developer should be able to:

- Understand the schema for every table, column, constraint, and index.
- Immediately translate specifications into TypeORM entity classes and migrations.
- Implement Row-Level Security (RLS) for tenant isolation.
- Configure TimescaleDB hypertables for time-series data.
- Set up pgvector for AI-powered retrieval-augmented generation (RAG).
- Execute common queries with confidence that performance is addressed through proper indexing.

1.2 Scope

The database design covers all data entities required to support the full application lifecycle, including:

- **Bulk/Tramp Shipping Domain**: Voyages, vessels, Charter Parties, SOF documents, NORs, laytime calculations, demurrage/despatch claims.
- **Container/Liner Domain**: Containers, shipments, milestones, carrier tariffs, free-time clocks, D&D invoices and disputes.

- **Cross-Cutting Domains**: Tenants, users, audit logs, AI interactions, feedback signals, predictive model artifacts, and knowledge base chunks for RAG.
- **Time-Series Data**: Vessel AIS positions and container milestone events, stored as TimescaleDB hypertables for efficient range queries and retention management.

Out of scope are OLAP cubes for the analytics dashboards (which will be built on top of this transactional schema) and specific caching strategies (handled by Redis).

1.3 Guiding Principles

The database design adheres to these principles:

- **Developer Productivity First**: The schema is designed to work seamlessly with TypeORM's Active Record and Repository patterns. Naming conventions, column types, and relationships are chosen to minimize impedance mismatch.
- **Multi-Tenancy by Design**: Every row in tenant-scoped tables carries a tenant_id. Row-Level Security policies enforce this at the database level, and a TypeORM subscriber automates tenant context injection in the application layer.
- **Performance at Scale**: Indexing strategies are defined upfront for the most critical query patterns (dashboard filters, time-range lookups, joins). Time-series data uses TimescaleDB hypertables to maintain query speed as volume grows.
- **Unbounded Data Isolation**: Large, append-only datasets (AIS positions, container milestones) are partitioned by time via hypertables to keep active working sets small.
- **Auditability & Immutability**: Core operational records (SOF events, milestone events, audit logs) are append-only. Audit log entries are cryptographically chained to provide tamper evidence for arbitration.
- **Simplicity over Premature Optimization**: We default to normal forms and add denormalization only when required by query patterns. The use of PostgreSQL extensions (TimescaleDB, pgvector) is deliberate and justified.

1.4 Conventions Used in This Document

Element	Convention
Table Names	Lowercase, snake_case, pluralized

Element	Convention	E
Column Names	Lowercase, snake_case	c
Primary Keys	id (UUID, generated client-side or by DB)	i
Foreign Keys	Singular table name + _id	v
Data Types	Use precise PostgreSQL types; avoid generic varchar without length	V
Indexes	Named idx_{table}_{column(s)}	i
Constraints	Named explicitly for migration management	c
Default Values	Defined in the database where sensible	s
TypeORM Annotations	Decorators shown inline in entity code blocks	@

2. Technology Stack & Rationale

2.1 Core Database: PostgreSQL

PostgreSQL was selected as the primary relational database for its unmatched combination of reliability, extensibility, and advanced feature set that directly addresses the platform's requirements:

- **Multi-Tenancy with Row-Level Security (RLS)**: PostgreSQL's built-in RLS policies allow us to enforce tenant data isolation at the database level, complementing application-level guards. By setting app.current_tenant_id via a runtime parameter, all queries automatically filter by tenant_id, providing defense-in-depth.
- **Rich Data Types**: Native support for UUID, JSONB, TIMESTAMPTZ, and TSVECTOR enables precise modeling of our domain (document metadata, flexible clause storage, timezone-aware events, full-text search on SOF remarks).
- **Performance & Maturity**: A proven optimizer, robust indexing (B-tree, GIN, BRIN), and mature tooling make it suitable for both OLTP workloads (voyage/container transactions) and analytical queries (optimize dashboards) without requiring a separate database.

- **Extensibility**: The ability to add extensions like TimescaleDB and pgvector directly into the same instance eliminates the operational complexity of managing multiple database systems, a critical advantage for a small team.

2.2 Time-Series Extension: TimescaleDB

Two core data streams in Demurrage Defender are inherently time-series: vessel AIS position reports and container milestone events. These are high-volume, append-only, and predominantly queried by time range (e.g., "show vessel positions for the last 7 days").

- **Automatic Partitioning via Hypertables**: Tables like vessel_positions and container_milestones are converted into hypertables, chunked by time (e.g., 1-day intervals). This ensures that queries for recent data touch only the relevant chunks, maintaining consistent performance even as terabytes of historical data accumulate.
- **Built-in Compression**: Older chunks can be compressed (often 10-20x reduction) without losing the ability to query them, drastically reducing storage costs for data that must be retained for years.
- **Continuous Aggregates**: For dashboards showing hourly or daily summary metrics (e.g., average anchorage waiting time), continuous aggregates can materialize the results and keep them automatically updated, making real-time analytics instantaneous.
- **Native PostgreSQL Integration**: Because TimescaleDB is an extension, standard SQL, TypeORM, and all PostgreSQL tooling (backups, replication, RLS) work out-of-the-box. No separate skills or operations are required.

2.3 Vector Search Extension: pgvector

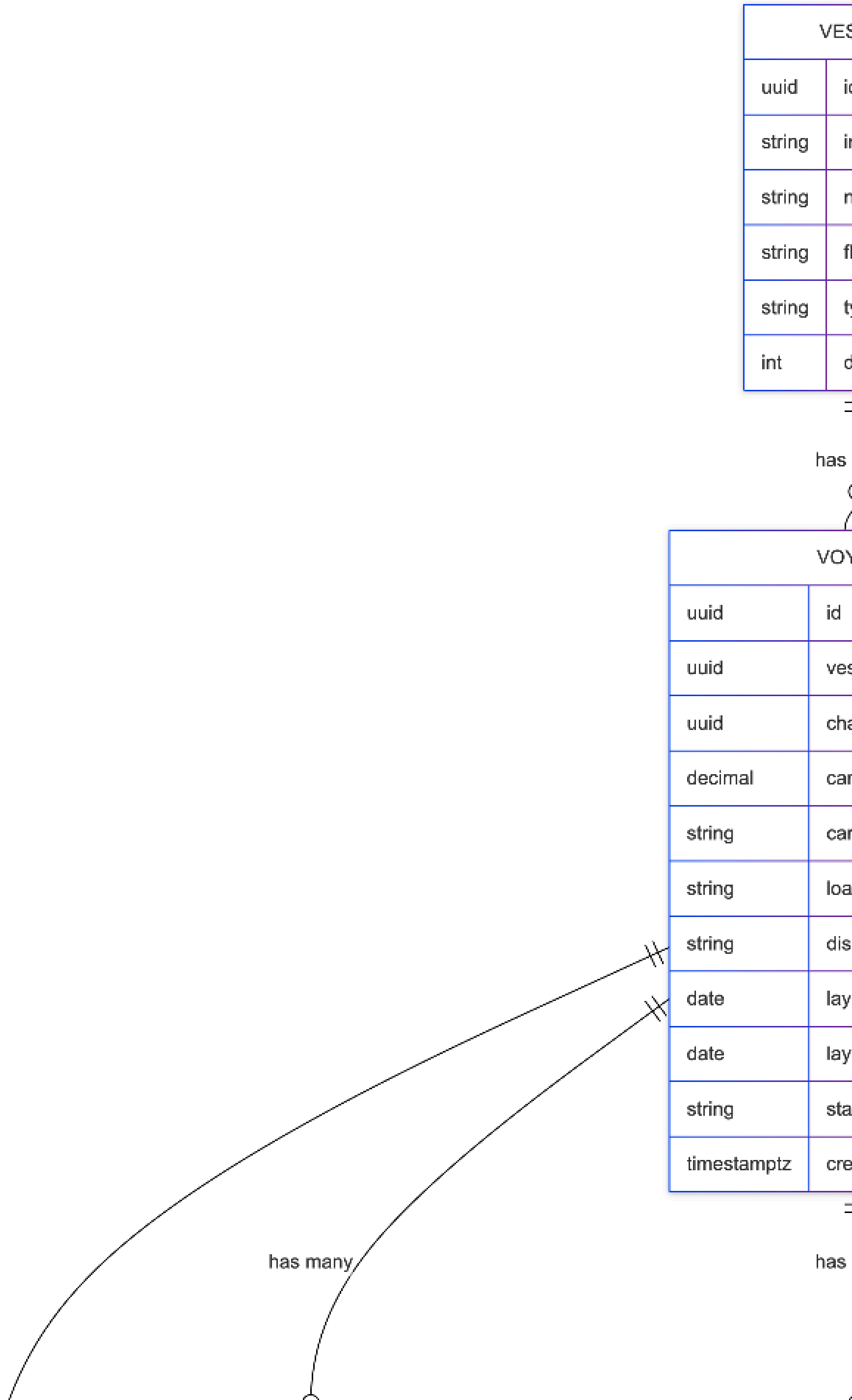
The Defender AI's Retrieval-Augmented Generation (RAG) pipeline requires storing and querying high-dimensional vector embeddings of documents (Charter Party clauses, case law, carrier tariffs, internal SOPs). These embeddings must be searched for semantic similarity.

- **Single Database Instance**: Embeddings are stored in a standard table with a VECTOR column, alongside the relational metadata (source document, tenant). This eliminates the need for a separate vector database like Pinecone or Weaviate, simplifying architecture and reducing operational overhead.

- **Adequate Performance for Initial Scale**: For a corpus of hundreds of thousands of chunks per tenant, pgvector's IVFFlat or HNSW indexes provide sub-second similarity search, which is sufficient for the AI chatbot's response time requirements.
- **Seamless TypeORM Support**: A community package (pgvector) extends TypeORM with a custom column type, allowing vector operations to be used directly within repository methods.
- **Migration Path**: If, at massive scale, a dedicated vector store becomes necessary, the abstraction provided by the knowledge_base_chunks table and its service layer allows for a smooth swap without refactoring the core application logic.

2. ERD

2.2 Bulk / Tramp Shipping Domain

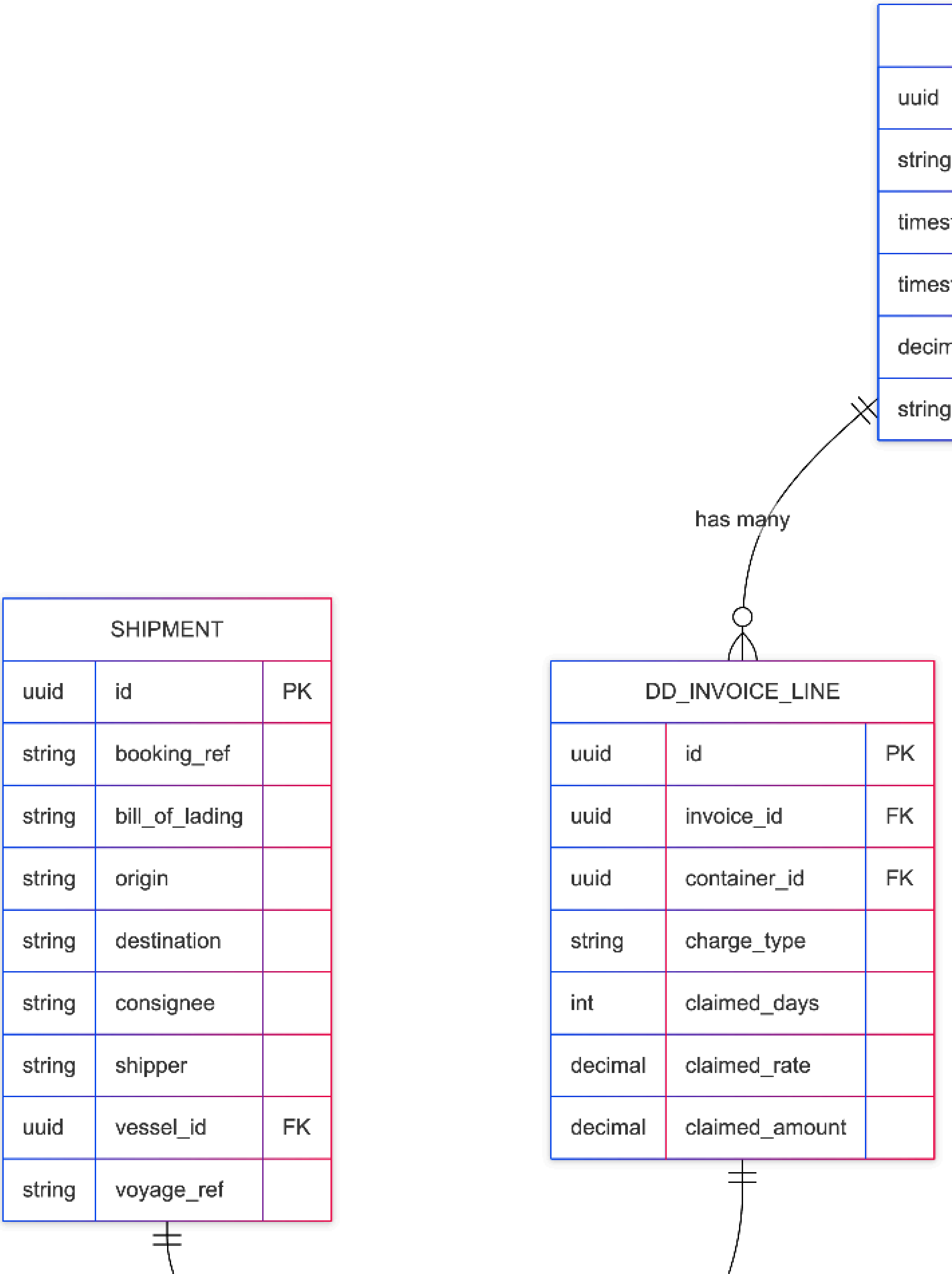


uuid
string
string
string
string
int
has
uuid
id
uuid
uuid
decimal
string
string
string
date
date
string
timestamptz
has many
has

Key Relationships

- A VOYAGE is the central entity. It links to one VESSEL, one CHARTER_PARTY, and can have multiple SOF_DOCUMENTs, NOR_DOCUMENTs, and LAYTIME_CALCULATIONs (versioned).
- SOF_EVENTs are extracted from a SOF_DOCUMENT and form the raw input to the laytime engine.
- CALCULATION_PERIODs are segments of time within a LAYTIME_CALCULATION, each potentially referencing a CP_CLAUSE that justified the period type (e.g., rain stoppage due to weather clause).

2.2 Container / Liner Domain

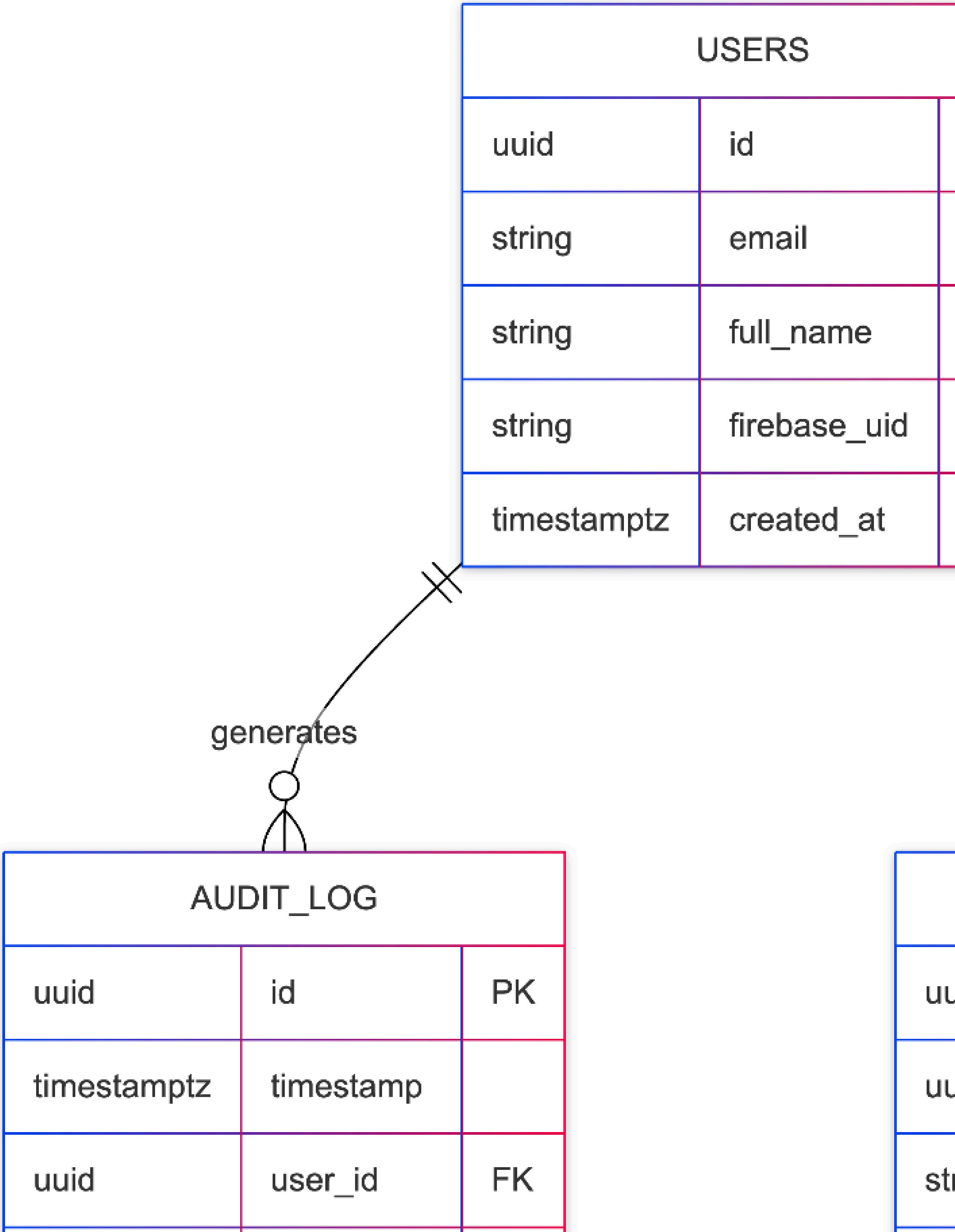


has many
SHIPMENT
uuid
id
PK
string
booking_ref
string
bill_of_lading
string
origin
string
destination
string
consignee
string
shipper
uuid
vessel_id
FK
string
voyage_ref
DD_INVOICE_LINE
uuid
id
PK
uuid
invoice_id
FK
uuid
container_id
FK
string
charge_type
int
claimed_days
decimal
claimed_rate
decimal
claimed_amount
uuid
string
string

Key Relationships

- SHIPMENT groups multiple CONTAINERs under one booking.
- Each CONTAINER generates many CONTAINER_MILESTONEs and is tracked by a FREE_TIME_CLOCK that references a specific CARRIER_TARIFF.
- DD_INVOICEs contain DD_INVOICE_LINEs, which can be disputed via DISPUTE_CASE_CONTAINER.

2.2 Cross cutting & AI Domain


USERS
uuid
id
string
email
string
full_name
string
firebase_uid
timestamptz
created_at
generates
AUDIT_LOG
uuid
id
PK
timestamptz
timestamp
uuid
user_id
FK

Key Relationships

- USERS is a simple table linked to Firebase authentication. Each user can have many audit log entries and AI interactions.
- AUDIT_LOG records every significant action in the system, with a reference to the user and entity changed.
- AI_INTERACTION stores the full conversation for audit and improvement purposes.
- KNOWLEDGE_BASE_CHUNK is a standalone table storing embeddings for RAG; it is not directly linked to other entities.

5. Table Definitions

5.1 Bulk / Tramp Shipping Domain

5.1.1 `vessels`

Purpose: Stores the core details of a physical vessel (ship). Reused across multiple voyages.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	Unique i
	`VARCHAR(7)`	`UNIQUE, NOT NULL`		IMO shi
	`VARCHAR(100)`	`NOT NULL`		Vessel r Harvest'
	`VARCHAR(50)`	`NOT NULL`		Flag sta
	`VARCHAR(50)`	`NOT NULL`		Vessel t "VLCC")
	`INTEGER`	`NOT NULL`		Deadwe
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	Auto-up

Indexes:

- `idx_vessels_imo` on `(imo)` – lookup by IMO.
- `idx_vessels_type` on `(type)` – fleet analytics.

5.1.2 voyages

Purpose: Master record for a single chartered voyage from load port to discharge port.

Data Type	Constraints	Default	Descrip
UUID	PRIMARY KEY	uuid_generate_v4()	
UUID	NOT NULL, FK -> vessels.id		The ves voyage.
UUID	FK -> charter_parties.id	NULL	Linked C
DECIMAL(10,2)	NOT NULL		Quantity
VARCHAR(100)	NOT NULL		e.g., "Sc
VARCHAR(10)	NOT NULL		UN/LOC
VARCHAR(10)	NOT NULL		UN/LOC
DATE	NOT NULL		Earliest
DATE	NOT NULL		Cancelli
VARCHAR(20)	NOT NULL, CHECK(status IN ('Planned','Active','Completed','Cancelled'))	'Planned'	
TIMESTAMPTZ	NOT NULL	NOW()	
TIMESTAMPTZ	NOT NULL	NOW()	Auto-up

Indexes:

- `idx_voyages_vessel` on `(vessel_id)` – fetch all voyages for a vessel.
- `idx_voyages_status` on `(status)` – dashboard filtering.
- `idx_voyages_ports` on `(load_port, discharge_port)` – route analysis.

5.1.3 charter_parties

Purpose: Stores the original text and metadata of a Charter Party agreement.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> voyages.id		Parent v
	VARCHAR(50)	NOT NULL		e.g., "GE "NORGR
	TEXT	NOT NULL		Complet
	DATE	NOT NULL		
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_charter_parties_voyage on (voyage_id) – get C/P for a voyage.

5.1.4 cp_clauses

Purpose: Individual tagged clauses extracted from the Charter Party, each with structured parameters that drive the laytime engine.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> charter_parties.id		
	VARCHAR(50)	NOT NULL		e.g., "lay "wibon", "despato
	TEXT	NOT NULL		Original
	JSONB	NOT NULL		Structur {"rate "MT",

Indexes:

- `idx_cp_clauses_cp` on `(charter_party_id)`.
- `idx_cp_clauses_type` on `(clause_type)`.

5.1.5 `sof_documents`

Purpose: Metadata for an uploaded Statement of Facts document.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`UUID`	`NOT NULL, FK -> voyages.id`		
	`VARCHAR(500)`	`NOT NULL`		S3 objec
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	
	`VARCHAR(20)`	`NOT NULL, DEFAULT 'Draft'`	`'Draft'`	e.g., "Dr

Indexes:

- `idx_sof_documents_voyage` on `(voyage_id)`.

5.1.6 `sof_events`

Purpose: Each timestamped event extracted from the SOF. The input to the laytime engine.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`UUID`	`NOT NULL, FK -> sof_documents.id`		
	`TIMESTAMPTZ`	`NOT NULL`		UTC eve
	`VARCHAR(50)`	`NOT NULL`		e.g., "NO "RAIN_S
	`TEXT`	`NULL`		Original
	`DECIMAL(3,2)`	`NULL`		OCR co

	Data Type	Constraints	Default	Descrip
	BOOLEAN	NOT NULL	false	
	TEXT	NULL		Justifica
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_sof_events_sof_time on (sof_id, event_time) – sequential scan per SOF.
- idx_sof_events_type on (event_type) – analytics.

5.1.7 nor_documents

Purpose: Notice of Readiness document and its key timestamps.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> voyages.id		
	VARCHAR(500)	NOT NULL		
	TIMESTAMPTZ	NOT NULL		Time NC
	TIMESTAMPTZ	NULL		Time NC

Indexes:

- idx_nor_voyage on (voyage_id).

5.1.8 laytime_calculations

Purpose: The result of a laytime computation for a voyage. Supports versioning for recalculations.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	

	Data Type	Constraints	Default	Descrip
	`UUID`	`NOT NULL, FK -> voyages.id`		
	`INTEGER`	`NOT NULL`	`1`	Increme
	`INTERVAL`	`NOT NULL`		Allowed
	`INTERVAL`	`NOT NULL`		Time ac
	`DECIMAL(12,2)`	`NOT NULL`	`0`	
	`DECIMAL(12,2)`	`NOT NULL`	`0`	
	`VARCHAR(20)`	`NOT NULL, DEFAULT 'Draft'`	`'Draft'`	"Draft", "
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	

Indexes:

- `idx_laytime_calc_voyage` on `(voyage_id)`.

5.1.9 `calculation_periods`

Purpose: A time segment within a laytime calculation, capturing whether the clock was running, stopped, or on demurrage, and which clause justified it.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`UUID`	`NOT NULL, FK -> laytime_calculations.id`		
	`TIMESTAMPTZ`	`NOT NULL`		
	`TIMESTAMPTZ`	`NOT NULL`		
	`VARCHAR(20)`	`NOT NULL`		"laytime "demurr

	Data Type	Constraints	Default	Descrip
	UUID	FK -> cp_clauses.id	NULL	The clau period (e

Indexes:

- `idx_calc_periods_calc` on `(calculation_id)`.

5.1.10 `dispute_cases_bulk`

Purpose: A formal dispute or despatch claim related to a voyage.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> voyages.id		
	VARCHAR(20)	NOT NULL		"demurr "despatc
	DECIMAL(12,2)	NOT NULL		Total am
	VARCHAR(30)	NOT NULL, DEFAULT 'Open'	'Open'	"Open", "In Nego
	TIMESTAMPTZ	NOT NULL	NOW()	
	TIMESTAMPTZ	NULL		
ount	DECIMAL(12,2)	NULL		

Indexes:

- `idx_disputes_voyage` on `(voyage_id)`.
- `idx_disputes_status` on `(status)`.

5.1.11 `counterparties`

Purpose: A shipowner or charterer entity that the company interacts with.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	VARCHAR(200)	NOT NULL		Compar
	VARCHAR(20)	NOT NULL		"owner",
	VARCHAR(50)	NULL		IMO cor

Indexes:

- idx_counterparties_name on (name).

5.2 Container / Liner Domain

5.2.1 containers

Purpose: Reference table for individual containers. Reused across multiple shipments.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	Internal
	VARCHAR(11)	UNIQUE, NOT NULL		ISO 634 MSKU1
	VARCHAR(20)	NOT NULL		e.g., "20
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_containers_number on (container_number) – lookup.

5.2.2 shipments

Purpose: Logical grouping of containers under one Bill of Lading or booking.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	

Data Type	Constraints	Default	Descrip
`VARCHAR(100)`	`NOT NULL`		Carrier b
`VARCHAR(100)`	`NULL`		Master c
`VARCHAR(10)`	`NOT NULL`		Origin p
`VARCHAR(10)`	`NOT NULL`		Destinat
`VARCHAR(200)`	`NULL`		Consign
`VARCHAR(200)`	`NULL`		Shipper
`UUID`	`NULL, FK -> vessels.id`		Vessel c known).
`VARCHAR(50)`	`NULL`		Voyage 0WW5A
`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	

Indexes:

- `idx_shipments_booking` on `(booking_ref)`.
- `idx_shipments_consignee` on `(consignee)`.

5.2.3 `shipment_containers`

Purpose: Join table linking containers to shipments (a container can move across multiple shipments over time, but is typically linked to one active shipment).

Data Type	Constraints	Default	Descrip
`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
`UUID`	`NOT NULL, FK -> shipments.id`		
`UUID`	`NOT NULL, FK -> containers.id`		
`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	

Indexes:

- `idx_ship_cont_shipment` on `(shipment_id)`.
- `idx_ship_cont_container` on `(container_id)`.

5.2.4 `container_milestones`

Purpose: Time-series events for a container (gate-in, gate-out, discharge, etc.). Converted to a **TimescaleDB hypertable** for efficient time-range queries.

	Data Type	Constraints	Default	Descrip
	`UUID`	`NOT NULL`	`uuid_generate_v4()`	Unique
	`TIMESTAMPTZ`	`NOT NULL`		Event oc
	`UUID`	`NOT NULL, FK -> containers.id`		
	`VARCHAR(30)`	`NOT NULL`		e.g., "DI "GATE_ "GATE_
	`VARCHAR(100)`	`NULL`		Termina
	`VARCHAR(50)`	`NOT NULL`		"carrier_ "manual
	`BOOLEAN`	`NOT NULL`	`false`	

Indexes:

- Composite primary key: `(time, id)` (required for hypertable).
- `idx_milestones_container` on `(container_id, time)` – container timeline.

5.2.5 `carrier_tariffs`

Purpose: Stores the D&D free-time rules for a carrier, often specific to a service contract and port.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	

	Data Type	Constraints	Default	Descrip
	VARCHAR(100)	NOT NULL		e.g., "Ma
	VARCHAR(100)	NULL		Service
	DATE	NOT NULL		
	VARCHAR(10)	NOT NULL		Port UN
	INTEGER	NOT NULL		Combin
	JSONB	NOT NULL		Array of day_ra
	JSONB	NULL		Same st demurra

Indexes:

- idx_tariffs_carrier_port on (carrier_name, port).

5.2.6 free_time_clocks

Purpose: Tracks the real-time free-time status and accrued charges for a container.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> containers.id		
	UUID	NOT NULL, FK -> carrier_tariffs.id		
	TIMESTAMPTZ	NOT NULL		When fr (dischar
	TIMESTAMPTZ	NOT NULL		Calculat
	INTERVAL	NOT NULL	'0'	Time us allowanc

	Data Type	Constraints	Default	Descrip
	DECIMAL(10,2)	NOT NULL	0	
	DECIMAL(10,2)	NOT NULL	0	
	VARCHAR(20)	NOT NULL, DEFAULT 'Active'	'Active'	"Active",

Indexes:

- idx_clocks_container on (container_id).
- idx_clocks_status_expiry on (status, expiry_time) – for alert queries.

5.2.7 dd_invoices

Purpose: Stores carrier-issued D&D invoice headers.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	VARCHAR(100)	NOT NULL		e.g., "Ma
	TIMESTAMPTZ	NOT NULL		
	TIMESTAMPTZ	NOT NULL		
	DECIMAL(12,2)	NOT NULL		
	VARCHAR(500)	NULL		S3 key f
	TIMESTAMPTZ	NOT NULL	NOW()	

5.2.8 dd_invoice_lines

Purpose: Individual charge lines on a D&D invoice.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	

	Data Type	Constraints	Default	Descrip
	UUID	NOT NULL, FK -> dd_invoices.id		
	UUID	NOT NULL, FK -> containers.id		
	VARCHAR(20)	NOT NULL		"detentio
	INTEGER	NOT NULL		
	DECIMAL(10,2)	NOT NULL		
	DECIMAL(10,2)	NOT NULL		

Indexes:

- idx_invoice_lines_invoice on (invoice_id).
- idx_invoice_lines_container on (container_id).

5.2.9 dispute_cases_container

Purpose: Formal dispute of D&D charges, linked to an invoice.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	UUID	NOT NULL, FK -> dd_invoices.id		
	DECIMAL(12,2)	NOT NULL		
	VARCHAR(30)	NOT NULL, DEFAULT 'Open'	'Open'	"Open", "In Nego
	TIMESTAMPTZ	NOT NULL	NOW()	
	TIMESTAMPTZ	NULL		
ount	DECIMAL(12,2)	NULL		

Indexes:

- `idx_container_disputes_invoice` on `(invoice_id)`.
- `idx_container_disputes_status` on `(status)`.

5.3 Cross-Cutting & AI Domain

5.3.1 `users`

Purpose: Stores user accounts for the platform. Authentication is delegated to Firebase Auth; this table mirrors essential identity information and links to the Firebase UID.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	Internal
	`VARCHAR(128)`	`UNIQUE, NOT NULL`		Firebase
	`VARCHAR(255)`	`UNIQUE, NOT NULL`		User's e
	`VARCHAR(200)`	`NOT NULL`		Display
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	
	`TIMESTAMPTZ`	`NULL`		Updated

Indexes:

- `idx_users_firebase` on `(firebase_uid)` – lookup after authentication.
- `idx_users_email` on `(email)`.

5.3.2 `audit_log`

Purpose: Immutable record of every significant system and user action. Supports arbitration and compliance. This table is append-only (no UPDATEs or DELETEs). Optionally converted to a TimescaleDB hypertable for efficient retention management.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`TIMESTAMPTZ`	`NOT NULL`	`NOW()`	When th

Data Type	Constraints	Default	Descrip
`UUID`	`NULL, FK -> users.id`		Null for s
`VARCHAR(100)`	`NOT NULL`		e.g., "SC "EVENT "LAY_C
`VARCHAR(50)`	`NOT NULL`		e.g., "Vo
`UUID`	`NOT NULL`		ID of the
`JSONB`	`NULL`		Previous
`JSONB`	`NULL`		New sta
`INET`	`NULL`		Client IF
`JSONB`	`NULL`		Addition agent).

Indexes:

- `idx_audit_timestamp` on `(timestamp DESC)` – time-range queries.
- `idx_audit_entity` on `(entity_type, entity_id)` – find all changes to a record.

5.3.3 `ai_interactions`

Purpose: Stores full prompts, responses, and context for every Defender AI chat interaction. Used for audit, feedback, and model improvement.

Data Type	Constraints	Default	Descrip
`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
`UUID`	`NOT NULL, FK -> users.id`		
`VARCHAR(100)`	`NOT NULL`		Groups
`TEXT`	`NOT NULL`		Full pror (includir
`TEXT`	`NOT NULL`		LLM res

	Data Type	Constraints	Default	Descrip
	JSONB	NULL		Array of RAG.
	VARCHAR(50)	NOT NULL		e.g., "gp "claude-
	INTEGER	NULL		User-pro
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_ai_interactions_user on (user_id, created_at DESC).
- idx_ai_interactions_session on (session_id).

5.3.4 feedback_signals

Purpose: Explicit corrective feedback from users on AI predictions, recommendations, or answers. Feeds into the Feedback & Learning Loop.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	VARCHAR(30)	NOT NULL		e.g., "pre "recomm
	UUID	NOT NULL		ID of the score, A
	UUID	NOT NULL, FK -> users.id		Who pro
	VARCHAR(10)	NOT NULL		"positive
	TEXT	NULL		User's e value.
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_feedback_source on (source_type, source_id).

5.3.5 knowledge_base_chunks

Purpose: Stores text chunks and their vector embeddings for Retrieval-Augmented Generation (RAG). Supports the Defender AI's ability to query maritime documents. Uses the **pgvector** extension.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	VARCHAR(30)	NOT NULL		e.g., "cp "carrier_
	VARCHAR(500)	NOT NULL		Referen docume "CP-123
	TEXT	NOT NULL		The raw
	VECTOR(1536)	NOT NULL		Vector e depends
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_kb_source on (source_type, source_ref).
- **Vector Index:** Created via raw SQL for similarity search (e.g., IVFFlat).

dsgsgdsdg

5.3 Cross-Cutting & AI Domain

5.3.1 users

Purpose: Stores user accounts for the platform. Authentication is delegated to Firebase Auth; this table mirrors essential identity information and links to the Firebase UID.

Data Type	Constraints	Default	Descrip
UUID	PRIMARY KEY	uuid_generate_v4()	Internal
VARCHAR(128)	UNIQUE, NOT NULL		Firebase
VARCHAR(255)	UNIQUE, NOT NULL		User's e
VARCHAR(200)	NOT NULL		Display
TIMESTAMPTZ	NOT NULL	NOW()	
TIMESTAMPTZ	NULL		Updated

Indexes:

- `idx_users_firebase` on `(firebase_uid)` – lookup after authentication.
- `idx_users_email` on `(email)`.

TypeORM Entity:

```
import { Entity, Column, PrimaryGeneratedColumn, CreateDateColumn, Index } from 'typeorm';

@Entity('users')

export class User {

  @PrimaryGeneratedColumn('uuid')

  id: string;

  @Index({ unique: true })

  @Column('varchar', { length: 128 })

  firebaseUid: string;
```

```
  @Index({ unique: true })

  @Column('varchar', { length: 255 })

  email: string;

  @Column('varchar', { length: 200 })

  fullName: string;

  @CreateDateColumn({ type: 'timestamptz' })

  createdAt: Date;

  @Column('timestamptz', { nullable: true })

  lastLogin: Date;

}
```

5.3.2 audit_log

Purpose: Immutable record of every significant system and user action. Supports arbitration and compliance. This table is append-only (no UPDATEs or DELETEs). Optionally converted to a TimescaleDB hypertable for efficient retention management.

	Data Type	Constraints	Default	Descrip
	UUID	PRIMARY KEY	uuid_generate_v4()	
	TIMESTAMPTZ	NOT NULL	NOW()	When th
	UUID	NULL, FK -> users.id		Null for s
	VARCHAR(100)	NOT NULL		e.g., "SC "EVENT "LAY_C/
	VARCHAR(50)	NOT NULL		e.g., "Vo
	UUID	NOT NULL		ID of the

	Data Type	Constraints	Default	Descrip
	JSONB	NULL		Previous
	JSONB	NULL		New sta
	INET	NULL		Client IP
	JSONB	NULL		Addition agent).

Indexes:

- `idx_audit_timestamp` on `(timestamp DESC)` – time-range queries.
- `idx_audit_entity` on `(entity_type, entity_id)` – find all changes to a record.

TypeORM Entity:

import { Entity, Column, PrimaryGeneratedColumn, ManyToOne, JoinColumn } from 'typeorm';

import { User } from './users.entity';

@Entity('audit_log')

export class AuditLog {

 @PrimaryGeneratedColumn('uuid')

 id: string;

 @Column('timestamptz', { default: () => 'NOW()' })

 timestamp: Date;

 @Column('uuid', { nullable: true })

 userId: string;

 @ManyToOne(() => User, { nullable: true })

 @JoinColumn({ name: 'userId' })

 user: User;

```
  @Column('varchar', { length: 100 })
  action: string;
  @Column('varchar', { length: 50 })
  entityType: string;
  @Column('uuid')
  entityId: string;
  @Column('jsonb', { nullable: true })
  oldValue: any;
  @Column('jsonb', { nullable: true })
  newValue: any;
  @Column('inet', { nullable: true })
  ipAddress: string;
  @Column('jsonb', { nullable: true })
  metadata: any;
}
```

5.3.3 `ai_interactions`

Purpose: Stores full prompts, responses, and context for every Defender AI chat interaction. Used for audit, feedback, and model improvement.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`UUID`	`NOT NULL, FK -> users.id`		
	`VARCHAR(100)`	`NOT NULL`		Groups

	Data Type	Constraints	Default	Descrip
	TEXT	NOT NULL		Full pror (includir
	TEXT	NOT NULL		LLM res
	JSONB	NULL		Array of RAG.
	VARCHAR(50)	NOT NULL		e.g., "gp "claude-
	INTEGER	NULL		User-pro
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- `idx_ai_interactions_user` on `(user_id, created_at DESC)`.
- `idx_ai_interactions_session` on `(session_id)`.

TypeORM Entity:

```
import { Entity, Column, PrimaryGeneratedColumn, CreateDateColumn, ManyToOne,
JoinColumn } from 'typeorm';

import { User } from './users.entity';

@Entity('ai_interactions')

export class AiInteraction {

  @PrimaryGeneratedColumn('uuid')

  id: string;

  @Column('uuid')

  userId: string;

  @ManyToOne(() => User)

  @JoinColumn({ name: 'userId' })
```

```
  user: User;

  @Column('varchar', { length: 100 })

  sessionId: string;

  @Column('text')

  prompt: string;

  @Column('text')

  response: string;

  @Column('jsonb', { nullable: true })

  retrievedContext: any;

  @Column('varchar', { length: 50 })

  modelVersion: string;

  @Column('int', { nullable: true })

  feedbackScore: number;

  @CreateDateColumn({ type: 'timestamptz' })

  createdAt: Date;

}
```

5.3.4 `feedback_signals`

Purpose: Explicit corrective feedback from users on AI predictions, recommendations, or answers. Feeds into the Feedback & Learning Loop.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	

Data Type	Constraints	Default	Descrip
VARCHAR(30)	NOT NULL		e.g., "pr "recomn
UUID	NOT NULL		ID of the score, A
UUID	NOT NULL, FK -> users.id		Who pro
VARCHAR(10)	NOT NULL		"positive
TEXT	NULL		User's e value.
TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- idx_feedback_source on (source_type, source_id).

TypeORM Entity:

```
import { Entity, Column, PrimaryGeneratedColumn, CreateDateColumn, ManyToOne,
JoinColumn } from 'typeorm';

import { User } from './users.entity';

@Entity('feedback_signals')

export class FeedbackSignal {

  @PrimaryGeneratedColumn('uuid')

  id: string;

  @Column('varchar', { length: 30 })

  sourceType: string;

  @Column('uuid')

  sourceId: string;
```

```
  @Column('uuid')
  userId: string;
  @ManyToOne(() => User)
  @JoinColumn({ name: 'userId' })
  user: User;
  @Column('varchar', { length: 10 })
  rating: string;
  @Column('text', { nullable: true })
  correctionText: string;
  @CreateDateColumn({ type: 'timestamptz' })
  createdAt: Date;
}
```

5.3.5 `knowledge_base_chunks`

Purpose: Stores text chunks and their vector embeddings for Retrieval-Augmented Generation (RAG). Supports the Defender AI's ability to query maritime documents. Uses the **pgvector** extension.

	Data Type	Constraints	Default	Descrip
	`UUID`	`PRIMARY KEY`	`uuid_generate_v4()`	
	`VARCHAR(30)`	`NOT NULL`		e.g., "cp "carrier_
	`VARCHAR(500)`	`NOT NULL`		Referen docume "CP-123
	`TEXT`	`NOT NULL`		The raw

	Data Type	Constraints	Default	Descrip
	VECTOR(1536)	NOT NULL		Vector e depends
	TIMESTAMPTZ	NOT NULL	NOW()	

Indexes:

- `idx_kb_source` on `(source_type, source_ref)`.
- **Vector Index:** Created via raw SQL for similarity search (e.g., IVFFlat).

TypeORM Entity:

```
import { Entity, Column, PrimaryGeneratedColumn, UpdateDateColumn } from 'typeorm';

import { Vector } from 'pgvector/typeorm'; // assuming pgvector TypeORM support

@Entity('knowledge_base_chunks')

export class KnowledgeBaseChunk {

  @PrimaryGeneratedColumn('uuid')

  id: string;

  @Column('varchar', { length: 30 })

  sourceType: string;

  @Column('varchar', { length: 500 })

  sourceRef: string;

  @Column('text')

  textChunk: string;

  @Column('vector', { length: 1536 }) // Adjust dimension as needed

  embedding: Vector;

  @UpdateDateColumn({ type: 'timestamptz' })
```

lastUpdated: Date;

}

That completes all table definitions for the three domains. The Database Design Document now covers every entity from the SRS with precise column types, constraints, indexes, and TypeORM code. Next, I can proceed to **Section 6: Extensions & Special Data Types** (TimescaleDB hypertable setup, vector index configuration) and **Section 7: Data Retention & Archiving**. Shall I continue?

Continuing the Database Design Document, here are **Section 6: Extensions & Special Data Types** and **Section 7: Data Retention & Archiving**.

---

6. Extensions & Special Data Types

This section details how PostgreSQL extensions and specialised data types are configured and utilised to meet the platform's requirements.

6.1 Enabling Extensions

All required extensions must be enabled in the target database, typically via the initial migration:

CREATE EXTENSION IF NOT EXISTS "uuid-ossp"; -- UUID generation

CREATE EXTENSION IF NOT EXISTS "timescaledb"; -- Time-series hypertables

CREATE EXTENSION IF NOT EXISTS "vector"; -- pgvector for embeddings

6.2 TimescaleDB – Hypertables for Time-Series Data

Two tables store high-volume, append-only time-series data:

- `container_milestones` (defined in 5.2.4)
- `vessel_positions` (to be added for AIS tracking, similar structure)

Both are converted into hypertables to optimise storage and query performance.

Conversion to Hypertable:

After the standard TypeORM migration creates the base table, a separate raw SQL migration executes:

-- For container milestones

SELECT create_hypertable('container_milestones', 'time', chunk_time_interval => INTERVAL '1 day');

-- For vessel positions (identical pattern)

SELECT create_hypertable('vessel_positions', 'time', chunk_time_interval => INTERVAL '1 day');

Hypertable Configuration:

- **Chunk interval:** 1 day – balances write throughput with query pruning for typical dashboards (7-30 day lookbacks).
- **Compression:** Enabled after 7 days on chunks to reduce storage by 10-20x. Compression is applied via a scheduled policy:

  ALTER TABLE container_milestones SET (

    timescaledb.compress,

    timescaledb.compress_segmentby = 'container_id',

    timescaledb.compress_orderby = 'time DESC'

  );

  SELECT add_compression_policy('container_milestones', INTERVAL '7 days');

- **Continuous Aggregates (future):** For high-traffic dashboards, materialised views can be defined (e.g., hourly milestone counts per terminal).

6.3 pgvector – Vector Search for RAG

The `knowledge_base_chunks` table (5.3.5) stores text embeddings using the `VECTOR` type.

Creating the Vector Column:

CREATE TABLE knowledge_base_chunks (

```
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),

    -- ... other columns

    embedding VECTOR(1536) NOT NULL

);
```

Vector Index for Similarity Search:

An index is essential for sub-second retrieval. Use the IVFFlat index (or HNSW) after the table is sufficiently populated:

```
-- Build IVFFlat index with cosine distance (recommended for OpenAI embeddings)

CREATE INDEX ON knowledge_base_chunks USING ivfflat (embedding vector_cosine_ops) WITH (lists = 100);
```

- **Dimension:** 1536 (OpenAI `text-embedding-3-small`). Adjust if using another model.
- **Lists:** Set to `sqrt(n)` where *n* is the number of chunks; update as data grows.

All vector operations (similarity search) are performed through the TypeORM repository using raw SQL queries or the `pgvector` TypeORM extension.

6.4 JSONB Columns – Flexible Structure

Several tables use `JSONB` to store semi-structured data without requiring schema changes:

- `cp_clauses.parameters` – stores clause-specific parameters like `{"rate": 10000, "unit": "MT", "shex": true}`.
- `carrier_tariffs.detention_tiers` / `demurrage_tiers` – arrays of tier objects.
- `audit_log.old_value` / `new_value` – captures row snapshots.

TypeORM maps these natively:

```
@Column('jsonb')

parameters: Record<string, any>;
```

JSONB allows indexing and querying within the JSON structure using GIN indexes where needed (e.g., `CREATE INDEX idx_clause_params ON cp_clauses USING GIN (parameters);`).

6.5 INTERVAL Data Type

PostgreSQL's native `INTERVAL` type is used for:

- `laytime_calculations.allowed_laytime`
- `laytime_calculations.used_laytime`
- `free_time_clocks.free_time_used`

This enables direct arithmetic (e.g., `used_laytime - allowed_laytime`) to determine overruns without manual conversion to hours/minutes. TypeORM handles `INTERVAL` as a string (e.g., `'6 days 3 hours'`). Input validation in the application layer must ensure ISO 8601 duration format or PostgreSQL-compatible strings.

6.6 TIMESTAMPTZ

All timestamp columns use `TIMESTAMPTZ` (timestamp with time zone). Values are stored in UTC internally. The application must convert to/from UTC. This avoids ambiguity across vessels in different time zones and simplifies audit logging.

---

7. Data Retention & Archiving

7.1 General Retention Policy

To meet maritime arbitration requirements (typical time bars of up to 6 years), the following retention periods apply from the date of voyage completion or container return:

- **Operational data:** 7 years
- **Audit logs:** 7 years (matched to operational data)
- **AI interaction logs:** 7 years (legal/quality review)

7.2 Time-Series Data Management

Hypertables (`container_milestones`, `vessel_positions`) leverage TimescaleDB's built-in data management features.

Automated Dropping of Old Chunks: After the retention period, data can be automatically purged:

SELECT add_retention_policy('container_milestones', INTERVAL '7 years');

SELECT add_retention_policy('vessel_positions', INTERVAL '7 years');

Compression: Already applied to older chunks (see 6.2) to minimise storage costs.

7.3 Relational Data Archival

For standard tables (e.g., `voyages`, `sof_events`, `dd_invoices`), the primary strategy is to keep all data online within PostgreSQL, relying on proper indexing and partitioning (if necessary) for performance. Given the expected data volumes per single-tenant instance, full-scale archiving to cold storage is not required initially.

If archival to S3 becomes necessary in the future, the following approach is recommended:

1. Identify entities with a `status` of 'Completed' or 'Closed' older than 7 years.
2. Export related records as JSON/CSV to S3 with a lifecycle policy (Glacier).
3. Delete the records from the operational database.

This process will be managed by a scheduled job.

7.4 Legal Hold

Any record (voyage, container, dispute) can be placed under legal hold. This is implemented by adding a `legal_hold` flag (`BOOLEAN DEFAULT false`) to core entities that may be subject to litigation. When set to `true`, automated retention policies and deletion jobs must skip that record and all related child records.

7.5 Deletion & Cascading

Hard deletes are the default. Foreign key constraints are defined with `ON DELETE CASCADE` where appropriate (e.g., deleting a `sof_document` removes its `sof_events`). Application-level checks prevent accidental deletion of active records.

Mermaid Diagrams

Context Diagram (SRS)

```
---
config:
 layout: elk
---
flowchart TB
subgraph Modules["Core Modules"]
       LD["Laytime Defender<br>Bulk"]
       DD["Detention Defender<br>Container"]
       AI["Defender AI<br>Co-pilot"]
 end
subgraph Infrastructure["Shared Infrastructure"]
       Ingest["Data Ingestion & Integration Hub"]
       Vault["Document Vault & Audit Trail"]
       Analytics["Analytics & Feedback Engine"]
 end
subgraph Platform["Demurrage Defender Platform"]
   direction TB
       Modules
       Infrastructure
 end
   AIS["AIS Data Providers"] -- Vessel positions, port calls<br><i>Predict, Prevent,
Manage</i> --> Ingest
   Carrier["Carrier APIs / EDI"] -- Container milestones, tariffs<br><i>Predict,
Prevent, Manage</i> --> Ingest
   Weather["Weather Services"] -- Weather forecasts, historical<br><i>Predict,
Manage</i> --> Ingest
   Port["Port / Terminal Systems"] -- Berth schedules, gate events<br><i>Predict,
Manage</i> --> Ingest
   User["User / Company Subscriber"] -- Uploads: SOF, NOR, C/P, EIR,
invoices<br>Queries, approvals<br><i>Manage, Dispute, Optimize</i> --> Ingest
   ERP["ERP / CTRM Systems"] -- Cargo, contract data<br><i>Predict, Manage,
Optimize</i> --> Ingest
   Ingest -- Structured data --> LD & DD
   Ingest -- Documents, events --> Vault
   LD -- Laytime calculations, alerts --> AI
   DD -- "Free-time clocks, D&D charges" --> AI
   AI == Recommendations, answers ==> User
   LD == Time sheets, disputes ==> User
   DD -- Dispute packages, reports --> User
```

```
User -- Dispute instructions --> LD & DD
Analytics -- Benchmarks, risk scores<br><i>Predict, Optimize</i> --> LD & DD
LD == Voyage outcomes, claims ==> Analytics
DD -- Shipment outcomes, D&D data --> Analytics
Analytics -- Feedback loop<br><i>Optimize → Predict</i> --> Analytics
LD == Alerts, notifications ==> Comms["Email / Slack / Teams"]
DD -- Alerts, notifications --> Comms
AI == Drafted emails, alerts ==> Comms
LD == Final statements, evidence ==> Legal["Legal / Arbitration Teams"]
DD -- Dispute packages --> Legal
ERP <== Cost data, accruals ==> LD & DD

 LD:::module
 DD:::module
 AI:::module
 Ingest:::module
 Vault:::module
 Analytics:::module
 AIS:::external
 Carrier:::external
 Weather:::external
 Port:::external
 User:::external
 ERP:::external
 Comms:::external
 Legal:::external
classDef external fill:#e0e0e0,stroke:#333,stroke-width:2px
classDef platform fill:#d0e0f0,stroke:#369,stroke-width:2px
classDef module fill:#fff,stroke:#369
style Modules fill:#E1BEE7
style Infrastructure fill:#FFE0B2
style User fill:#BBDEFB
style ERP fill:#BBDEFB
style Platform fill:#C8E6C9
linkStyle 11 stroke:#D50000,fill:none
linkStyle 12 stroke:#00C853
linkStyle 18 stroke:#00C853,fill:none
linkStyle 21 stroke:#00C853,fill:none
linkStyle 23 stroke:#D50000,fill:none
linkStyle 24 stroke:#00C853,fill:none
linkStyle 26 stroke:#369,fill:none
linkStyle 27 stroke:#369,fill:none
```

Architecture Diagram

```
---
config:
 layout: elk
---
flowchart TB
subgraph Client["Client"]
       Browser["Next.js Web App"]
       Mobile["Mobile Browser"]
 end
subgraph Firebase_GCP["Firebase GCP"]
       Auth["Firebase Auth"]
       FCM["Firebase Cloud Messaging"]
 end
subgraph Networking["Networking"]
       APIGW["API Gateway / Load Balancer"]
 end
subgraph Compute_NestJS["NestJS Main API"]
       Gateway["API Gateway Service"]
       LaytimeSvc["Laytime Service"]
       DetentionSvc["Detention Service"]
       DisputeSvc["Dispute Service"]
       OptimizeSvc["Optimize Service"]
       DocSvc["Document Service"]
 end
subgraph Compute_AI["AI Python Services"]
       OCR["OCR & Extraction"]
       PredictML["Predictive Models"]
       Chatbot["Defender AI Chatbot"]
 end
subgraph Data_Stores["Data_Stores"]
       RDS[("PostgreSQL / RDS<br>+ TimescaleDB + pgvector")]
       S3[("S3 - Documents")]
       ElastiCache["ElastiCache / Redis<br>Queue + Cache"]
 end
subgraph External["External"]
       AIS["AIS Providers"]
```

```
        Carriers["Carrier APIs"]
        Weather["Weather Services"]
        ERP["ERP Systems"]
        Email["Email/Slack/Teams"]
 end
subgraph AWS_Cloud["AWS"]
        Networking
        Compute_NestJS
        Compute_AI
        Data_Stores
        External
 end
   Browser --> APIGW
   Mobile --> APIGW
   Browser <--> Auth
   Mobile <--> Auth
   APIGW --> Gateway
   Gateway --> LaytimeSvc & DetentionSvc & DisputeSvc & OptimizeSvc & DocSvc & Chatbot
& ElastiCache & AIS & Carriers & Weather & ERP & Email & FCM
   LaytimeSvc --> OCR & PredictML & RDS & ElastiCache
   DetentionSvc --> PredictML & RDS & ElastiCache
   DisputeSvc --> Chatbot & RDS
   OptimizeSvc --> PredictML & RDS
   DocSvc --> S3
   Chatbot --> RDS
   PredictML --> RDS
   OCR --> S3
   FCM --> Mobile & Browser
```

ERD

Bulk

```
erDiagram
   VESSEL {
       uuid id PK
       string imo
       string name
       string flag
       string type
       int dwt
```

```
    }
    VOYAGE {
        uuid id PK
        uuid vessel_id FK
        uuid charter_party_id FK
        decimal cargo_quantity
        string cargo_type
        string load_port
        string discharge_port
        date laycan_start
        date laycan_end
        string status
        timestamptz created_at
    }
    CHARTER_PARTY {
        uuid id PK
        uuid voyage_id FK
        string form_type
        text full_text
        timestamptz effective_date
    }
    CP_CLAUSE {
        uuid id PK
        uuid charter_party_id FK
        string clause_type
        text raw_text
        jsonb parameters
    }
    SOF_DOCUMENT {
        uuid id PK
        uuid voyage_id FK
        string file_path
        timestamptz upload_date
        string status
    }
    SOF_EVENT {
        uuid id PK
        uuid sof_id FK
        timestamptz event_time
        string event_type
        text remarks
        decimal confidence_score
        bool is_manual_override
        text override_reason
    }
    NOR_DOCUMENT {
        uuid id PK
        uuid voyage_id FK
        string file_path
        timestamptz tender_time
        timestamptz accepted_time
    }
```

```
    LAYTIME_CALCULATION {
        uuid id PK
        uuid voyage_id FK
        int version
        interval allowed_laytime
        interval used_laytime
        decimal demurrage_amount
        decimal despatch_amount
        string status
        timestamptz calculated_at
    }
    CALCULATION_PERIOD {
        uuid id PK
        uuid calculation_id FK
        timestamptz start_time
        timestamptz end_time
        string period_type
        uuid applied_clause_id FK
    }
    DISPUTE_CASE_BULK {
        uuid id PK
        uuid voyage_id FK
        string type
        decimal amount_disputed
        string status
        timestamptz created_date
        timestamptz resolved_date
        decimal final_settlement_amount
    }
    COUNTERPARTY {
        uuid id PK
        string name
        string type
        string imo_company_id
    }

    VESSEL ||--o{ VOYAGE : "has many"
    VOYAGE ||--o| CHARTER_PARTY : "has one"
    VOYAGE ||--o{ SOF_DOCUMENT : "has many"
    VOYAGE ||--o{ NOR_DOCUMENT : "has many"
    VOYAGE ||--o{ LAYTIME_CALCULATION : "has many"
    VOYAGE ||--o{ DISPUTE_CASE_BULK : "has many"
    CHARTER_PARTY ||--o{ CP_CLAUSE : "has many"
    SOF_DOCUMENT ||--o{ SOF_EVENT : "has many"
    LAYTIME_CALCULATION ||--o{ CALCULATION_PERIOD : "has many"
    CALCULATION_PERIOD }o--|| CP_CLAUSE : "references"
```

Container

```
erDiagram
```

```
CONTAINER {
    uuid id PK
    string number UK
    string type
}
SHIPMENT {
    uuid id PK
    string booking_ref
    string bill_of_lading
    string origin
    string destination
    string consignee
    string shipper
    uuid vessel_id FK
    string voyage_ref
}
CONTAINER_MILESTONE {
    uuid id PK
    uuid container_id FK
    string event_type
    timestamptz event_time
    string location
    string source
    bool is_correction
}
CARRIER_TARIFF {
    uuid id PK
    string carrier_id
    string contract_ref
    timestamptz effective_date
    string port
    int free_time_days
    jsonb detention_tiers
    jsonb demurrage_tiers
}
FREE_TIME_CLOCK {
    uuid id PK
    uuid container_id FK
    uuid tariff_id FK
    timestamptz start_time
    timestamptz expiry_time
    interval free_time_used
    decimal demurrage_accrued
    decimal detention_accrued
    string status
}
DD_INVOICE {
    uuid id PK
    string carrier_id
    timestamptz invoice_date
    timestamptz due_date
    decimal total_amount
```

```
        string file_path
    }
    DD_INVOICE_LINE {
        uuid id PK
        uuid invoice_id FK
        uuid container_id FK
        string charge_type
        int claimed_days
        decimal claimed_rate
        decimal claimed_amount
    }
    DISPUTE_CASE_CONTAINER {
        uuid id PK
        uuid invoice_id FK
        decimal amount_disputed
        string status
        timestamptz created_date
        timestamptz resolved_date
        decimal final_settlement_amount
    }

    SHIPMENT ||--o{ CONTAINER : "contains"
    CONTAINER ||--o{ CONTAINER_MILESTONE : "has many"
    CONTAINER ||--o| FREE_TIME_CLOCK : "tracks"
    CARRIER_TARIFF ||--o{ FREE_TIME_CLOCK : "applied to"
    DD_INVOICE ||--o{ DD_INVOICE_LINE : "has many"
    DD_INVOICE_LINE ||--o{ CONTAINER : "charged for"
    DD_INVOICE ||--o| DISPUTE_CASE_CONTAINER : "disputed in"
```

Cross Cutting & AI Domain

```
erDiagram
    USERS {
        uuid id PK
        string email
        string full_name
        string firebase_uid UK
        timestamptz created_at
    }
    AUDIT_LOG {
        uuid id PK
        timestamptz timestamp
        uuid user_id FK
        string action
        string entity_type
        uuid entity_id
        jsonb old_value
        jsonb new_value
        string ip_address
    }
    AI_INTERACTION {
```

```
    uuid id PK
    uuid user_id FK
    string session_id
    text prompt
    text response
    jsonb retrieved_context
    string model_version
    int feedback_score
    timestamptz created_at
}
FEEDBACK_SIGNAL {
    uuid id PK
    string source_type
    uuid source_id
    string rating
    text correction_text
    timestamptz created_at
}
KNOWLEDGE_BASE_CHUNK {
    uuid id PK
    string source_type
    string source_ref
    text text_chunk
    vector embedding(1536)
    timestamptz last_updated
}

USERS ||--o{ AUDIT_LOG : "generates"
USERS ||--o{ AI_INTERACTION : "has"
AI_INTERACTION ||--o{ FEEDBACK_SIGNAL : "receives"
```